

# Strict convexity and differentiability of the planar exponential first-passage limit shape

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September 24, 2026

## Abstract

We prove that the limit shape of undirected nearest-neighbor first-passage percolation on  $\mathbb{Z}^2$  with independent exponential edge weights is strictly convex and has a  $C^1$  boundary. This resolves the strict convexity and differentiability conjectures for the planar exponential model. More generally, we prove differentiability of the time-constant norm for every Gamma edge-weight law with positive shape and rate.

# Contents

|    |                                                             |    |
|----|-------------------------------------------------------------|----|
| 1  | Introduction                                                | 3  |
| I  | Differentiability for Gamma weights                         | 6  |
| 2  | Shape estimates and supporting functionals                  | 6  |
| 3  | Localized crossings                                         | 7  |
| 4  | Characteristic lengths                                      | 13 |
| 5  | Endpoint extensions                                         | 16 |
| 6  | Disjoint witnesses and exclusion of corners                 | 20 |
| 7  | Differentiability and the boundary                          | 26 |
| II | Strict convexity for exponential weights                    | 28 |
| 8  | Exponential passage times and the strict-convexity argument | 28 |
| 9  | Calibration and local roads                                 | 29 |
| 10 | Local gauges for nearly calibrated roads                    | 35 |
| 11 | Phase conventions and locality of the comparison fields     | 43 |
| 12 | Uniform estimates for selected local observations           | 45 |
| 13 | Selected access and charged tunnels                         | 52 |
| 14 | Horizontal links extracted from a long geodesic             | 64 |
| 15 | Improving a road while preserving locality                  | 78 |
| 16 | Excluding flat faces                                        | 88 |
| 17 | Geometric and geodesic consequences                         | 89 |

# 1 Introduction

Give each unoriented nearest-neighbor edge  $e$  of  $\mathbb{Z}^2$  an independent positive weight  $\tau_e$ . The cost  $\tau(P)$  of a finite path is the sum of its edge weights, counted with traversal multiplicity. Write  $\Delta P$  for its terminal vertex minus its initial vertex. Then

$$T(x, y) = \inf_{P: x \rightarrow y} \tau(P)$$

is its passage time. All lattice paths in this definition are undirected. For the laws considered here, the shape theorem gives a deterministic norm  $\mu$  on  $\mathbb{R}^2$ , characterized by

$$\frac{T(0, \lfloor tv \rfloor)}{t} \rightarrow \mu(v) \quad \text{almost surely for each fixed } v \in \mathbb{R}^2.$$

Its unit ball  $\mathcal{B} = \{v : \mu(v) \leq 1\}$  is the limit shape. We write  $|\cdot|$  for the Euclidean norm unless another norm is specified.

**Theorem 1.1** (The exponential limit shape). *For independent exponential edge weights of any rate  $\lambda > 0$ , the norm  $\mu$  is Fréchet differentiable on  $\mathbb{R}^2 \setminus \{0\}$ , and its unit ball is strictly convex. In particular,*

$$\mu((1-t)x + ty) < 1 \quad \text{if } x, y \in \partial\mathcal{B}, \ x \neq y, \ 0 < t < 1,$$

and  $\partial\mathcal{B}$  is a  $C^1$  curve with a unique supporting line at every point.

The differentiability argument applies to a larger family. For  $\kappa, \lambda > 0$ , the Gamma law of shape  $\kappa$  and rate  $\lambda$  has density

$$p_{\kappa, \lambda}(t) = \frac{\lambda^\kappa}{\Gamma(\kappa)} t^{\kappa-1} e^{-\lambda t}, \quad t > 0.$$

The exponential law is the case  $\kappa = 1$ .

**Theorem 1.2** (Differentiability for Gamma weights). *For independent Gamma edge weights with any fixed shape  $\kappa > 0$  and rate  $\lambda > 0$ , the norm  $\mu$  is Fréchet differentiable at every nonzero vector, and  $\partial\mathcal{B}$  is a  $C^1$  curve.*

Strict convexity and differentiability exclude different features of a convex boundary: a flat face has more than one point on a supporting line, whereas a corner has more than one supporting line at a point (Figure 1). Theorem 1.2 does not assume strict convexity. The proof of strict convexity in Part II uses the exponential law and is independent of the differentiability proof.

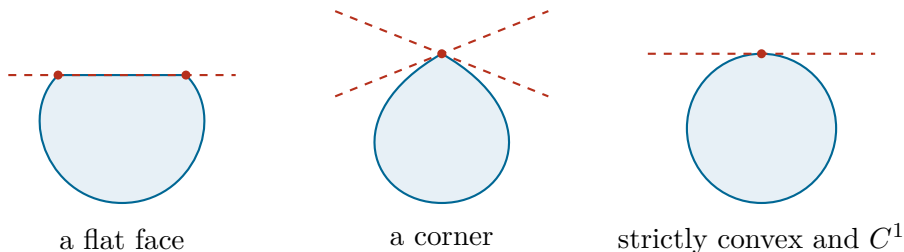


Figure 1: The two obstructions to regularity. These are schematic convex bodies; the circle is not a proposed formula for the limit shape.

## 1.1 History and significance

First-passage percolation was introduced by Hammersley and Welsh [19]. The asymptotic-shape theory developed through the work of Richardson [31] and Cox and Durrett [8], alongside Kingman’s ergodic theory of subadditive processes [23]. Although the shape theorem gives a convex deterministic body, its fine geometry is much harder to determine; see the survey of Auffinger, Damron, and Hanson [5]. Theorem 1.1 resolves positively the strict-convexity and differentiability conjectures for the planar exponential model.

There are important partial results for both questions. Lalley [25] gave sufficient conditions for strict convexity in the standard lattice model, involving asymptotic straightness and mean-zero scaling limits. In Richardson’s site growth model, Durrett and Liggett [14] established a flat edge for parameters sufficiently close to one and identified its onset with the threshold of a related contact process. For planar edge first-passage percolation with support minimum normalized to one and an atom of mass  $p$  there, Marchand [27, Theorem 1.3] determined the exact flat segment on the first-quadrant unit  $\ell^1$  boundary when  $p > \vec{p}_c$ , where  $\vec{p}_c$  is the oriented-percolation threshold, and a single diagonal contact point when  $p = \vec{p}_c$ . For finite-mean laws in this class with  $\vec{p}_c \leq p < 1$ , Auffinger and Damron [4, Theorem 2.1] proved differentiability at the endpoints, including the critical contact point. These last two results concern an atom at a positive minimum; exponential edge weights are nonatomic and have essential infimum zero.

The connection between shape geometry and geodesics was developed by Newman and Licea–Newman [29, 26]. Curvature hypotheses and fixed-direction uniqueness play distinct roles in those results. Damron and Hanson [10] proved convergence and coalescence toward a specified supporting sector under differentiability assumptions. Our two regularity conclusions make that sector a single direction. Section 17 states the resulting geodesic and Busemann conclusions with the direction fixed before the probability-one event is chosen. The midpoint question of Benjamini, Kalai, and Schramm [6] is already answered qualitatively without shape regularity by Ahlberg and Hoffman [2]. We explain its relation to the directional results. For all Gamma laws, differentiability supplies infinitely many extreme points, making the quantitative midpoint estimate of Dembin, Elboim, and Peled [13] applicable as well.

## 1.2 Proof strategy

**Excluding a corner.** Suppose two distinct extreme supporting functionals  $f_-, f_+$  touch the norm in a direction  $u$ , and let  $f = (f_- + f_+)/2$ . Their gap gives a linear penalty for transverse motion. Part I turns this penalty into a positive excess over  $f$  for every sufficiently long simple path, contradicting the shape theorem along  $u$ .

First, a crossing confined to a long narrow strip is unlikely to save a fixed multiple of the strip width over either extreme support. Otherwise independent parallel trials, connected using one-sided first-order contact with the norm, would yield macroscopic travel below that support. Second, a finite-region change of measure produces characteristic lengths  $L$  for widths  $s$ , with  $L/s \rightarrow \infty$  and  $L \leq s^3$ . Nested grids then extend the crossing estimate to arbitrary endpoints, with a summable cost. Third, every cheap path chunk contains a witness for a rare decreasing event. The van den Berg–Kesten inequality [32], applied to disjoint edge witnesses, handles repeated visits to the same coarse box and yields a bound for all long simple paths at once. The counting builds on Kesten’s rectangle itineraries and crossing-multiplicity estimates [22, Section 3]; here the local witnesses certify negative excess relative to a supporting functional.

The Gamma extension is built into this proof. Its distributional inputs are exponential tails for short coordinate paths, a continuous quantile representation for the witness inequality, and an

exact likelihood calculation: scaling one Gamma weight by  $1 - \delta$  gives likelihood second moment  $(1 - \delta^2)^{-\kappa}$ . Scaling  $O(ns)$  weights with  $\delta$  of order  $s/n$  has bounded second moment when  $n = s^3$ . No differentiability or curvature estimate for  $\mu$  enters these inputs.

**Excluding a flat face.** Under a hypothetical flat face, the stationary cocycle construction of Damron and Hanson [9] supplies an additive random function  $B(x, y)$ , with mean the supporting functional and with  $|B(x, y)| \leq T(x, y)$ . Thus a path from  $x$  to  $y$  has nonnegative defect  $\tau(P) - B(x, y)$ . Part II considers local random paths, called roads, tracking two directions near the ends of the face. At lattice scale  $n$ , let  $b(n)$  be the infimum of a common upper bound for their mean defect per period. A local perturbation gives  $b(n) \geq c_0 > 0$ . The main construction gives a fixed integer  $D$  and  $\rho < 1$  such that

$$b(Dn) \leq \rho b(n).$$

Iteration contradicts the positive lower bound.

A finite-cell projection constructs local endpoint potentials approximating increments of the generally nonlocal cocycle; interpolation controls every possible attachment in a bounded road interval (Section 10). A physical routing argument bounds the savings obtainable from a selected collection of local shortcuts (Proposition 13.5). An area-preserving deformation then controls those savings along paths concentrated near an extreme direction (Lemma 12.7). Passage-time concentration and mean approximation estimates of Damron–Hanson–Sosoe [11] and Damron–Kubota [12] supply the quantitative metric input.

These estimates let a long geodesic toward an extreme endpoint supply many crossings between neighboring walls while controlling the discarded pieces. The extraction gives a small upper bound for the mean clipped cost of a local crossing. Comparing this crossing with the current road, including the short wall connections, transfers the clipped bound to a primary candidate. To control the remaining large positive costs, a local rule groups their flags into short clusters and traverses an independently built spare road across each accepted cluster, with forward connections at both ends (Lemma 15.1). A matched cap controls the frequency of flags, while separation and first-moment estimates control the added cost. This reduces the full expected excess per road period. A fixed rescaling restores the required tracking and locality, so the improved paths belong to the class defining  $b(n)$ .

The two arguments use different endpoint corrections. In Part I, the deterministic defect  $\tau(P) - f(\Delta P)$  can be negative. In Part II, the cocycle defect is nonnegative on every forward subpath. Each part introduces its own local coordinates and parameters.

## Part I

# Differentiability for Gamma weights

## 2 Shape estimates and supporting functionals

Fix  $\kappa, \lambda > 0$  throughout this part and write  $\mu = \mu_{\kappa, \lambda}$ . All constants and scale thresholds may depend on this fixed law. We shall compare path costs with the two extreme supporting functionals at a prescribed direction. The crossing estimates use only their one-sided contact with  $\mu$ ; the assumption that they differ enters the final contradiction.

### 2.1 Uniform shape and local sums

We first record the precise form of the standard shape theorem that will be used. It holds for the fixed Gamma law:

$$\sup_{z \in \mathbb{Z}^2: |z| \geq R} \frac{|T(0, z) - \mu(z)|}{|z|} \longrightarrow 0 \quad \text{almost surely as } R \rightarrow \infty. \quad (2.1)$$

This is the shape theorem of [24, Theorem A.1 and Equation (A.3)], at weight shift zero. The Gamma law has essential infimum zero and no atom there. For four independent copies of an edge weight,

$$\mathbb{E}[\min\{\tau_1, \tau_2, \tau_3, \tau_4\}^2] \leq \mathbb{E}[\tau_1^2] = \frac{\kappa(\kappa + 1)}{\lambda^2} < \infty.$$

These facts verify the cited hypotheses. The source uses  $\ell^1$  distance, which gives (2.1) by equivalence of norms. There are deterministic constants  $0 < \alpha \leq \beta < \infty$  such that

$$\alpha|z| \leq \mu(z) \leq \beta|z| \quad (z \in \mathbb{R}^2). \quad (2.2)$$

An event expressed relative to a deterministic lattice origin has the same probability at every translate. This observation will allow us to use shape estimates in probability at origins that vary with the scale. Equation (2.1) itself is an almost-sure statement simultaneously over all endpoints from the fixed origin.

Almost surely every edge weight is finite and positive, since there are only countably many edges. In particular, loop erasure does not increase cost. A path constrained to a set will mean a path all of whose vertices belong to that set; such path-existence events use only its internal edges. Concatenated paths may retrace edges, with each traversal contributing to the cost.

The following bound controls the vertical attachments used below.

**Lemma 2.1** (Gamma exponential moments). *For  $\tau \sim \text{Gamma}(\kappa, \lambda)$  and  $\theta < \lambda$ ,*

$$\mathbb{E}e^{\theta\tau} = \left(\frac{\lambda}{\lambda - \theta}\right)^\kappa.$$

*If  $V$  is the sum over a deterministic set of  $M$  distinct edges and  $M \leq Av$ , where  $A, v > 0$ , then*

$$\mathbb{P}\left(V > \frac{2A\kappa}{\lambda}v\right) \leq \exp\{-A\kappa(1 - \log 2)v\}. \quad (2.3)$$

*Proof.* Integrating the Gamma density with the substitution  $y = (\lambda - \theta)t$  gives the moment identity. At  $\theta = \lambda/2$  it reads  $\mathbb{E}e^{\lambda\tau/2} = 2^\kappa$ . Independence and exponential Markov inequality give

$$\mathbb{P}\left(V > \frac{2A\kappa}{\lambda}v\right) \leq e^{-A\kappa v} 2^{\kappa M} \leq e^{-A\kappa(1 - \log 2)v}.$$

□

## 2.2 The two extreme supports

Signed coordinate permutations preserve the law and hence  $\mu$ . Together with positive homogeneity, these symmetries reduce the proof to directions

$$u = (1, b), \quad -1 \leq b \leq 1.$$

Fix such a deterministic direction for the rest of the argument, and write

$$x(z) = z_1, \quad r(z) = z_2 - bz_1, \quad m_0 = \mu(u) > 0.$$

The function  $t \mapsto \mu(u + te_2)$  is convex and Lipschitz. Let

$$a_- = \lim_{t \uparrow 0} \frac{\mu(u + te_2) - m_0}{t}, \quad a_+ = \lim_{t \downarrow 0} \frac{\mu(u + te_2) - m_0}{t}.$$

These finite limits satisfy  $a_- \leq a_+$ . Define

$$\begin{aligned} f_{\pm}(z) &= m_0 x(z) + a_{\pm} r(z), \\ a &= \frac{a_- + a_+}{2}, \quad c = \frac{a_+ - a_-}{2}, \quad f(z) = m_0 x(z) + ar(z). \end{aligned} \tag{2.4}$$

**Lemma 2.2** (One-sided contact). *The linear functionals  $f_-$ ,  $f_+$ , and  $f$  are bounded above by  $\mu$  on  $\mathbb{R}^2$  and equal  $\mu$  at  $u$ . For  $\sigma \in \{-, +\}$ , with the corresponding numerical sign when multiplied by a scalar,*

$$\mu(u + te_2) - f_{\sigma}(u + te_2) = o(|t|) \quad (t \rightarrow 0, \sigma t > 0). \tag{2.5}$$

*Proof.* Each of  $a_-$  and  $a_+$  is a subgradient at zero of the convex function  $t \mapsto \mu(u + te_2)$ . If  $x(z) > 0$ , write  $z = x(z)(u + r(z)x(z)^{-1}e_2)$  and use positive homogeneity to get  $f_{\sigma}(z) \leq \mu(z)$ . If this inequality failed at any other point  $z$ , then a sufficiently small positive convex combination of  $z$  with  $u$  would still violate it, by convexity, and would have positive first coordinate. This is impossible. The average  $f$  is also bounded above by  $\mu$ . Equation (2.5) is the respective one-sided derivative expansion.  $\square$

The *defect* of a path is its cost minus the value of a supporting functional on its displacement. We shall use

$$H(P) = \tau(P) - f(\Delta P), \quad H_{\sigma}(P) = \tau(P) - f_{\sigma}(\Delta P). \tag{2.6}$$

They are additive under concatenation, and

$$\min\{H_+(P), H_-(P)\} = H(P) - c|r(\Delta P)|. \tag{2.7}$$

The crossing and endpoint estimates will hold for both signs whether or not  $c$  is positive. In Section 6, the gap assumption  $c > 0$  and (2.7) will make transverse exits expensive relative to the average support  $f$ .

## 3 Localized crossings

The main estimate in this section is Lemma 3.4: a crossing confined to a strip of width of order  $s$  is unlikely to have defect below  $-\varepsilon s$  when its length is much larger than  $s$ . The proof amplifies any positive probability of such a crossing into an infinite concatenation whose defect decreases linearly, contradicting the shape theorem.

We prepare three ingredients for this construction. The shape theorem first gives short paths confined near their joining segments. A finite net then makes this assertion simultaneous over all endpoints in two intervals. Finally, an oriented-percolation argument joins sufficiently likely local tests even when nearby tests overlap. Throughout the section, distances are Euclidean unless indicated otherwise. A path constrained to a region has all its vertices there, so its existence and cost are tested using only edges with both endpoints in that region.

**Lemma 3.1** (Localized travel). *Fix  $D, \eta, \delta > 0$ . Uniformly over deterministic vertices  $p, q \in \mathbb{Z}^2$  with  $|q - p| \leq Ds$ , the probability tends to one as  $s \rightarrow \infty$  that there is a path from  $p$  to  $q$  with cost at most*

$$\mu(q - p) + \eta s$$

*whose vertices are within distance  $\delta s$  of the segment  $[p, q]$ .*

*Proof. Uniform shape estimates.* Use the constants  $\alpha, \beta$  from (2.2). Equation (2.1) implies, for each fixed  $C > 0$ ,

$$\frac{1}{s} \max_{\substack{z \in \mathbb{Z}^2 \\ |z| \leq Cs}} |T(0, z) - \mu(z)| \longrightarrow 0 \quad \text{almost surely.} \quad (3.1)$$

Indeed, outside any sufficiently large fixed ball the error is bounded by an arbitrarily small multiple of  $|z|$ , while the finitely many passage times inside that ball contribute  $o(s)$ . The same shape theorem gives

$$\mathbb{P} \left( \inf_{\substack{z \in \mathbb{Z}^2 \\ |z| \geq \delta s/3}} T(0, z) \geq \alpha \delta s/6 \right) \longrightarrow 1. \quad (3.2)$$

Translation invariance gives the same probability bounds at every deterministic lattice origin. This uniformity will allow the origins to depend on the endpoints and on  $s$ .

*Short pieces cannot escape.* Choose a fixed integer  $B$  so large that  $\beta D/B < \alpha \delta/24$ . Round the  $B + 1$  equally spaced points of  $[p, q]$  to vertices  $p_0 = p, p_1, \dots, p_B = q$ . Uniformly in the endpoints,

$$\mu(p_{j+1} - p_j) = \frac{\mu(q - p)}{B} + O(1).$$

Apply (3.1) and (3.2) at these finitely many deterministic vertices. With probability tending to one, uniformly in  $p, q$ , every short passage time is at most

$$\frac{\mu(q - p)}{B} + \min \left\{ \frac{\eta}{2B}, \frac{\alpha \delta}{24} \right\} s,$$

and escape from the ball of radius  $\delta s/3$  about its starting vertex costs at least  $\alpha \delta s/6$ . For each  $j = 0, \dots, B - 1$ , choose a path from  $p_j$  to  $p_{j+1}$  with cost at most  $T(p_j, p_{j+1}) + 1$ . For all sufficiently large  $s$ , each chosen path has cost strictly below the corresponding escape bound. Nonnegativity of the weights forces it to stay inside the ball about  $p_j$ : its initial portion up to a vertex outside would already cost at least the escape bound.

*Concatenation.* Concatenating the  $B$  paths gives total cost at most  $\mu(q - p) + \eta s/2 + B \leq \mu(q - p) + \eta s$ . Each ball is centered within bounded distance of the original segment, so it lies in its  $\delta s$ -neighborhood for large  $s$ . All probability bounds used above are uniform in the deterministic vertices, proving the stated uniformity.  $\square$

The next form permits the endpoints to be selected after the environment is observed. The intervals themselves remain deterministic, and the assertion holds for every endpoint pair on one event of high probability.

**Lemma 3.2** (Travel between intervals). *Fix  $D, \eta, \delta > 0$ . Let  $I_1, I_2$  be deterministic vertical intervals on lattice columns, and suppose that the diameter of  $I_1 \cup I_2$  is at most  $Ds$ . Uniformly over their locations, with probability tending to one as  $s \rightarrow \infty$ , every pair  $p \in I_1 \cap \mathbb{Z}^2$ ,  $q \in I_2 \cap \mathbb{Z}^2$  can be joined by a path of cost at most  $\mu(q - p) + \eta s$  within distance  $\delta s$  of  $[p, q]$ .*

*Proof.* The assertion is vacuous if either interval contains no vertex.

*Finite nets and vertical attachments.* Let  $\beta$  be the upper norm bound in (2.2), and choose a fixed  $\gamma > 0$  so small that

$$\left(\frac{4\kappa}{\lambda} + 2\beta\right)\gamma \leq \eta/2, \quad \gamma \leq \delta/2.$$

In each interval, choose a net of lattice vertices containing its extreme lattice vertices and having consecutive gaps at most  $\lfloor \gamma s \rfloor$ . For all large  $s$ , each net needs at most  $2D/\gamma + 2$  vertices. Assign every interval vertex to a net vertex at vertical distance at most  $\gamma s$ , using only edges of its intervening gap.

For any such gap, the total weight  $V$  is a sum of at most  $\lfloor \gamma s \rfloor$  independent edge weights. Lemma 2.1, with  $A = \gamma$  and  $v = s$ , gives

$$\mathbb{P}\left(V > \frac{2\kappa}{\lambda}\gamma s\right) \leq e^{-\kappa(1-\log 2)\gamma s}.$$

There are only boundedly many gaps. Thus, with probability tending to one, all vertical connections from interval vertices to their chosen net vertices cost at most  $2\kappa\gamma s/\lambda$ .

*Connections between net vertices.* Apply Lemma 3.1, with errors  $\eta/2$  and  $\delta/2$ , to every pair of net vertices. Its uniformity and the bounded number of pairs give these connections simultaneously.

*All endpoint pairs on the same event.* On the intersection of these high-probability events, take arbitrary  $p \in I_1 \cap \mathbb{Z}^2$  and  $q \in I_2 \cap \mathbb{Z}^2$ . Concatenate their vertical attachments with the path between the assigned net vertices  $\hat{p}, \hat{q}$ . The resulting cost is at most

$$\mu(\hat{q} - \hat{p}) + \eta s/2 + \frac{4\kappa}{\lambda}\gamma s \leq \mu(q - p) + \eta s/2 + \left(\frac{4\kappa}{\lambda} + 2\beta\right)\gamma s.$$

The segments  $[p, q]$  and  $[\hat{p}, \hat{q}]$  have Hausdorff distance at most  $\gamma s$ . The choice of  $\gamma$  therefore gives both the cost bound and the required localization, including the vertical attachments. The same event works for every  $p, q$ , and its probability bound is uniform in the deterministic locations of the intervals.  $\square$

We will place local crossing and connection tests on an auxiliary copy of  $\mathbb{Z}^2$ . The following elementary lemma needs only a uniform bound on their failure probabilities and independence at sufficiently separated sites.

**Lemma 3.3** (High-density oriented paths). *For every integer  $k_0 \geq 0$  there is  $\zeta = \zeta(k_0) > 0$  with the following property. Let  $(G_v)_{v \in \mathbb{Z}^2}$  be events such that any finite family indexed by sites at pairwise sup distance greater than  $k_0$  is independent. If  $\mathbb{P}(G_v^c) \leq \zeta$  for every  $v$ , then with positive probability there is an infinite path starting at 0, using steps  $(1, 0)$  and  $(0, 1)$ , all of whose sites satisfy  $G_v$ .*

*Proof.* A finite reachable set if there is no infinite path. Call a site good when  $G_v$  occurs. Let  $S$  consist of the endpoints of finite oriented paths from 0 for which every site except possibly the last is good. In particular  $0 \in S$ . If  $S$  were infinite, the finitely branching tree of these paths would have an infinite branch; every site on that branch would be good. Hence, on the event of no infinite good path,  $S$  is finite.

*A surrounding contour forces bad sites.* For such a finite  $S$ , orient the boundaries of the unit squares centered at its sites counterclockwise and cancel shared sides. The remaining edges form a balanced directed graph and hence decompose into closed directed trails, with no edge repeated in a trail. Repeated vertices are allowed, so corner contacts cause no difficulty. Every edge has an  $S$ -square on its left and a complementary square on its right. Winding number is additive under cancellation and decomposition. Its total about 0 is one, contributed by the square centered at 0; therefore some trail has nonzero winding.

Let  $m$  be the length of this trail. Closedness implies that exactly  $m/2$  of its steps are upward or leftward. For these steps, the site on the right is respectively an east or north neighbor of the site on the left. The left-hand site must be bad: if it were good, extending a path that reaches it would put this neighbor in  $S$  as well.

*Independent bad sites along a fixed contour.* A site borders at most four edges of the trail, so at least  $m/8$  distinct sites are forced to be bad. For any fixed candidate trail these sites are deterministic. Greedy selection in a fixed order gives at least

$$\frac{m}{8(2k_0 + 1)^2}$$

of them at pairwise sup distance greater than  $k_0$ : each selected site removes at most  $(2k_0 + 1)^2$  candidates. Their tests are independent, so the probability of all the badness conditions forced by this particular trail is at most  $\zeta^{m/[8(2k_0+1)^2]}$ .

*Summing over contours.* Nonzero winding forces 0 into the coordinate bounding box of the trail. A length- $m$  trail has coordinate spans at most  $m$ , so it lies in a square of side  $O(m)$  about 0. Choosing its starting vertex and then its step sequence bounds the number of candidates by  $C(m+1)^2 4^m$ . Consequently the probability of no infinite good path is at most

$$\sum_{m \geq 4} C(m+1)^2 \left( 4\zeta^{1/[8(2k_0+1)^2]} \right)^m.$$

This is less than one when  $\zeta$  is sufficiently small. The argument uses no stationarity assumption on the tests.  $\square$

We now combine these ingredients. The relevant defect is measured against one of the extremal supports  $f_\sigma$ . Its one-sided contact with  $\mu$  allows successive crossings to be connected with defect as small a multiple of  $s$  as needed, by drifting consistently to the corresponding side of the direction  $u$ .

**Lemma 3.4** (Rare fast localized crossings). *Fix  $A, R, \varepsilon > 0$  and  $\sigma \in \{-, +\}$ . As  $s \rightarrow \infty$  and  $n/s \rightarrow \infty$ , with  $s$  real and  $n$  a positive integer,*

$$\mathbb{P} \left( \begin{array}{l} \text{there is a path } P \text{ from } x = 0 \text{ to } x = n \text{ with } H_\sigma(P) < -\varepsilon s, \\ \text{all of whose vertices satisfy } -An \leq x \leq (1+A)n, \quad |r| \leq Rs \end{array} \right) \rightarrow 0. \quad (3.3)$$

*The same assertion holds, with the same probability, for every exact lattice translate of the event.*

*Proof.* Suppose that the probability in (3.3) is at least  $p_0 > 0$  along a sequence satisfying the stated limits. At each auxiliary site we will try several disjoint translates of this crossing event and require connections to both forward neighbors. Enough trials make a successful crossing likely; the one-sided contact estimate makes the connections likely without spending the whole crossing deficit.

1. *Parameters and site geometry.* The dependence radius must be fixed before we choose how many trials to make. Set

$$k_0 = \lceil 4(A + 3) + 8 \rceil + 1$$

and let  $\zeta$  be supplied by Lemma 3.3. Choose a fixed integer  $M$  so large that  $(1 - p_0)^M < \zeta/2$ . For each member of the sequence put

$$S_0 = 100M(R + 1)\lceil s \rceil, \quad d = \lceil D_0 s \rceil,$$

where the fixed constant  $D_0$  will be chosen below, after  $M$ .

Identify the signs  $+$  and  $-$  with 1 and  $-1$  when they multiply coordinates. For a test site  $v = (v_1, v_2) \in \mathbb{Z}^2$ , write

$$i = v_1 + v_2, \quad X_i = i(n + d), \quad Y_v = \sigma(i + v_2)S_0.$$

An oriented step sends  $i$  to  $i + 1$  and changes  $Y_v$  by  $\sigma S_0$  or  $2\sigma S_0$ . Thus every step advances to the next longitudinal block and moves transversely to the same prescribed side.

2. *Independent crossing trials within a site.* At site  $v$ , make  $M$  trials using exact lattice translates of the event in (3.3), all with initial column  $X_i$ , and with their strips disjoint and contained in  $|r - Y_v| \leq S_0/4$ . Here is an explicit choice, which also accounts for lattice phases. Put

$$h_j = (2j - M - 1)3(R + 1)\lceil s \rceil, \quad t_{v,j} = \left( X_i, \left\lfloor bX_i + Y_v + h_j + \frac{1}{2} \right\rfloor \right), \quad 1 \leq j \leq M.$$

The translated strip has center  $r(t_{v,j})$ , within  $1/2$  of  $Y_v + h_j$ . For  $s \geq 1$ , consecutive centers are separated by more than  $2Rs$ , and every strip of half-width  $Rs$  lies inside the designated  $S_0/4$  window. Thus the trials use disjoint internal edge sets. Translation leaves both displacement and its supporting functional unchanged, so each trial has probability at least  $p_0$ , exactly as in the original event. Their probability of all failing is at most  $(1 - p_0)^M$ .

3. *Connection tests for every endpoint pair.* For each of the two forward neighbors  $v'$  of  $v$ , impose an additional test. Every vertex on

$$x = X_i + n, \quad |r - Y_v| \leq S_0/4$$

must be connectible to every vertex on

$$x = X_{i+1}, \quad |r - Y_{v'}| \leq S_0/4$$

by a path of cost at most  $f_\sigma$  of its displacement plus  $\varepsilon s/2$ , lying in

$$X_i + n - s \leq x \leq X_{i+1} + s, \quad |r - Y_v| \leq 4S_0. \quad (3.4)$$

These are existence tests using only the internal edges of this region. Requiring every endpoint pair will permit the successful trials to be chosen later, after the environment has been observed.

4. *Choosing the connector length.* We next choose  $D_0$  and verify that both connection tests pass with probability tending to one uniformly in  $v$ . A connector displacement  $z$  satisfies

$$x(z) = d, \quad S_0/2 \leq \sigma r(z) \leq 5S_0/2 \leq 3S_0.$$

Set  $C_0 = 200M(R + 1)$ , so that  $S_0/s \leq C_0$  for  $s \geq 1$ . The one-sided expansion (2.5) supplies  $h_0 > 0$  such that

$$0 \leq \mu(u + he_2) - f_\sigma(u + he_2) \leq \frac{\varepsilon}{12C_0}|h| \quad \text{if } 0 < \sigma h \leq h_0.$$

Choose  $D_0 \geq 1$  with  $3C_0/D_0 \leq h_0$ . For every possible connector,  $h = r(z)/d$  is on the required side and has absolute value at most  $h_0$ . Homogeneity therefore gives

$$\mu(z) - f_\sigma(z) \leq \frac{\varepsilon}{12C_0} |r(z)| \leq \varepsilon s/4.$$

This leaves another  $\varepsilon s/4$  for the travel error.

The two endpoint intervals have union diameter at most  $Ds$  for a fixed  $D$  depending only on  $M, R, D_0$ . Their joining segments have longitudinal coordinates between the two endpoint columns and satisfy  $|r - Y_v| \leq 9S_0/4$ . Since  $|b| \leq 1$ , moving Euclidean distance  $\delta s$  changes  $x$  by at most  $\delta s$  and  $r$  by at most  $\sqrt{2}\delta s$ . Thus a tube with any fixed  $0 < \delta \leq 1/2$  fits inside (3.4). Lemma 3.2, with cost error  $\varepsilon s/4$ , proves the required connection assertions simultaneously for all endpoint pairs. Its probability estimate is uniform in the intervals' locations, including their lattice phases.

5. *A dependence radius independent of the later choices.* Call  $v$  good when at least one of its trials succeeds and both connection tests pass. Its bad probability is at most  $(1 - p_0)^M + o(1)$ , uniformly in  $v$ . We verify that the initially chosen  $k_0$  is a dependence radius. Take a sufficiently large member of the sequence that  $n \geq d + s$ . The entire test at  $v$  is supported on internal edges in

$$|x - X_i| \leq (A + 3)n, \quad |r - Y_v| \leq 4S_0.$$

To compare two sites, use the integer coordinates  $i = v_1 + v_2$  and  $j = i + v_2$ . If their support regions share a vertex, the longitudinal and transverse bounds give, respectively,

$$|\Delta i| \leq 2(A + 3), \quad |\Delta j| \leq 8,$$

where  $\Delta$  denotes the difference between the two sites. Since  $v_2 = j - i$  and  $v_1 = 2i - j$ , this implies

$$|\Delta v_2| \leq 2(A + 3) + 8, \quad |\Delta v_1| \leq 4(A + 3) + 8 < k_0.$$

Tests at sites at pairwise sup distance greater than  $k_0$  therefore use pairwise disjoint edge sets and are jointly independent. In particular, the dependence radius does not increase when  $M$  or  $D_0$  is chosen. The order of choices was consequently legitimate:  $k_0$  and  $\zeta$  first, then  $M$ , then  $D_0$ , and finally a sufficiently large member of the sequence. For that member the bad probability is at most  $\zeta$ .

6. *Concatenation at one fixed scale.* By Lemma 3.3, with positive probability there is an infinite oriented path of good sites starting at 0. Fix this one scale  $s, n, d, S_0$  and a configuration with such a path. At each visited site choose a successful trial and join it to the next by a tested connector. Prepend a vertical path from the lattice origin to the initial trial's starting vertex; its defect  $H_\sigma$  is a finite random quantity  $C$ . If  $z_t$  is the endpoint of the trial at the  $t$ th site, with the initial site numbered 0, additivity gives a path from 0 to  $z_t$  of defect at most

$$C - (t + 1)\varepsilon s + t\varepsilon s/2. \tag{3.5}$$

The trials and connectors need not be disjoint: costs are counted with edge multiplicities, so their concatenation still bounds passage time from above.

7. *The shape-theorem contradiction.* At that site,  $i = t$  and  $0 \leq v_2 \leq t$ . Hence

$$x(z_t) = t(n + d) + n, \quad |r(z_t)| \leq 2tS_0 + S_0/4.$$

Thus  $|z_t| \rightarrow \infty$  and  $|z_t| \leq C'(t + 1)$  for a deterministic constant  $C'$  at this fixed scale. The limit here is  $t \rightarrow \infty$ ; the parameters  $s, n, d, S_0$  remain fixed. On the probability-one event of the origin

shape theorem, its uniform outside-ball estimate applies also to these adaptively chosen endpoints and yields

$$T(0, z_t) \geq \mu(z_t) - o(t) \geq f_\sigma(z_t) - o(t),$$

where the supporting inequality is from Lemma 2.2. This contradicts (3.5), whose negative part grows at least as  $t\epsilon s/2$ . The infinite-path event consequently has probability zero, contradicting its construction. No sequence with crossing probability bounded below by  $p_0$  exists, proving the asserted joint limit. Exact lattice-translation invariance proves the last assertion.  $\square$

## 4 Characteristic lengths

The rare-fast estimate concerns crossings with a negative defect. We next identify a length at which crossings are rare even with a positive allowance of order the strip width. A finite-region change of measure supplies an upper bound on that length. Its definition then guarantees enough shorter crossings to construct the endpoint extensions in the next section.

Small multiplicative changes of many independent weights, with a controlled quadratic probability cost, also appear in fluctuation arguments; see Chatterjee [7]. Here the exact Gamma likelihood calculation is valid for every positive shape parameter.

### 4.1 A first crossing threshold

Set

$$K = 100(1 + |a_-| + |a_+|).$$

For real  $s \geq 1$  and integer  $n \geq 1$ , let  $E(s, n)$  be the event that there is a finite path  $P$  from the column  $x = 0$  to the column  $x = n$ , with every vertex in

$$\{-n \leq x \leq 2n, |r| \leq 8s\},$$

and with

$$\tau(P) < m_0 n + Ks.$$

Endpoints on the two columns are unrestricted within this region. As throughout, the event tests only the internal edges of the indicated region. The notation  $E(s, n)$  will also be used with an explicitly specified lattice translation.

**Lemma 4.1** (Rarity at cubic length). *For  $s = 2^k$  and  $k \rightarrow \infty$ ,*

$$\mathbb{P}(E(s, s^3)) \rightarrow 0.$$

*Proof. Turning an allowed crossing into a fast crossing.* Put  $n = s^3$ . Choose a fixed  $\eta > 0$  such that

$$\eta m_0 > K + 16|a_+| + 1,$$

and set  $\delta = \eta s/n$  and  $\alpha = 1 - \delta$ . Take  $s$  sufficiently large that  $0 < \delta \leq 1/2$ . Multiply every internal edge weight of the region defining  $E(s, n)$  by  $\alpha$ , leaving the other weights unchanged. If  $P$  witnesses  $E(s, n)$  before this operation, its transverse displacement obeys  $|r(\Delta P)| \leq 16s$ , and in the modified environment

$$\begin{aligned} H_+(P) &< \alpha(m_0 n + Ks) - m_0 n - a_+ r(\Delta P) \\ &\leq (K - \eta m_0 + 16|a_+|)s - \eta K \frac{s^2}{n} < -s. \end{aligned}$$

This computation includes every traversal if  $P$  repeats an edge. Let  $F(s, n)$  be the crossing event  $H_+(P) < -s$  in the same region, and let  $Q$  be the law of the modified environment. The deterministic implication gives

$$\mathbb{P}(E(s, n)) \leq Q(F(s, n)). \quad (4.1)$$

*The cost of the change of measure.* Under  $Q$ , a modified edge has distribution  $\text{Gamma}(\kappa, \lambda/\alpha)$ . On  $t > 0$  its likelihood ratio with respect to the original edge law is

$$L_\delta(t) = \frac{p_{\kappa, \lambda/\alpha}(t)}{p_{\kappa, \lambda}(t)} = \alpha^{-\kappa} \exp\{-\lambda(\alpha^{-1} - 1)t\}.$$

Both densities are positive on  $(0, \infty)$ , including when  $0 < \kappa < 1$ . Direct integration gives

$$\begin{aligned} \mathbb{E}[L_\delta(\tau_e)^2] &= \alpha^{-2\kappa} \frac{\lambda^\kappa}{\Gamma(\kappa)} \int_0^\infty t^{\kappa-1} e^{-\lambda(2/\alpha-1)t} dt \\ &= \alpha^{-2\kappa} (2/\alpha - 1)^{-\kappa} = (\alpha(2 - \alpha))^{-\kappa} = (1 - \delta^2)^{-\kappa}. \end{aligned}$$

The region has at most  $(3n + 1)(16s + 2)$  vertices and at most twice that many internal unoriented edges. Thus the number  $M$  of modified coordinates is at most  $144ns$ . The likelihood ratio is a product over these distinct coordinates, regardless of path multiplicity. By independence,

$$\begin{aligned} \mathbb{E} \left[ \left( \frac{dQ}{d\mathbb{P}} \right)^2 \right] &= (1 - \delta^2)^{-\kappa M} \leq \exp(2\kappa M \delta^2) \\ &\leq \exp \left( 288\kappa \eta^2 \frac{s^3}{n} \right) = \exp(288\kappa \eta^2). \end{aligned}$$

*Transfer back to the original law.* Lemma 3.4, applied under the original law with  $A = 1$ ,  $R = 8$ , and  $\varepsilon = 1$ , gives  $\mathbb{P}(F(s, n)) \rightarrow 0$  because  $n/s = s^2 \rightarrow \infty$ . Cauchy–Schwarz therefore gives

$$\mathbb{P}(E(s, n)) \leq Q(F(s, n)) \leq e^{144\kappa \eta^2} \mathbb{P}(F(s, n))^{1/2} \rightarrow 0.$$

All supports and the norm in  $F(s, n)$  are those of the original fixed law; no shape theorem for the modified environment is used.  $\square$

Fix any probability threshold  $q_0 \in (0, 1)$  for this section and the next. It may be arbitrarily small; Section 6 will specify its value. Write  $s_k = 2^k$ .

**Lemma 4.2** (Characteristic lengths). *For every fixed  $C > 0$ ,*

$$\mathbb{P}(E(s, n)) \rightarrow 1 \quad \text{as } s \rightarrow \infty, \quad 1 \leq n \leq Cs, \quad n \in \mathbb{N}. \quad (4.2)$$

*There is therefore an integer  $k_0 \geq 0$  such that, for every  $k \geq k_0$ , the first dyadic integer  $L_k \in \{1, 2, 4, \dots\}$  satisfying*

$$\mathbb{P}(E(s_k, L_k)) \leq q_0 \quad (4.3)$$

*exists. These lengths satisfy*

$$L_k \leq 2^{3k}, \quad \frac{L_k}{2^k} \rightarrow \infty. \quad (4.4)$$

*Moreover, every dyadic integer  $l < L_k$  satisfies  $\mathbb{P}(E(s_k, l)) > q_0$ .*

*Proof. Short crossings are likely.* First suppose  $n \rightarrow \infty$  with  $n \leq Cs$ , and put  $q = (n, \lfloor bn \rfloor)$ . Then

$$\mu(q) \leq m_0 n + \mu(e_2), \quad -1 < r(q) \leq 0.$$

Apply Lemma 3.1 at scale  $n$ , with endpoints  $0, q$ , with cost error  $\eta n$  where  $\eta = K/(4C)$ , and with tube radius  $\delta_0 n$ , where

$$0 < \delta_0 \leq \min\{1/4, 1/(2C)\}.$$

The endpoints have Euclidean distance at most  $3n$ . Since  $|b| \leq 1$ , changing a point by Euclidean distance  $t$  changes its  $r$ -coordinate by at most  $\sqrt{2}t$ . The resulting tube is inside  $-n \leq x \leq 2n$ , and inside  $|r| \leq 8s$  for all sufficiently large  $s$ : its transverse extent is at most  $1 + \sqrt{2}\delta_0 n$ . With probability tending to one its path cost is at most

$$m_0 n + \mu(e_2) + \eta n \leq m_0 n + \mu(e_2) + Ks/4 < m_0 n + Ks.$$

This proves (4.2) when  $n \rightarrow \infty$ .

If  $n$  remains bounded, direct lattice paths from  $0$  to  $(n, \lfloor bn \rfloor)$  use a bounded number of edges and remain in a bounded transverse region with  $0 \leq x \leq n$ . Their costs are bounded in probability, whereas  $Ks \rightarrow \infty$ . This proves the same assertion for bounded  $n$ . The two cases give the joint limit in (4.2) by the subsequence criterion.

*The first dyadic threshold.* Lemma 4.1 shows that the dyadic integer  $2^{3k}$  satisfies (4.3) for all sufficiently large  $k$ . Thus  $L_k$  exists and has the stated upper bound. If  $L_k/2^k$  failed to tend to infinity, a subsequence would satisfy  $L_k \leq C2^k$  for some fixed  $C$ , contradicting (4.2) and (4.3). The last assertion follows from the definition of the first dyadic length; it uses no monotonicity in  $n$  of the events  $E(s, n)$ .  $\square$

## 4.2 Scales compatible with endpoint extensions

The sequence  $L_k$  need not be monotone. The endpoint construction only requires a lower bound for earlier lengths relative to the one currently in use. Record minima give this on an infinite subsequence.

**Lemma 4.3** (Record scales). *There is an infinite set  $\mathcal{K} \subset \{k_0, k_0 + 1, \dots\}$  such that, for  $k \in \mathcal{K}$  and every integer  $0 \leq j \leq k - k_0$ ,*

$$L_{k-j} \geq 2^{-4j} L_k. \tag{4.5}$$

*Proof.* The positive sequence  $L_k 2^{-4k}$  is at most  $2^{-k}$ , so it tends to zero and has infinitely many strict record minima. Take these indices for  $\mathcal{K}$ . At a record index  $k$ , comparison with  $k - j$  gives

$$L_k 2^{-4k} \leq L_{k-j} 2^{-4(k-j)},$$

which is (4.5).  $\square$

Henceforth  $k$  tends to infinity through  $\mathcal{K}$ , and we abbreviate

$$s = 2^k, \quad L = L_k = 2^m.$$

In particular  $m \leq 3k$  and  $m - k \rightarrow \infty$ .

## 5 Endpoint extensions

Lemma 3.4 concerns crossings between two prescribed columns. We now extend it to every path contained in a region of longitudinal size  $O(L)$  and transverse size  $O(s)$ , including paths whose endpoints are chosen after observing the edge weights. The main step is to construct connections that work simultaneously for all endpoints. These connections move each endpoint to a coarse column at a defect cost that is an arbitrarily small multiple of  $s$ . Only boundedly many pairs of coarse columns then remain to be tested.

We continue along the record subsequence  $k \in \mathcal{K}$ , with  $s = 2^k$ ,  $L = L_k = 2^m$ ,  $m \leq 3k$ , and  $L/s \rightarrow \infty$ . Fix  $A, R > 0$ . An integer  $J \geq 1$  will specify the coarsest mesh. In the probability argument we first hold  $J$  fixed and let  $k \rightarrow \infty$ , and then let  $J \rightarrow \infty$ . Use levels

$$J \leq j \leq j_* := \left\lceil \frac{m-1}{4} \right\rceil,$$

with longitudinal mesh and transverse trial scale

$$l_j = 2^{\max\{0, m-4j-1\}}, \quad w_j = 2^{k-j}. \quad (5.1)$$

For every fixed  $J$ , these levels are well defined for all sufficiently large  $k \in \mathcal{K}$ . Indeed  $j_* \leq 3k/4 + 1$ , so all indices  $k-j$  are at least  $k/4 - 1$  and eventually exceed  $k_0$ . For  $l_j > 1$ , (4.5) gives

$$l_j = 2^{-4j-1}L \leq \frac{1}{2}L_{k-j} < L_{k-j}.$$

For  $l_j = 1$ , the same strict inequality follows from  $k-j \geq k/4 - 1 \rightarrow \infty$  and Lemma 4.2. Thus, uniformly over the levels being used,

$$\mathbb{P}(E(w_j, l_j)) > q_0. \quad (5.2)$$

The meshes are nested integer powers of two, the ratio of two successive distinct meshes is at most 16, and  $l_{j_*} = 1$ . The longitudinal meshes therefore reach every integer column, while the transverse scales  $w_j = 2^{-j}s$  make the eventual connection errors summable over the levels.

**Lemma 5.1** (Simultaneous grid connections). *There is a constant  $C_1$ , independent of  $A, R, J, j, k$ , and for each fixed  $A, R$  a sequence  $\beta_J \rightarrow 0$ , such that the following holds. For every fixed  $J \geq 1$  and all sufficiently large  $k \in \mathcal{K}$ , there is an event  $G_{k,J}$  with*

$$\mathbb{P}(G_{k,J}^c) \leq \beta_J$$

*on which the following connections exist simultaneously. For every level  $J \leq j \leq j_*$ , every grid interval*

$$[X, X + l_j] \subset [-(A+4)L, (A+4)L], \quad X \in l_j\mathbb{Z},$$

*every real  $r_0$  with  $|r_0| \leq Rs$ , and every pair of lattice vertices  $p, q$  on its left and right endpoint columns with*

$$|r(p) - r_0| \leq 1, \quad |r(q) - r_0| \leq 1,$$

*there is a path from  $p$  to  $q$  whose cost is at most*

$$m_0 l_j + C_1 j^2 w_j, \quad (5.3)$$

*and whose vertices satisfy*

$$X - l_j \leq x \leq X + 2l_j, \quad |r - r_0| \leq C_1 j^2 w_j. \quad (5.4)$$

*One may take  $C_1 = K + 200\kappa/\lambda$ . Thus  $C_1$  may depend on  $K, \kappa, \lambda$ ; the sequence  $\beta_J$  may depend on  $A, R, q_0, \kappa, \lambda$ .*

*Proof. Independent crossing trials.* Bin the values of  $r_0$  in intervals  $[hw_j, (h+1)w_j]$ , taking those bins with  $h \in \mathbb{Z}$  that meet  $[-Rs, Rs]$ . Fix  $j, X, h$ , and write  $w = w_j, l = l_j$ . Make  $j^2$  trials, using the lattice translates of  $E(w, l)$  by

$$v_t = (X, \lfloor bX \rfloor + (h + 30t)w), \quad 1 \leq t \leq j^2.$$

Here  $w$  is an integer, so each  $v_t$  is a lattice vector. The transverse center of its strip is

$$r(v_t) = \lfloor bX \rfloor - bX + (h + 30t)w.$$

Thus the lattice phase contributes the same offset, of absolute value less than one, to every trial. Consecutive centers differ by exactly  $30w$ , and each strip has half-width  $8w$ , so their vertex sets and internal edge sets are disjoint. Each trial is an exact lattice translate of  $E(w, l)$  and has the same probability as that event. By independence and (5.2), all trials fail with probability at most

$$(1 - q_0)^{j^2}. \quad (5.5)$$

*Vertical attachments for every endpoint.* On each column  $x = X, X + l$ , consider the deterministic vertical interval whose vertices satisfy

$$hw - 2 \leq r \leq hw + (30j^2 + 9)w + 2.$$

It contains every vertex whose transverse coordinate is within one unit of the chosen bin, as well as every possible trial endpoint. There are at most  $50j^2w$  internal vertical edges. If  $V$  is their total weight, Lemma 2.1, with  $A = 50$  and  $v = j^2w$ , gives

$$\mathbb{P}\left(V > \frac{100\kappa}{\lambda}j^2w\right) \leq e^{-c_0j^2w},$$

$$c_0 = 50\kappa(1 - \log 2) > 0.$$

Except with probability  $2e^{-c_0j^2w}$ , vertical travel between any vertices in either of these intervals costs at most  $100\kappa j^2w/\lambda$ .

*Cost and localization.* If a trial succeeds and the vertical bounds hold, join  $p$  vertically to that trial, follow its crossing, and join its other endpoint vertically to  $q$ . The total cost is at most

$$m_0l + Kw + \frac{200\kappa}{\lambda}j^2w \leq m_0l + \left(K + \frac{200\kappa}{\lambda}\right)j^2w.$$

The path stays in the longitudinal range  $[X - l, X + 2l]$ , and all its vertices are within  $50j^2w$  transversely of every  $r_0$  in the chosen bin. Since  $K = 100(1 + |a_-| + |a_+|) \geq 100$ ,  $C_1 = K + 200\kappa/\lambda$  supplies both bounds. This argument does not require independence between a successful trial and its attachments.

*One event for all grid connections.* Let  $G_{k,J}$  be the event that, for every tested triple  $j, X, h$ , some crossing trial succeeds and both vertical bounds hold. There are at most  $C_{A,R}2^{5j}$  pairs  $X, h$  to test at level  $j$ : the grid interval count is bounded by a constant times  $L/l_j \leq 2^{4j+1}$  and the bin count by a constant times  $s/w_j = 2^j$ . The first bound also holds at the terminal mesh  $l_j = 1$ , because then  $m \leq 4j + 1$ . Taking the union over these pairs and the levels, and using  $w_j \geq 1$ , gives

$$\begin{aligned} \mathbb{P}(G_{k,J}^c) &\leq C_{A,R} \sum_{j=J}^{j_*} 2^{5j} ((1 - q_0)^{j^2} + 2e^{-c_0j^2w_j}) \\ &\leq C_{A,R} \sum_{j \geq J} 2^{5j} (e^{-q_0j^2} + 2e^{-c_0j^2}) =: \beta_J. \end{aligned}$$

For every fixed  $q_0 > 0$  and  $\kappa > 0$  this series is convergent, and its tail tends to zero as  $J \rightarrow \infty$ , independently of  $k$ . Whole vertical intervals were tested, so the event gives every allowed endpoint pair simultaneously.  $\square$

We next use these simultaneous connections to extend any path with a negative defect. The coarse mesh leaves only boundedly many column pairs, and the finer meshes attach the actual endpoints at a total error controlled by the tail  $\sum_{j \geq J} j^2 2^{-j}$ . Since the connection event already covers every endpoint, this reduction also applies when the original path is chosen from the observed environment.

**Lemma 5.2** (Arbitrary endpoints). *For every fixed  $A, R, \varepsilon > 0$ ,*

$$\mathbb{P} \left( \begin{array}{c} \text{there is a finite path } P \text{ in } \{|x| \leq AL, |r| \leq Rs\} \\ \text{with } H_+(P) < -\varepsilon s \text{ or } H_-(P) < -\varepsilon s \end{array} \right) \rightarrow 0 \quad (5.6)$$

*as  $k \rightarrow \infty$  through  $\mathcal{K}$ . The endpoints of  $P$  may be any vertices of the region. The convergence is uniform over lattice translations of the region.*

*Proof.* Fix  $A, R, \varepsilon$ , and let  $J \geq 1$  be a fixed integer, to be chosen large. We first work on the event  $G_{k,J}$  and construct extensions for all vertices in the region. Set  $l = l_J$ .

*Connections to the coarse columns.* For a vertex  $p$  in

$$\mathcal{D} = \{|x| \leq AL, |r| \leq Rs\},$$

define coarse columns

$$X_p^- = (\lfloor x(p)/l \rfloor - 1)l, \quad X_p^+ = (\lceil x(p)/l \rceil + 1)l.$$

Each of  $x(p) - X_p^-$  and  $X_p^+ - x(p)$  lies in  $[l, 2l)$ .

Decompose  $[X_p^-, x(p)]$  into grid intervals by taking one full interval of length  $l_J$ , then filling the remaining interval greedily with successively finer meshes. There are at most 16 intervals of each level: at finer levels this follows from the ratio bound 16, and the mesh of one reaches the integer endpoint exactly. At every integer column  $X$  that occurs, use the vertex

$$z_p(X) = (X, \lfloor bX + r(p) \rfloor).$$

It satisfies  $|r(z_p(X)) - r(p)| < 1$ , and  $z_p(x(p)) = p$ . Apply Lemma 5.1 with the same transverse target  $r_0 = r(p)$  to every interval, and concatenate its paths. This gives an entrance path  $P_p^-$  from the column  $X_p^-$  to  $p$ . Decomposing  $[x(p), X_p^+]$  from its right endpoint in the same way, and then concatenating the resulting intervals in left-to-right order, gives an exit path  $P_p^+$  from  $p$  to the column  $X_p^+$ .

*The cost and region of the extensions.* Each path costs at most  $m_0$  times its longitudinal displacement plus

$$16C_1 s \sum_{j \geq J} j^2 2^{-j}. \quad (5.7)$$

The  $r$ -coordinate of every intermediate vertex used to join successive paths differs from  $r(p)$  by less than 1. In particular the *whole* entrance and the whole exit each have transverse displacement of absolute value less than one. Thus, for either sign,

$$H_\sigma(P_p^\pm) \leq 16C_1 s \sum_{j \geq J} j^2 2^{-j} + |a_\sigma|. \quad (5.8)$$

Only the displacement of the whole extension enters this bound; no rounding error is accumulated over the levels.

The grid intervals used here lie inside  $[-(A+4)L, (A+4)L]$ . Their paths have longitudinal range within  $3l$  of  $x(p)$ , so they also lie in this enlarged longitudinal interval. Since  $\sup_{j \geq 1} j^2 2^{-j} < 2$ , their transverse localization in (5.4) gives

$$|x| \leq (A+4)L, \quad |r| \leq (R+C_2)s, \quad C_2 = 2C_1. \quad (5.9)$$

All these statements hold simultaneously for every  $p \in \mathcal{D}$ .

*Preserving a negative defect.* Now suppose a path from  $p$  to  $z$  in  $\mathcal{D}$  satisfies  $H_\sigma(P) < -\varepsilon s$ . Put  $a_* = \max\{|a_-|, |a_+|\}$ . Nonnegativity of its cost and  $|r(z) - r(p)| \leq 2Rs$  imply

$$m_0(x(z) - x(p)) > \varepsilon s - a_\sigma(r(z) - r(p)) \geq \varepsilon s - 2Ra_*s.$$

In particular

$$x(z) - x(p) \geq -C_3s, \quad C_3 = 2Ra_*/m_0. \quad (5.10)$$

Thus the final endpoint can lie at most  $C_3s$  to the left of the initial endpoint. This bound lets the extensions produce a left-to-right crossing even when the original endpoints occur in reverse order.

Choose  $J$  so large that

$$32C_1 \sum_{j \geq J} j^2 2^{-j} \leq \varepsilon/4.$$

Then take  $k \in \mathcal{K}$  large enough that  $2a_* \leq \varepsilon s/4$ . By (5.8), the concatenation

$$\tilde{P} = P_p^- P P_z^+$$

satisfies  $H_\sigma(\tilde{P}) < -\varepsilon s/2$ . It starts on the coarse column  $X_p^-$ , ends on the coarse column  $X_z^+$ , and stays in (5.9).

*A long crossing between coarse columns.* For this fixed  $J$ , eventually

$$l_J = 2^{-4J-1}L, \quad \frac{l_J}{s} \longrightarrow \infty.$$

Using (5.10) and the two attachments of longitudinal length at least  $l_J$ , the new span satisfies

$$d' := X_z^+ - X_p^- \geq 2l_J - C_3s \geq l_J \quad (5.11)$$

for all sufficiently large  $k$ . Thus  $d'$  is a positive integer and  $d'/s \rightarrow \infty$ , uniformly over the possible endpoint choices.

*Reduction to finitely many crossing tests.* Both coarse columns lie in  $[-(A+2)L, (A+2)L]$  and are multiples of  $l_J$ . There are at most a constant depending on  $A, J$  possible pairs, because  $L/l_J = 2^{4J+1}$ . For each pair obeying (5.11), translate its first column to zero by the lattice vector  $(X_p^-, \lfloor bX_p^- \rfloor)$ . The transverse center changes by less than one. Also  $L/d' \leq 2^{4J+1}$ . It follows that the translated region in (5.9) is contained in a region of the form

$$-A_J d' \leq x \leq (1 + A_J)d', \quad |r| \leq (R + C_2 + 1)s,$$

where, for example,  $A_J = (2A + 8)2^{4J+1}$  is independent of  $k$ . Lemma 3.4 therefore shows that the probability of such an extended crossing tends to zero. This application is uniform over the bounded number of pairs: otherwise a sequence of violating pairs would contradict that lemma's joint limit, since every admissible pair has  $d'/s \geq l_J/s \rightarrow \infty$ . Taking the union over the pairs and both signs still gives a probability tending to zero.

*Order of limits and translations.* Write  $B_k$  for the event in (5.6). We have proved, for every fixed sufficiently large  $J$ ,

$$\limsup_{\substack{k \rightarrow \infty \\ k \in \mathcal{K}}} \mathbb{P}(B_k) \leq \limsup_{\substack{k \rightarrow \infty \\ k \in \mathcal{K}}} \mathbb{P}(G_{k,J}^c) \leq \beta_J.$$

All constants in the crossing tests above were fixed before taking the limit in  $k$ . We may now let  $J \rightarrow \infty$ , so that  $\beta_J \rightarrow 0$ , proving the assertion. Every exact lattice translate of  $B_k$  has exactly the same probability by translation invariance of the edge-weight law and of path displacements, giving the stated uniformity.  $\square$

## 6 Disjoint witnesses and exclusion of corners

We now turn the crossing estimates into an obstruction to a corner of the limit shape. Assume that the two extreme supporting slopes in a fixed direction differ. We will choose one deterministic scale  $(s, L)$  at which certain local path events are rare, and then divide every simple path from the origin into chunks at that scale. A full chunk with small defect supplies a witness for one of these events. More generally, the number of disjoint witnesses in a chunk controls how negative its defect can be.

These deterministic bounds force any long path with small total defect to contain many edge-disjoint witnesses. The BK inequality and a count of coarse itineraries make the existence of such paths summably unlikely. At the chosen scale this almost surely yields a positive average defect per full chunk, contradicting the shape theorem along the direction of contact. The argument allows the paths, the chunks, and their witnesses to be chosen after the weights are observed.

Kesten's path-length estimates [22, Section 3] already count rectangle itineraries, record repeated rectangles through multiple vertex-disjoint crossings, and use independence over spatially disjoint rectangles. Here the witnesses certify negative excess relative to a supporting functional, and BK controls their disjoint occurrence even when the coarse test regions overlap.

### 6.1 Disjoint occurrence for strict path witnesses

The required form of BK concerns the edges that certify the events. Test regions may overlap, and the same event may appear several times in the list. This is essential when a path returns to a coarse cell.

Continuous-product extensions of disjoint occurrence are treated more generally by Arratia, Garibaldi, and Hales [3]. We give the finite-bit reduction for the strict path events used here.

**Lemma 6.1** (BK for strict path witnesses). *Let  $A_1, \dots, A_q$  be a finite list of events. For each  $j$ , suppose  $A_j$  is the existence of a path  $P$  in a prescribed family of finite lattice paths, all of whose edges belong to a fixed finite edge set, satisfying*

$$\tau(P) < b_j(P),$$

*where  $b_j(P)$  is deterministic. A family may include several labeled copies of a path with different bounds. Let  $D(A_1, \dots, A_q)$  be the event that one can choose a witnessing path for each event with pairwise disjoint edge sets. Then*

$$\mathbb{P}(D(A_1, \dots, A_q)) \leq \prod_{j=1}^q \mathbb{P}(A_j). \quad (6.1)$$

*The events in the list need not be distinct.*

*Proof.* Let  $F_{\kappa,\lambda}$  be the Gamma distribution function and define its quantile function by

$$\Phi(u) = F_{\kappa,\lambda}^{-1}(u) \quad (0 \leq u < 1), \quad \Phi(1) = +\infty.$$

Here  $\Phi(0) = 0$ . The density is positive for every  $t > 0$ , so the distribution function is continuous and strictly increasing from zero to one. Consequently  $\Phi$  is increasing and continuous on  $[0, 1)$ . Represent the independent weights as

$$\tau_e = \Phi(U_e), \quad U_e = \sum_{h \geq 1} 2^{-h} B_{e,h},$$

where all  $B_{e,h}$  are independent Bernoulli variables of parameter  $1/2$ . At precision  $t$  put

$$U_e^{(t)} = 2^{-t} + \sum_{h=1}^t 2^{-h} B_{e,h}, \quad \tau_e^{(t)} = \Phi(U_e^{(t)}).$$

Then  $U_e^{(t)} \downarrow U_e$ ; because  $U_e < 1$  almost surely, continuity of  $\Phi$  gives  $\tau_e^{(t)} \downarrow \tau_e$  almost surely on the finite union of the edge sets under consideration. Define  $A_j^{(t)}$  and  $D^{(t)}$  by using these rounded weights.

At each fixed precision, the events  $A_j^{(t)}$  are decreasing events of finitely many independent bits. Fixing all precision- $t$  bits of the edges of a witnessing path forces its strict cost inequality regardless of the other bits. Edge-disjoint path witnesses thus give disjoint bit certificates. Complement the bits so that the events are increasing. For each certificate, retain its realized one-bits and set every other bit to zero. This configuration still belongs to the event, because it agrees with the realized configuration on the certificate, and its one-set lies inside that certificate. Disjoint certificates therefore give disjoint positive witnesses in the convention of [32]; any unused one-bits can be assigned by monotonicity. The finite Bernoulli BK inequality [32, Theorem 3.3, Equation (3.4)], iterated over the list, gives

$$\mathbb{P}(D^{(t)}) \leq \prod_{j=1}^q \mathbb{P}(A_j^{(t)}) \leq \prod_{j=1}^q \mathbb{P}(A_j).$$

Monotone disjoint occurrence is associative, so iteration allows repeated events. The second inequality follows from  $\tau_e^{(t)} \geq \tau_e$ .

The events  $D^{(t)}$  increase with  $t$ . A fixed finite list of original witnessing paths involves finitely many traversals. Its rounded path costs decrease to the original costs, so every strict inequality holds at all sufficiently fine precisions. Conversely, every rounded witness remains a witness with the original weights. Therefore, up to a null set,

$$D(A_1, \dots, A_q) = \bigcup_{t \geq 1} D^{(t)}.$$

Continuity of probability from below proves (6.1). No bound on the Gamma density or on the derivative of its quantile function is used, so the proof includes  $0 < \kappa < 1$ .  $\square$

## 6.2 Choosing one scale under a corner assumption

Fix  $b \in [-1, 1]$  and use the notation of Lemma 2.2. For the rest of the construction suppose, towards a contradiction, that

$$c = \frac{a_+ - a_-}{2} > 0, \quad \eta = \min\{1, c/4\}.$$

Choose  $q_0 \in (0, 1)$  so small that

$$400(2q_0)^{1/16} < \frac{1}{2}. \quad (6.2)$$

Use this value of  $q_0$  in Lemmas 4.2 and 4.3, and write  $s = 2^k$  and  $L = L_k$  along the resulting subsequence. In particular,  $s$  and  $L$  are integers and  $L/s \rightarrow \infty$ .

For  $(i, h) \in \mathbb{Z}^2$ , set

$$v_{i,h} = (iL, \lfloor biL \rfloor + hs), \quad \mathcal{B}_{i,h} = v_{i,h} + \{z \in \mathbb{Z}^2 : |x(z)| \leq 4L, |r(z)| \leq 4s\}.$$

At each index  $(i, h)$  we test for either a saving below an extreme support or a cheap forward crossing. Precisely, define  $\mathcal{A}_{i,h}$  to be the union of the following events:

1. Some path  $P$  with all its vertices in  $\mathcal{B}_{i,h}$  satisfies  $H_\sigma(P) < -\eta s$  for at least one sign  $\sigma \in \{-, +\}$ .
2. The event  $E(s, L)$  holds at the lattice translate  $v_{i+1,h}$ .

These are decreasing events of finitely many edge weights. Throughout this section a *path witness* for  $\mathcal{A}_{i,h}$  means a path satisfying one of these two defining conditions, including its strict cost inequality. All such events have the form in Lemma 6.1.

The first event has probability  $o(1)$  by Lemma 5.2, and the second has probability at most  $q_0$  by the definition of  $L$ . Translation invariance makes these bounds uniform in  $(i, h)$ . We may therefore fix one sufficiently large index  $k$  on the subsequence such that

$$\mathbb{P}(\mathcal{A}_{i,h}) \leq 2q_0 \quad \text{for every } (i, h) \in \mathbb{Z}^2. \quad (6.3)$$

We also take this same index large enough that  $s \geq 2$  and

$$\begin{aligned} C_f &:= \max\{|m_0 - ab|, |a|\} \leq \eta s, \\ 2m_0L - |a|(s+1) &> \eta s, \quad 3\eta s + 2|a|(s+1) < Ks. \end{aligned} \quad (6.4)$$

The first two requirements follow from  $s \rightarrow \infty$  and  $L/s \rightarrow \infty$ ; the third follows from the definition of  $K$  and  $\eta \leq 1$ . The scale  $k$ , and hence  $s$  and  $L$ , stays fixed from now on. The probability bound (6.3) holds uniformly at this one scale; only the length of the path will subsequently tend to infinity. This is the finite-scale obstruction we will establish: a positive slope gap and these rare local events almost surely give a positive lower bound on  $H(P)/N$  for every simple path from the origin with sufficiently many full chunks  $N$ . In particular,  $s/L$  is now a fixed positive number, and the final contradiction uses this positivity.

### 6.3 Extracting witnesses from path chunks

Let  $P$  be a finite simple lattice path starting at the origin. Decompose it into chunks as follows. At the start of a chunk, with coordinates  $(x_0, r_0)$ , follow the path until it first reaches a vertex for which

$$|x - x_0| \geq 2L \quad \text{or} \quad |r - r_0| \geq s.$$

Cut there and start the next chunk. A chunk ending at such a cut is called *full*. The decomposition consists of  $N$  full chunks and a final remainder, which may be empty. Associate to each chunk, including the remainder, the index

$$(i, h) = (\lfloor x_0/L \rfloor, \lfloor r_0/s \rfloor). \quad (6.5)$$

The cuts and indices depend on the candidate path. All the estimates below hold for each path individually and require no independence of its chunks.

For a nearest-neighbor step,  $|\Delta x| \leq 1$  and  $|\Delta r| \leq 1$ . Thus every vertex of a chunk has  $|x - x_0| \leq 2L$  and  $|r - r_0| \leq s + 1$ . Moreover,

$$0 \leq x_0 - iL < L, \quad 0 \leq r_0 - r(v_{i,h}) < s + 1.$$

The entire chunk, and each of its subpaths, therefore lies in  $\mathcal{B}_{i,h}$ : its longitudinal offsets from  $v_{i,h}$  belong to  $[-2L, 3L)$ , and its transverse offsets have absolute value at most  $2s + 2 \leq 4s$ .

**Lemma 6.2** (A cheap full chunk supplies a witness). *If a full chunk  $Q$  with index  $(i, h)$  satisfies  $H(Q) < \eta s$ , then its edges contain a path witness for  $\mathcal{A}_{i,h}$ .*

*Proof.* There are three possible exit behaviors. A transverse exit uses the slope gap, a backward exit has large defect by nonnegativity, and a forward exit reduces to the crossing event after controlling its outer pieces.

If  $|r(\Delta Q)| \geq s$ , the support identity gives

$$\min_{\sigma \in \{-, +\}} H_\sigma(Q) = H(Q) - c|r(\Delta Q)| < (\eta - c)s \leq -3\eta s < -\eta s.$$

The whole chunk is then a witness for the first defining event.

Otherwise a longitudinal exit occurred, and  $x(\Delta Q) \in \{-2L, 2L\}$ . In the negative case, nonnegative weights and (6.4) imply

$$H(Q) \geq 2m_0L - |a|(s + 1) > \eta s,$$

contrary to the hypothesis.

It remains to consider  $x(\Delta Q) = 2L$ . The initial column is strictly to the left of  $(i + 1)L$ , and the final column is at or to the right of  $(i + 2)L$ . Let  $z$  be the first vertex of  $Q$  on column  $(i + 2)L$ , and let  $p$  be the last vertex on column  $(i + 1)L$  preceding  $z$ . The subpath  $Q'$  from  $p$  to  $z$  stays between these two columns. Its transverse offsets from  $v_{i+1,h}$  have absolute value at most  $2s + 2 \leq 8s$ . Consequently  $Q'$  fits inside the translate of the block defining  $E(s, L)$  at  $v_{i+1,h}$ .

Write  $Q = Q_{\text{in}} * Q' * Q_{\text{out}}$ , allowing an empty outer part. If either outer part has  $H < -\eta s$ , it witnesses the first defining event, because it lies in  $\mathcal{B}_{i,h}$  and  $\min_\sigma H_\sigma \leq H$ . Otherwise additivity gives  $H(Q') < 3\eta s$ . Since  $|r(\Delta Q')| \leq 2(s + 1)$ , we obtain

$$\tau(Q') < m_0L + 3\eta s + 2|a|(s + 1) < m_0L + Ks.$$

Thus  $Q'$  witnesses the second defining event. □

## 6.4 Controlling negative defect by witness multiplicity

To control chunks whose defects are very negative, we count how many edge-disjoint witnesses their edges support. A stopping rule will split off a new witness each time a prefix accumulates a defect below  $-\eta s$ ; the bound on a single step limits the defect lost at each split.

For any chunk  $Q$  with index  $(i, h)$ , let  $M_Q$  be the maximum number of pairwise edge-disjoint path witnesses for  $\mathcal{A}_{i,h}$  whose edges all belong to  $Q$ . This maximum is finite. Every witness uses at least one edge: the first defining event has a strictly negative defect bound, and the second crosses two distinct columns. Hence  $M_Q$  is bounded by the number of edges of  $Q$ .

**Lemma 6.3** (Defect bound from disjoint witnesses). *Every chunk, including the remainder, satisfies*

$$H(Q) \geq -\eta s - 2\eta s M_Q. \tag{6.6}$$

A full chunk satisfies the stronger bound

$$H(Q) \geq \eta s - 4\eta s M_Q. \quad (6.7)$$

If  $M_{\text{tot}}$  denotes the sum of  $M_Q$  over the  $N$  full chunks and the remainder of  $P$ , then

$$H(P) \geq (N - 1)\eta s - 4\eta s M_{\text{tot}}. \quad (6.8)$$

The path contains  $M_{\text{tot}}$  pairwise edge-disjoint witnesses, assigned to the corresponding chunk indices, even if some indices repeat.

*Proof.* For each oriented nearest-neighbor step  $e$ ,

$$H(e) = \tau_e - f(\Delta e) \geq -C_f.$$

Starting at the beginning of  $Q$ , split off the first prefix with  $H < -\eta s$ , if one exists, and repeat with the remaining path. At the last step of each split prefix, its defect decreases from a value at least  $-\eta s$  to a value below  $-\eta s$ . By (6.4), the defect of this prefix is at least  $-\eta s - C_f \geq -2\eta s$ . Every split prefix is a witness for the original chunk's event  $\mathcal{A}_{i,h}$ : it remains in the same box, and  $\min_{\sigma} H_{\sigma} \leq H < -\eta s$ .

If this procedure splits off  $t$  prefixes, simplicity of  $P$  makes their edge sets disjoint, so  $t \leq M_Q$ . The unsplit final part has  $H \geq -\eta s$ . Additivity now gives

$$H(Q) \geq -\eta s - 2\eta s t \geq -\eta s - 2\eta s M_Q,$$

which proves (6.6).

For a full chunk with  $M_Q = 0$ , Lemma 6.2 gives  $H(Q) \geq \eta s$ . If  $M_Q \geq 1$ , then

$$-\eta s - 2\eta s M_Q \geq \eta s - 4\eta s M_Q.$$

This proves (6.7). Apply that bound to all full chunks and use (6.6), weakened to  $-\eta s - 4\eta s M_Q$ , on the remainder. Their sum is (6.8).

Finally, choose a family attaining  $M_Q$  in each chunk. Distinct chunks of a simple path have disjoint edge sets, so the union of these families is still edge-disjoint. The assertion does not require their associated events or their boxes to be distinct.  $\square$

## 6.5 Counting itineraries and witness multiplicities

We now apply BK to the witnesses supplied by the deterministic bounds. The counting records the coarse itinerary and the number of witnesses assigned to each of its positions. For each fixed number of chunks this produces a finite collection of deterministic event lists covering all possible choices of the path and its witnesses, including paths that revisit a coarse cell.

**Proposition 6.4** (Uniform lower bound for long paths). *At the fixed scale chosen above, almost surely there exists a finite  $N_0$  such that every finite simple path  $P$  starting at the origin and having  $N \geq N_0$  full chunks satisfies*

$$H(P) \geq \frac{1}{2} N \eta s. \quad (6.9)$$

*Proof.* First quantify how many witnesses a path of small defect must supply. Let  $B_N$  be the event that there exists such a simple path with  $N$  full chunks and  $H(P) < N\eta s/2$ . If it occurs, (6.8) implies

$$M_{\text{tot}} > \frac{N}{8} - \frac{1}{4}.$$

For all sufficiently large  $N$ , we may therefore select exactly  $t_N = \lfloor N/16 \rfloor$  of the edge-disjoint witnesses supplied by Lemma 6.3.

Next count the possible event lists. List the chunk indices in order, including the index of the remainder. The first is  $(0, 0)$ . Each coordinate of consecutive indices differs by at most 3, by the displacement bounds for a full chunk and (6.5). There are consequently at most  $49^N$  possible index itineraries. For a fixed itinerary, the numbers of selected witnesses in its  $N + 1$  positions form a weak composition of  $t_N$ . The number of these compositions is

$$\binom{N + t_N}{t_N} \leq 2^{N+t_N} \leq 4^N.$$

The allocation is indexed by positions along the itinerary, so repeated visits to one coarse cell retain separate positions. Each composition specifies a list of  $t_N$  events  $\mathcal{A}_{i,h}$ , with repetitions when appropriate. Its selected witnesses certify disjoint occurrence of that list. Lemma 6.1 and (6.3) bound its probability by  $(2q_0)^{t_N}$ . There is no additional count of witness paths: that lemma bounds the existence of all choices of disjoint witnessing paths at once. Thus the path-dependent selection is covered by a union over deterministic itineraries and compositions. Taking this union yields

$$\mathbb{P}(B_N) \leq 196^N (2q_0)^{\lfloor N/16 \rfloor} \quad (6.10)$$

for all sufficiently large  $N$ . All the events involved are measurable; in particular, finite lattice paths form a countable family.

Put  $p = 2q_0$ . Condition (6.2) implies  $0 < p < 1$  and  $196p^{1/16} < 1$ , while the right-hand side of (6.10) is at most

$$p^{-1} (196p^{1/16})^N.$$

Thus  $\sum_N \mathbb{P}(B_N) < \infty$ . The first Borel–Cantelli lemma proves the simultaneous eventual bound (6.9) for all the paths in the proposition.  $\square$

## 6.6 Contradiction with the shape theorem

The bound just proved applies simultaneously to all sufficiently long simple paths from the origin. In particular it applies to approximate minimizers selected from the realized environment. Along the contact direction  $u$ , their number of chunks must grow with their longitudinal displacement, whereas the shape theorem makes their defect sublinear in that displacement.

**Proposition 6.5** (Absence of corners). *For every  $b \in [-1, 1]$ , the two one-sided transverse derivatives at  $u = (1, b)$  agree:*

$$a_- = a_+.$$

*Proof.* Fix an arbitrary deterministic  $b \in [-1, 1]$  and suppose that  $c > 0$ . Make all the preceding choices for this  $b$ , including the single fixed scale  $k$ . For positive integers  $t$ , let

$$z_t = (t, \lfloor bt \rfloor).$$

A path from 0 to  $z_t$  with  $N$  full chunks has  $N + 1$  chunks when the remainder is included. Each changes the  $x$ -coordinate by at most  $2L$ , so

$$N + 1 \geq \frac{t}{2L}. \quad (6.11)$$

Work on the probability-one event where the shape theorem and Proposition 6.4 both hold. For each  $t$ , choose a finite path from 0 to  $z_t$  with cost at most  $T(0, z_t) + 1$  and erase its loops. The

resulting simple path  $P_t$  still has cost between  $T(0, z_t)$  and  $T(0, z_t) + 1$ , since the edge weights are nonnegative. The shape theorem gives  $T(0, z_t)/t \rightarrow m_0$ , and

$$f(z_t) = m_0 t + a(\lfloor bt \rfloor - bt) = m_0 t + O(1).$$

It follows that

$$\frac{H(P_t)}{t} \rightarrow 0. \quad (6.12)$$

On the other hand, the numbers  $N_t$  of full chunks of  $P_t$  tend to infinity by (6.11). Proposition 6.4 therefore applies for all sufficiently large  $t$  and gives

$$\liminf_{t \rightarrow \infty} \frac{H(P_t)}{t} \geq \frac{\eta s}{2} \liminf_{t \rightarrow \infty} \frac{N_t}{t} \geq \frac{\eta s}{4L} > 0.$$

This contradicts (6.12). Hence  $c > 0$  is impossible, and convexity gives  $a_- = a_+$ .

The choice of  $b$  was arbitrary and deterministic. The corner gap of the deterministic norm is itself deterministic, and the argument excludes a positive gap for each such  $b$ . Accordingly the conclusion holds for every  $b \in [-1, 1]$ ; no intersection of uncountably many probability-one events is being asserted or needed.  $\square$

## 7 Differentiability and the boundary

The preceding section has ruled out distinct extreme supports at every prescribed direction. We now pass from this directional statement to Fréchet differentiability and identify its equivalent boundary formulation.

**Lemma 7.1** (Convex boundary regularity). *For a norm on  $\mathbb{R}^2$ , Fréchet differentiability at every nonzero vector is equivalent to uniqueness of the supporting line at every point of its unit sphere. These conditions are also equivalent to the unit sphere being a regular  $C^1$  curve.*

*Proof. Differentiability and a unique support.* Fix a nonzero vector  $u$  and choose a vector  $e$  linearly independent of it. The function  $t \mapsto \mu(u + te)$  has finite one-sided derivatives  $a_- \leq a_+$ . As in Lemma 2.2, the linear functionals taking the value  $\mu(u)$  on  $u$  and the values  $a_-$  and  $a_+$  on  $e$  support  $\mu$  globally. Any other supporting functional at  $u$  must have its value on  $e$  between these two numbers, by the inequalities for positive and negative  $t$ .

If the endpoints agree with value  $a$ , the two-sided expansion gives, as  $xu + re \rightarrow u$  with  $x > 0$ ,

$$\mu(xu + re) = x\mu(u + (r/x)e) = x\mu(u) + ar + o(|r|).$$

Since  $x \rightarrow 1$  and  $|r| \leq C|xu + re - u|$ , this is Fréchet differentiability at  $u$ . The expression is exact when  $r = 0$ . Conversely, Fréchet differentiability makes the two one-sided derivatives equal. Thus differentiability is equivalent to a unique linear functional  $g \leq \mu$  satisfying  $g(u) = \mu(u)$ .

*Supports of the norm and of its unit ball.* Because zero lies in the interior of the unit ball, every supporting line at  $u/\mu(u)$  has a unique normalization  $g = 1$  with  $g \leq \mu$  and  $g(u) = \mu(u)$ . This identifies uniqueness of the functional with uniqueness of the line.

*Continuity of the derivative and the boundary.* Suppose now that this uniqueness holds everywhere off zero. All supporting functionals lie in the compact dual unit ball. If  $u_j \rightarrow u \neq 0$  and a subsequence of the supports at  $u_j$  converges to  $g$ , passing to the limit shows  $g \leq \mu$  and  $g(u) = \mu(u)$ . Uniqueness determines  $g$ , so the whole sequence of supports converges to the support at  $u$ . The

derivative of  $\mu$  is therefore continuous off zero. At a unit vector for this norm it is nonzero, since homogeneity gives

$$D\mu(u)[u] = \mu(u) = 1.$$

The implicit function theorem shows that the level set  $\mu = 1$  is a  $C^1$  curve.

Conversely, at any point of a regular  $C^1$  convex boundary a supporting functional has a local maximum along a local boundary parametrization. It therefore annihilates the tangent direction. Its supporting line is the tangent line, which is unique. This proves the converse and all the equivalences.  $\square$

*Proof of Theorem 1.2.* Proposition 6.5 gives  $a_- = a_+$  for every fixed  $b \in [-1, 1]$ . The expansion in the proof of Lemma 7.1, applied to  $u = (1, b)$  and  $e = e_2$ , gives Fréchet differentiability at  $u$ . Signed coordinate permutations and positive homogeneity cover every nonzero vector. Lemma 7.1 gives the remaining conclusions. The parameters  $\kappa, \lambda > 0$  were arbitrary and fixed throughout, so the conclusion holds for each Gamma law in the theorem.  $\square$

## Part II

# Strict convexity for exponential weights

## 8 Exponential passage times and the strict-convexity argument

We now restrict to independent exponential edge weights of mean one. The argument in this part uses no differentiability conclusion. Its starting point is a hypothetical flat face of the limit shape, and its contradiction will come from improving local paths in two directions inside that face.

In this part  $\tau_e$  has density  $e^{-t}$  on  $[0, \infty)$ , and  $T$ ,  $\mu$ , and  $\mathcal{B}$  refer to this rate-one law.

We regard an edge also as an interval, assigning a partial traversal the corresponding fraction of its cost and interpolating vertex potentials affinely. This permits cuts inside edges while leaving all vertex-to-vertex passage times unchanged.

The quantitative input is sublinear control of passage-time errors, uniformly over polynomially many possible endpoints. We state the forms used below for the exponential law.

**Lemma 8.1** (Uniform metric estimates). *There are constants  $C, c > 0$  such that, for  $x, y \in \mathbb{Z}^2$  and  $R = |x - y|_1 \geq 2$ ,*

$$0 \leq \mathbb{E}T(x, y) - \mu(y - x) \leq C\sqrt{R \log(2 + R)}, \quad (8.1)$$

$$\mathbb{P}(|T(x, y) - \mathbb{E}T(x, y)| \geq t\sqrt{R}) \leq Ce^{-ct}, \quad t \geq 0. \quad (8.2)$$

Consequently, for any fixed  $p, C_0 < \infty$  and  $\alpha \in (1/2, 1)$ , with probability tending to one all pairs in  $[-n^p, n^p]^2 \cap \mathbb{Z}^2$  at distance at most  $C_0 n$  satisfy

$$|T(x, y) - \mu(y - x)| \leq n^\alpha.$$

The expected maximum lower deficit over those pairs is at most  $C_{p, C_0} \sqrt{n} \log(2 + n)$ . The same conclusions hold for metric-graph endpoints after adding an  $O(\log n)$  term in expectation.

*Proof.* Equation (8.1) is the nonrandom-fluctuation estimate of Damron and Kubota [12, Proposition 1.1]. The two-sided concentration theorem of Damron, Hanson, and Sosoe [11, Theorem 1.1] gives the stronger scale  $\sqrt{R/\log R}$ ; enlarging constants gives (8.2). Its hypotheses hold since  $\mathbb{P}(\tau_e = 0) = 0$  and  $\mathbb{E}e^{a\tau_e} = (1 - a)^{-1} < \infty$  for  $a < 1$ . The finitely many small separations can be bounded by a coordinate path and the exponential tail of its cost.

There are polynomially many pairs in the displayed box. Since  $\sqrt{n} \log n = o(n^\alpha)$ , a union bound in (8.2) proves the simultaneous assertion. For the lower deficit, use  $\mu(y - x) \leq \mathbb{E}T(x, y)$  and integrate the bound  $\min\{1, Cn^{C_p}e^{-ct/\sqrt{n}}\}$ . For interior-edge endpoints, attach the endpoint of each incident edge, at a cost bounded by twice the maximum edge weight in a slightly larger box; changes in  $\mu$  are bounded by a fixed constant. The expected maximum of polynomially many exponential variables is  $O(\log n)$ . This proves the remaining statements.  $\square$

Lemma 9.3 will supply a linear geodesic-length bound with exponentially small failure probability by counting self-avoiding paths. Together these estimates give the required microscopic controls.

**A hypothetical flat and the quantity to improve.** Suppose  $\partial\mathcal{B}$  contains a nontrivial segment. On its supporting line take the maximal face, with endpoints  $a, b$ , and let  $h$  be the linear functional

equal to one there. Then  $h \leq \mu$ , and  $\mu = h$  on the positive cone generated by the face. The endpoints are extreme points of  $\mathcal{B}$  and are linearly independent.

Choose rational vectors  $v, w$  in this cone close to  $a, b$ . For a small rational  $s > 0$  put  $A = v + sw$  and  $B = w + sv$ . These slightly inward directions will carry the two road families. Tracking and input locality are measured by  $\|x\|_* = |[v \ w]^{-1}x|_\infty$ . Constants may depend on the fixed flat. The tracking and support constants of the final construction, however, must be independent of  $s$ , of all large auxiliary cutoffs, and of the input road rule; this is what will allow a fixed rescaling.

The variational quantity is the infimum  $b(n)$  of a common bound for the expected excess cost per period of the two local roads, with costs kept in their original lattice units. We will prove both  $0 < c_0 \leq b(n)$  and  $b(n) \leq C_s n^{3/4}$ . The upper bound provides inputs to an improvement; the lower bound makes indefinite improvement impossible.

The middle of the proof establishes that improvement. Local gauges make endpoint corrections computable from nearby inputs. Physical routing bounds the savings that selected local tunnels can offer, and a thin-diagram argument converts this into capped crossing estimates. An independent spare road then replaces rare large positive costs. The resulting pair satisfies  $b(Dn) \leq \rho b(n)$  for a fixed integer  $D$  and  $\rho < 1$ , contradicting the positive lower bound.

## 9 Calibration and local roads

Assume that the exponential time-constant ball has a nontrivial flat face. We associate a nonnegative defect with every path, then minimize the expected defect of roads that track either of two directions in the cone of that face. Two estimates begin the argument: there exist local roads with sublinear expected defect at large scales, and the infimum of the common period bound over admissible pairs is bounded away from zero. Both estimates will remain available when the road rules are changed later.

### 9.1 A dominated stationary calibration

Let  $h$  be the supporting functional equal to one on the hypothetical face. Write  $\mathcal{G}_n = n^{-1}\mathbb{Z}^2$ ; rescaling the lattice does not rescale its exponential edge weights. For  $z \in \mathcal{G}_n$ , the shift  $\theta_z$  translates microscopic inputs by  $nz$ .

The stationary Busemann-measure construction of Damron and Hanson [9, Section 3, Propositions 3.2 and 3.4, and Theorem 3.5] provides, on a stationary extension of the weight space, an additive cocycle  $\mathcal{B}(x, y)$  with

$$|\mathcal{B}(x, y)| \leq T(x, y), \quad \mathbb{E}\mathcal{B}(0, x) = h(x), \quad x, y \in \mathbb{Z}^2.$$

The construction allows the prescribed supporting functional  $h$  and requires no differentiability of the norm. Its moment and subcritical-zero-mass assumptions hold for exponential weights. For mean identification, Damron and Hanson use Hoffman's argument [21], developed from an averaging argument of Garet and Marchand [16], with a presentation inspired by Gou  r  's Lemma 2.6 [17].

Condition  $\mathcal{B}$  on the full edge environment. This preserves additivity, domination and the mean. The resulting cocycle also has a translation-covariant version: stationarity gives the conditional expectation identity for each fixed translation, and countability of the vertices and translations gives a common exceptional null set. After rescaling, we therefore have an edge-measurable cocycle

$B_n$  on  $\mathcal{G}_n$  satisfying

$$\begin{aligned} B_n(x, z) &= B_n(x, y) + B_n(y, z), \\ |B_n(x, y)| &\leq T(nx, ny), \\ \mathbb{E}B_n(x, y) &= nh(y - x), \\ B_{n,\omega}(x + z, y + z) &= B_{n,\theta_z\omega}(x, y). \end{aligned} \tag{9.1}$$

These identities persist after adjoining independent private inputs. For a path  $\gamma$  from  $x$  to  $y$ , put

$$P_n(x) = B_n(0, x) - nh(x), \quad Q_n(\gamma) = \tau(\gamma) - nh(y - x)$$

and define its *defect* by

$$D_n(\gamma) = Q_n(\gamma) - P_n(y) + P_n(x) = \tau(\gamma) - B_n(x, y) \geq 0. \tag{9.2}$$

Defect is additive under concatenation and nonnegative on every forward subpath, with repeated edge occurrences counted with multiplicity. This pathwise accounting will allow endpoints to be chosen after the environment has been inspected.

For cuts inside an edge, interpolate  $B_n(0, \cdot)$  affinely and charge the traversed fraction of the weight. Each orientation of the edge has nonnegative defect, so (9.2) also holds for these partial traversals. Passage times between lattice vertices are unchanged.

## 9.2 The class of local randomized roads

Choose rational  $v, w$  in the cone of the face, close to its two endpoints, and a rational  $0 < s < 1/4$ . Set

$$A = v + sw, \quad B = w + sv, \quad r = s/1000.$$

We restrict to  $n$  for which  $nA, nB$  are lattice vectors. All road tracking distances and input-dependence radii are measured in the fixed oblique norm

$$\|x\|_* = |[v \ w]^{-1}x|_\infty.$$

In these coordinates  $A = (1, s)$  and  $B = (s, 1)$ . In particular,

$$|[A \ B]^{-1}x|_\infty \leq 2\|x\|_*, \quad \|x\|_* \geq \frac{3}{4}|[A \ B]^{-1}x|_\infty. \tag{9.3}$$

The conversion to  $(v, B)$  coordinates has operator norm at most 2 as well. Thus changes between these oblique coordinates have uniform constants. Conversion to microscopic lattice coordinates may depend on the fixed face: restrict the neighborhoods of its endpoints once and choose  $K_{\text{lat}}, M \geq 1$  such that, for the standard basis vectors  $e_1, e_2$ ,

$$\max_{i=1,2} \|e_i\|_* \leq K_{\text{lat}}, \quad |A|_2, |B|_2 \leq M. \tag{9.4}$$

The constants depend only on the fixed face. Such uniform choices are possible because its endpoints, lying on  $h = 1$ , are linearly independent.

**Definition 9.1** (Admissible road). For  $d \in \{A, B\}$ , an admissible  $d$ -road is a measurable oriented two-sided lattice walk together with a continuous nondecreasing nominal parameter on its edge occurrences. The nominal parameter tends to  $-\infty$  and  $+\infty$  at the two ends. Every compact

parameter interval contains finitely many edge occurrences. A point with nominal parameter  $t$  satisfies  $\|x - td\|_* \leq r$ .

The construction uses the edge environment and a private product field  $U = (U_\iota)_{\iota \in \mathcal{I}}$  independent of it. Each coordinate has a spatial location and, when needed, a road-position or auxiliary type label. For every  $z \in \mathcal{G}_n$ , the shift  $\theta_z$  acts measurably by translating the spatial and road-position indices together and preserves the product law. Thus  $(\theta_z \omega, \theta_z U)$  has the same law as  $(\omega, U)$ . Marginals may depend on a type left unchanged by translations. The road's restriction to a parameter interval  $I$ , including all cuts and occurrences, is a function of inputs at  $\|\cdot\|_*$ -distance at most 10 from  $\{td : t \in I\}$ . The rule commutes with translation by  $d$  and a unit shift of nominal parameter. At an integer parameter level use the first occurrence of that level as the marked cut  $X_i$ . In particular,

$$X_i(\omega, U) = id + X_0(\theta_{id} \omega, \theta_{id} U). \quad (9.5)$$

The cost between consecutive marked cuts has finite expectation. Translated copies are obtained by translating this same rule, using fresh private inputs when independence of the private randomizations is required.

A pause at an integer level belongs to the following period under this convention; any other covariant marking gives the same accounting. Locality applies to private inputs as well as edge weights. In particular, it does not permit a global random phase shared by distant parts of a road.

Measurability of a parametrized walk is understood in the following standard Borel coding. In a compact parameter window, record the finite ordered list of labeled closed subintervals of oriented edge occurrences, their endpoint fractions in the original edges, and their continuous nominal maps after linear reparametrization to  $[0, 1]$ . Use the usual Borel structure of  $C([0, 1])$  with its uniform topology, together with the discrete edge and occurrence labels. Marked occurrences are part of this local record, and repeated visits retain distinct labels. This coding also applies to every bounded road portion below.

**Lemma 9.2** (Random-cut period identity). *For any admissible  $d$ -road,*

$$\mathbb{E}[P_n(X_{i+1}) - P_n(X_i)] = 0, \quad \mathbb{E}D_n(\gamma[i, i+1]) = \mathbb{E}Q_n(\gamma[i, i+1]) \geq 0. \quad (9.6)$$

Here  $\gamma[i, i+1]$  denotes the oriented road portion between its marked cuts.

*Proof.* By the joint stationarity of the edge and private inputs, it is enough to consider  $i = 0$ . Combining the centered cocycle identity with (9.5) gives

$$P_{n,\omega}(X_1) = P_{n,\omega}(d) + P_{n,\theta_d \omega}(X_0(\theta_d \omega, \theta_d U)).$$

The first term has mean zero; the second has the law of  $P_n(X_0)$ . All terms are integrable because each cut lies on one of finitely many deterministic edges, where domination bounds its calibration value by a finite sum of weights and a deterministic linear term. Taking expectations proves the first identity, and (9.2) proves the second. The mean of  $P_n(X_0)$  itself need not vanish.  $\square$

For a pair of admissible roads define its common period bound as

$$\beta = \max\{\mathbb{E}Q_n(\gamma_A[0, 1]), \mathbb{E}Q_n(\gamma_B[0, 1])\},$$

and let  $b(n)$  be the infimum over admissible pairs. The period identity gives  $b(n) \geq 0$ . More generally, a forward portion between levels  $u \leq v$  has defect bounded by the sum of the complete-period defects whose parameter intervals meet  $[u, v]$ . Nonnegativity makes this a pathwise bound, valid for environment-selected endpoints. We will repeatedly use it to bound a random road portion by a fixed number of periods.

### 9.3 Initialization and a strictly positive infimum

The two estimates for  $b(n)$  use the same elementary fact: a long crossing cannot be very cheap, uniformly over all choices of the crossing path. It first localizes geodesics between nearby anchors; later it guarantees a saving when all weights in a small region are decreased.

**Lemma 9.3** (No long cheap paths). *Let  $V$  be a finite set of vertices of  $\mathbb{Z}^2$ . For every integer  $R \geq 1$  and  $t \geq 0$ ,*

$$\mathbb{P} \left( \begin{array}{c} \text{some self-avoiding path starting in } V \text{ has} \\ \text{at least } R \text{ edges and cost at most } t \end{array} \right) \leq \frac{8}{3}|V|e^{5t}2^{-R}. \quad (9.7)$$

*Consequently there are numerical constants  $c, C > 0$  such that, for  $z \neq 0$  and  $R \geq 40|z|_1$ , the probability that a geodesic from 0 to  $z$  has at least  $R$  edges is at most  $Ce^{-cR}$ .*

*Proof.* A fixed vertex starts at most  $4 \cdot 3^{m-1}$  self-avoiding paths of length  $m$ . For each path, independence and the identity  $\mathbb{E}e^{-5\tau_e} = 1/6$  give

$$\mathbb{P}(\tau(\gamma) \leq t) \leq e^{5t}6^{-m}.$$

Sum over paths and over  $m \geq R$  to obtain (9.7).

For the geodesic estimate, compare with a deterministic coordinate path of length  $\ell = |z|_1$  and cost  $S$ . Its exponential moment gives  $\mathbb{P}(S > t) \leq 2^\ell e^{-t/2}$ . Since weights are positive, a geodesic is self-avoiding. If it has at least  $R$  edges and  $S \leq t$ , it is counted by (9.7) with  $V = \{0\}$ . Taking  $t = (\log 2)R/10$  makes both probabilities exponentially small in  $R$  when  $R \geq 40\ell$ . The same estimate excludes arbitrarily long paths of bounded cost. Comparing with  $S$  therefore also proves existence of a finite minimizing path.  $\square$

**Proposition 9.4** (Sublinear initialization). *For fixed  $s, v, w$  there are admissible pairs of roads with*

$$b(n) \leq C_s \sqrt{n} \log(2+n) \leq C'_s n^{3/4} \quad (9.8)$$

*for all sufficiently large multiples of a fixed integer. The constants may depend on  $s$ ; the exponent  $3/4$  does not.*

*Proof.* We concatenate geodesics between closely spaced deterministic anchors, restricting each geodesic to a small ball to make the rule local. Choose an integer  $m$  large enough that, with  $\delta = 1/m$ ,

$$\delta \max(|A|_1, |B|_1) \leq r/(1000K_{\text{lat}}).$$

Increase the fixed divisibility requirement so that  $n\delta A$  and  $n\delta B$  are lattice vectors. For  $d = A$  or  $B$ , join successive deterministic anchors  $k\delta d$  by a minimizing path restricted to the lattice portion of the  $\|\cdot\|_*$ -ball of scaled radius  $r/4$  about the first anchor. Each microscopic edge has  $\|\cdot\|_*$ -length at most  $K_{\text{lat}}/n$ , so a coordinate path between the anchors stays within  $K_{\text{lat}}\delta|d|_1 \leq r/1000$  of the first anchor. The restricted minimum therefore exists. Use a translation-covariant tie rule.

Restriction changes the minimum only if the unrestricted geodesic leaves the ball. Such a geodesic has at least  $rn/(4K_{\text{lat}})$  edges, so Lemma 9.3, with integer rounding, bounds this event by  $Ce^{-c_s n}$  for large  $n$ . On that event the increase is at most the coordinate-path cost  $S_k$ . Since  $(\mathbb{E}S_k^2)^{1/2} \leq C_s n$ , Cauchy-Schwarz bounds the expected increase by  $C_s n e^{-c_s n/2}$ .

For each patch, Lemma 8.1 bounds the mean unrestricted cost above its norm value by  $C_s \sqrt{n} \log(2+n)$ . The latter value is  $\mu(n\delta d) = n\delta h(d)$  because  $d$  lies in the cone of the face. Summing over the  $m$  patches in a unit period proves the first inequality in (9.8); increasing  $C'_s$  gives the second.

Parametrize each patch monotonically from  $k\delta$  to  $(k+1)\delta$ , for example proportionally to its number of edge occurrences. Its  $\|\cdot\|_*$ -distance from its nominal location is at most  $r/4 + \delta\|d\|_* \leq r/4 + K_{\text{lat}}\delta\|d\|_1 < r$ . The inputs for each patch lie within this same distance of every nominal point in the patch. Thus the radius-ten locality condition holds. The rule commutes with translation by  $d$ , and each period has finite expected cost. This proves admissibility.  $\square$

**Proposition 9.5** (Uniform positive lower bound). *There is  $c_0 > 0$ , depending only on the fixed flat and the dependence radius 10, such that*

$$b(n) \geq c_0 \tag{9.9}$$

*for all sufficiently large admissible  $n$ . The constant  $c_0$  is independent of the road rules, of sufficiently small  $s$ , and of the nearby rational directions. The lower threshold on  $n$  may depend on  $s$  and the divisibility requirements.*

*Proof.* Two routes around a large square have small expected defects if  $\beta$  is small. We decrease weights near the middle of one route, leaving the other route and their common endpoints fixed. The first route then becomes a cheaper competitor, while the change in law is small enough to transfer that saving back to the original environment.

*The two routes.* Fix an admissible pair with common bound  $\beta$ . Write  $(\xi, \eta) = [A \ B]^{-1}x$  for its oblique coordinates, and put  $\rho = 2r$ . Choose the fixed even integer

$$L = 200.$$

Use four translated roads, with nominal lines

$$\begin{aligned} \eta &= 0, & \eta &= L & \text{for the two } A\text{-roads,} \\ \xi &= 0, & \xi &= L & \text{for the two } B\text{-roads.} \end{aligned}$$

They use the same physical edge weights and independent private inputs. Their strips have oblique half-width at most  $\rho$ .

At each nominal corner, determine the intersection from parameter windows of half-width  $4\rho$  on the two incident roads. One road crosses the square of coordinate half-width  $2\rho$  from left to right inside its horizontal strip, and the other crosses from bottom to top inside its vertical strip. The elementary planar crossing property for a rectangle gives an intersection. Nearest-neighbor lattice edges can meet only at a common vertex or along a common edge. If  $K_{\text{lat}}/n \leq r$ , a common vertex can be chosen with coordinate distance at most  $2\rho$  from the corner and with both nominal parameters within  $3\rho$  of their corner values: moving from an interior intersection to an endpoint of its edge costs at most  $2K_{\text{lat}}/n \leq \rho$  in coordinates, after which tracking costs at most another  $\rho$  in nominal parameter. The chosen vertex remains in the inspected windows. Select the lexicographically first eligible vertex and then the first pair of eligible occurrences. These finite choices give a measurable local corner rule.

The bottom and right sides, traversed forward between these selected corners, form  $R_1$ ; the left and top sides form  $R_2$ . Their initial and terminal vertices agree. Since  $3\rho < 1$ , every side used is a subpath of the periods with indices  $-1, 0, \dots, L$ . Nonnegative defects and Lemma 9.2 give, with  $C_R = 2(L+2)$ ,

$$\mathbb{E}D_n(R_j) \leq C_R\beta, \quad j = 1, 2, \quad |\tau(R_1) - \tau(R_2)| \leq D_n(R_1) + D_n(R_2). \tag{9.10}$$

*A region traversed by only one route.* Consider the deterministic middle-side parallelogram

$$S = \{(L/2)A + uA + vB : |u| \leq 2, |v| \leq 2\},$$

and let  $\mathcal{E}_n(S)$  be the edges having both endpoints in  $S \cap \mathcal{G}_n$ . The entire input support of  $R_2$ , including its selected endpoints and the middle corner, is outside  $S$ . Indeed that support is contained in the  $\|\cdot\|_*$ -neighborhood of radius 10 of the deterministic union  $\mathcal{C}$  of the left and top nominal segments and all incident nominal corner windows of parameter half-width  $4\rho$ . The coordinate distance from  $S$  to  $\mathcal{C}$  is at least  $L/2 - 2 - 4\rho$ . By (9.3), its  $\|\cdot\|_*$ -distance is at least  $\frac{3}{4}(L/2 - 2 - 4\rho) > 10$ . The same separation holds for the unused corner window. Thus changing only weights in  $\mathcal{E}_n(S)$  leaves  $R_2$  and its endpoints unchanged, even when its rule and corner selections are recomputed.

The bottom-road portion with nominal parameters between  $L/2 - 1$  and  $L/2 + 1$  belongs to  $R_1$ . For sufficiently large  $n$ , every edge meeting this portion has both endpoints in  $S$ . Discarding the first and last partial edges leaves endpoints with coordinate separation at least  $2 - 2\rho - 4K_{\text{lat}}/n \geq 1$ . Each microscopic edge has  $(A, B)$ -coordinate length at most  $2K_{\text{lat}}/n$ , so the resulting lattice path has at least  $n/(2K_{\text{lat}})$  edges, up to integer rounding. Loop erasure preserves the endpoints and their separation, so the resulting self-avoiding path obeys the same length bound and has no greater cost.

There are at most  $C_V n^2$  vertices in  $S \cap \mathcal{G}_n$ , where one may take  $C_V = (8M + 3)^2$ . Apply Lemma 9.3 with

$$R = \lfloor n/(2K_{\text{lat}}) \rfloor, \quad a_* = (\log 2)/(40K_{\text{lat}}), \quad t = a_* n.$$

For  $n \geq 4K_{\text{lat}}$  the resulting bound is

$$\mathbb{P}(\text{the cost of } R_1 \text{ charged to } \mathcal{E}_n(S) \text{ is below } a_* n) \leq q_n := Cn^2 e^{-cn}, \quad (9.11)$$

The constants  $c, C > 0$  depend only on  $K_{\text{lat}}, M$ . Uniformity in the road rule follows because the cheap-path event includes every self-avoiding path starting at every vertex of  $S$ .

*Perturbation and transfer of the saving.* Keep the private inputs fixed. For  $0 < \varepsilon \leq 1/2$ , put  $a_n = 1 - \varepsilon/n$  and multiply every weight in  $\mathcal{E}_n(S)$  by  $a_n$ . Let  $\mathbb{P}$  be the original joint law of weights and private inputs, and  $\mathbb{Q}$  its pushforward under this map. Write  $m_n = |\mathcal{E}_n(S)|$ . Since each vertex is charged at most two undirected edges,  $m_n \leq C_E n^2$ , where  $C_E = 2C_V$ . The relative entropy of a scaled exponential variable with respect to an unscaled one is  $a_n - 1 - \log a_n$ . Product additivity, followed by Pinsker's inequality [33, Theorem 31, order one], with total variation taken as one half of the  $L^1$  distance, gives

$$\begin{aligned} D_{\text{KL}}(\mathbb{Q} \parallel \mathbb{P}) &= m_n(a_n - 1 - \log a_n) \leq C_E \varepsilon^2, \\ \|\mathbb{Q} - \mathbb{P}\|_{\text{TV}} &\leq C_{\text{TV}} \varepsilon, \quad C_{\text{TV}} = \sqrt{C_E/2}. \end{aligned} \quad (9.12)$$

Here we used  $-x - \log(1 - x) \leq x^2$  for  $0 \leq x \leq 1/2$ .

Set  $\lambda = a_* \varepsilon/2$ . Let  $E$  be the event that the cost of the current  $R_2$  exceeds passage time between its current endpoints by at least  $\lambda$ . This event uses only the current  $R_2$ , its endpoints and the passage metric. In the perturbation coupling, outside the event in (9.11), the old  $R_1$  saves at least  $a_* \varepsilon$ . It still joins the endpoints of the unchanged  $R_2$ . If its old cost exceeded that of  $R_2$  by at most  $\lambda$ , it is therefore a competitor cheaper by at least  $\lambda$  after perturbation. Equation (9.10) and Markov's inequality give

$$\mathbb{Q}(E) \geq 1 - q_n - \mathbb{P}(D_n(R_1) + D_n(R_2) > \lambda) \geq 1 - q_n - \frac{2C_R \beta}{\lambda}.$$

Only the old  $R_1$  is used as a competitor; its recomputed rule may produce a different path. Transferring  $E$  to the original law by (9.12) yields

$$\mathbb{P}(E) \geq 1 - q_n - C_{\text{TV}} \varepsilon - \frac{2C_R \beta}{\lambda}.$$

In the original environment domination implies  $D_n(R_2) \geq \lambda$  on  $E$ , also for its random endpoints. Hence

$$C_R \beta \geq \lambda \mathbb{P}(E) \geq \lambda(1 - q_n - C_{TV}\varepsilon) - 2C_R \beta.$$

Fix  $\varepsilon > 0$  with  $C_{TV}\varepsilon \leq 1/8$  and then take  $n$  large enough that  $q_n \leq 1/8$ . Rearranging yields

$$\beta \geq \frac{a_* \varepsilon}{8C_R} =: c_0 > 0.$$

The constants in  $c_0$  depend only on the uniform geometry bounds and the fixed locality radius. Taking the infimum over admissible pairs proves (9.9). The transfer used the event  $E$ , so no comparison of calibrations in the two environments was needed.  $\square$

## 10 Local gauges for nearly calibrated roads

The defect estimates above involve the generally nonlocal calibration  $P_n$ . The next step replaces its increments by increments of a *local gauge*: an endpoint potential computed from nearby inputs. For endpoints with specified road labels, the corrected cost is  $Q_n(\gamma) - g(y) + g(x)$ . These corrections telescope whenever successive pieces use the same endpoint label.

The construction has three stages. Short signed trips along crossing roads estimate calibration increments. A finite-cell projection turns these local estimates into local potentials. Interpolation then controls all attachment occurrences in a bounded window at once. Throughout, we estimate differences of calibration errors; we do not need a local approximation to the absolute value of  $P_n$ .

Expectations include the roads' private randomness. We may condition throughout on a fixed finite collection of phases, provided every hypothesis, including the period defect bound, holds uniformly over the conditioned phase types. We retain the norm  $\|\cdot\|_*$  and geometry constants of Section 9.2. Conversion to  $(A, B)$  or  $(v, B)$  coordinates has operator norm at most 2 uniformly for  $0 < s < 1/4$ ; the scaled edge length is at most  $K_{\text{lat}}/n$ , where  $K_{\text{lat}}$  may depend on the fixed flat. Suppress the subscript  $n$  in this section. We write  $Q(\gamma) = \tau(\gamma) - nh(y - x)$  for a path from  $x$  to  $y$ ,  $P(x) = B_n(0, x) - nh(x)$ , and

$$D(\gamma) = Q(\gamma) - P(y) + P(x) \geq 0.$$

A signed trip is a concatenation of forward road intervals with specified signs: both  $Q$  and the increments of  $P$  are summed with those signs. It estimates a potential difference. Physical path costs, by contrast, always charge every traversal positively.

### 10.1 A bounded network of signed comparisons

**Lemma 10.1** (Local increment comparisons). *Let  $A = v + sw$ ,  $B = w + sv$  with  $0 < s < 1/4$ , and suppose two road rules have tracking radius  $r$  and dependence radius 10 in  $\|\cdot\|_*$ , and expected defect at most  $\beta$  on each marked unit period. There exist numerical constants  $r_0, H, C > 0$  with the following property whenever  $r < r_0$  and  $n \geq K_{\text{lat}}/r$ .*

*One can choose a random anchor  $X_z$  near each point  $z_1 A + z_2 B$ ,  $z \in \mathbb{Z}^2$ , and, for  $i = 1, 2$ , a random variable  $E_{z,i}$  such that*

$$\mathbb{E} |E_{z,i} - P(X_{z+e_i}) + P(X_z)| \leq C\beta. \quad (10.1)$$

*The variable  $E_{z,i}$  uses only input cells at index distance at most  $H$  from  $z$ . Here an input cell contains all microscopic weights and private inputs assigned to the corresponding unit parallelogram.*

One may also prescribe any collection of additional marked anchors  $X_{z,\lambda}$ , each on a translated  $A$ - or  $B$ -road within a fixed distance of  $z_1A + z_2B$ . There are comparisons of the same kind between any two such anchors at the same or neighboring indices. Their constants depend on that distance, but not on the number of labels  $\lambda$ .

*Proof.* We build a grid from the two road families, then attach any additional anchor to that grid by a bounded signed trip. Put an  $A$ -road on each line  $jB + \mathbb{R}A$  and a  $B$ -road on each line  $iA + \mathbb{R}B$ . Write physical coordinates in the basis  $(A, B)$ . The coordinate-conversion bound implies that the first road lies in  $|x_2 - j| \leq \delta$  and the second in  $|x_1 - i| \leq \delta$ , where  $\delta = 2r$ . Take  $r_0 = 1/100$ , so  $8\delta < 1/2$ .

The portions whose nominal parameters are within  $4\delta$  of the nominal crossing connect opposite sides of a small rectangle. The planar crossing property gives an intersection of the two portions. An intersection lies in both strips and hence within coordinate distance  $\delta$  of the nominal crossing. Nearest-neighbor paths meet at a common vertex or along a common edge. Each scaled edge has  $(A, B)$  coordinate length at most  $2K_{\text{lat}}/n \leq \delta$ ; a vertex of that edge therefore has coordinate distance at most  $2\delta$  from the crossing and nominal parameter distance at most  $3\delta$  on either road. It lies within the inspected portions. Choose the first such vertex occurrence along the  $A$  portion, breaking any remaining tie by its occurrence on the  $B$  portion. This choice uses only those two bounded portions, is covariant under lattice translations, and specifies  $X_{(i,j)}$  together with its occurrence label on each road. Loops and repeated visits cause no ambiguity because occurrences, rather than only vertex names, are used.

The disjoint parameter windows order neighboring intersections along the relevant road. Define  $E_{z,1}$  as the  $Q$ -cost of the intervening forward  $A$  portion, and define  $E_{z,2}$  using the  $B$  portion. Each portion lies in a fixed number of marked periods, so additivity and nonnegativity bound its expected defect by  $C\beta$ . This proves (10.1).

Tracking and locality put all inputs in a fixed number of unit parallelograms. Assign boundary weights by a fixed half-open convention and private inputs to the cell of their nominal location. Since each microscopic input is assigned to one cell, the cell variables are independent.

To attach an additional anchor, follow its road with either sign to a nearby transverse network road, then follow that road to a network intersection. A bounded number of network intervals reaches any other anchor under consideration. All nominal distances are bounded, and crossings use the same local selection rule. The error on each signed interval is its signed defect, whose absolute value is bounded by complete-period defects. The triangle inequality gives the stated comparison. Its constant depends on the allowed anchor distance, independently of the number of other labels.  $\square$

## 10.2 An explicit finite-cell lemma

We now isolate the step from local increment estimates to a local potential. Only independence of coarse cells is needed. A cell may contain arbitrarily many microscopic variables, so the constants below do not depend on lattice scale or on the fineness of a fixed row mesh.

**Lemma 10.2** (Local increments admit a local potential). *Let  $(\eta_z)_{z \in \mathbb{Z}^2}$  be independent random elements and let  $V_{z,\lambda} \in L^1$ , where  $\lambda$  belongs to an arbitrary label set with a distinguished label 0. Set  $\mathcal{F}_I = \sigma(\eta_z : z \in I)$  and  $Q_H(z) = z + [-H, H]^2 \cap \mathbb{Z}^2$ , with integer  $H \geq 2$ . Suppose that whenever  $|z - z'|_1 \leq 1$ , and for every pair of labels in question, there is an  $\mathcal{F}_{Q_H(z)}$ -measurable variable  $E_{(z,\lambda),(z',\lambda')}$  such that*

$$\left\| E_{(z,\lambda),(z',\lambda')} - V_{z',\lambda'} + V_{z,\lambda} \right\|_1 \leq \varepsilon. \quad (10.2)$$

Then there are random variables  $g_{z,\lambda}$ , measurable with respect to  $\mathcal{F}_{Q_{R+2H}(z)}$ , where  $R = 8H + 20$ , such that

$$\mathbb{E}g_{z,\lambda} = \mathbb{E}V_{z,\lambda}, \quad (10.3)$$

$$\|g_{z',\lambda'} - g_{z,\lambda} - V_{z',\lambda'} + V_{z,\lambda}\|_1 \leq C_H \varepsilon. \quad (10.4)$$

The constant  $C_H$  depends only on  $H$ , not on the number of labels. Translation covariance is preserved whenever the true increment field  $V_{z',\lambda'} - V_{z,\lambda}$  is covariant under measure-preserving input shifts and the deterministic means have the required covariance.

*Proof.* We decompose a local increment according to the finite sets of cells on which it depends. Each component can be integrated locally; an exterior detour controls the error where that integration is cut off.

For a nonempty finite  $S \subset \mathbb{Z}^2$ , define

$$\Pi_S F = \sum_{I \subset S} (-1)^{|S|-|I|} \mathbb{E}[F \mid \mathcal{F}_I].$$

These are finite-product ANOVA (Hoeffding) projections; see Efron and Stein [15, Section 2] for the classical square-integrable decomposition. Here we use only finite inclusion–exclusion identities in  $L^1$ . Each projection has mean zero and  $L^1$  operator norm at most  $2^{|S|}$ . If  $F$  is measurable with respect to a finite cell set  $N$ , then  $\Pi_S F = 0$  unless  $S \subset N$ , and

$$F = \mathbb{E}F + \sum_{\emptyset \neq S \subset N} \Pi_S F. \quad (10.5)$$

Cancellation and independence prove both identities directly in  $L^1$ ; no infinite or  $L^2$  expansion is involved.

A comparison supported on  $Q_H(z)$  has components only on sets of at most  $m = (2H+1)^2$  cells and diameter at most  $2H$ . Accordingly, let  $\mathcal{S}$  be the collection of nonempty finite sets  $S$  with  $|S| \leq m$  and  $\text{diam}_\infty S \leq 2H$ . Let  $k(S)$  be the lexicographically least point of  $S$ , put

$$U_S = Q_R(k(S)), \quad z_S = k(S) + (R + 4H + 10)e_1,$$

and use  $z_S$  as a reference beyond the cutoff  $U_S$ . Define

$$g_{z,\lambda} = \mathbb{E}V_{z,\lambda} + \sum_{\substack{S \in \mathcal{S} \\ z \in U_S}} \Pi_S (V_{z,\lambda} - V_{z_S,0}). \quad (10.6)$$

There are only a bounded number  $N_H$  of summands:  $k(S)$  must lie in  $Q_R(z)$  and  $S \subset Q_{2H}(k(S))$ . For example  $N_H \leq (2R+1)^2 2^{(4H+1)^2}$  suffices. Every summand is measurable with respect to cells in  $Q_{R+2H}(z)$  and has mean zero. This proves locality and (10.3).

*The cutoff error.* Suppose  $z \in U_S$  but a neighbor  $z'$  lies outside. There is a lattice path from  $z$  to  $z_S$  of length at most  $20(R+4H+10)$  lying outside  $Q_{R-2}(k(S))$ : follow the boundary of  $Q_R(k(S))$  to its eastern side and then move east. Every local support  $Q_H(u)$  along this path is disjoint from  $S \subset Q_{2H}(k(S))$ . If necessary first switch from label  $\lambda$  to label 0 at  $z$ . By (10.2), telescoping and the projection norm,

$$\|\Pi_S (V_{z,\lambda} - V_{z_S,0})\|_1 \leq [20(R+4H+10) + 1] 2^m \varepsilon. \quad (10.7)$$

Each local comparison has zero  $\Pi_S$  projection because its support misses  $S$ . The connected exterior in two dimensions is used exactly at this step.

*Recovery of an increment.* Consider an edge  $(z, \lambda), (z', \lambda')$  from the hypothesis and write  $W = V_{z', \lambda'} - V_{z, \lambda}$  and  $E$  for its local comparison. The centered variables satisfy  $\|W - \mathbb{E}W - E + \mathbb{E}E\|_1 \leq 2\varepsilon$ . Every nonempty subset of  $Q_H(z)$  contributes to both endpoint gauges, since  $R > H + 1$ . A set contributing at both endpoints gives  $\Pi_S W$ , within  $2^m \varepsilon$  of  $\Pi_S E$ ; this latter projection vanishes when the set is not contained in  $Q_H(z)$ . A set contributing at only one endpoint is controlled by (10.7). Thus (10.5), the centered error bound above and at most  $2N_H$  component errors prove (10.4). For a same-index label change there are no cutoff terms.

Finally, the choices  $k(S)$ ,  $U_S$  and  $z_S$  commute with index translations. The random summands in (10.6) contain only differences of  $V$ , so a random common additive constant under translation cancels. Covariance of the increments and the deterministic means therefore passes to the gauge.  $\square$

*Remark 10.3* (Random-anchor means). In the application  $V_{z, \lambda} = P(X_{z, \lambda})$ , the deterministic mean in (10.6) need not be zero. It is, however, compatible with translation. If  $X_{z+k, \lambda}(\omega) = t_k + X_{z, \lambda}(\theta_{t_k} \omega)$ , the centered cocycle identity and  $\mathbb{E}P(t_k) = 0$  give  $\mathbb{E}P(X_{z+k, \lambda}) = \mathbb{E}P(X_{z, \lambda})$ . Thus it is a label mean, compatible with spatial covariance. A common deterministic constant may be subtracted from every label.

### 10.3 Interpolation and uniform envelopes

Apply Lemmas 10.1 and 10.2 with  $\varepsilon = C\beta$ . To extend the result from marked anchors to arbitrary attachments, let  $g_i$  be the gauge at the  $i$ th marked cut of a road and put

$$C_i = Q(i, i+1) - g_{i+1} + g_i.$$

Additivity and nonnegativity of defect give

$$\mathbb{E}|C_i| \leq C\beta. \quad (10.8)$$

For an occurrence between the marked cuts  $i$  and  $i+1$ , with nominal parameter  $t \in [i, i+1]$ , set

$$g(t) = g_i + Q(i, t) - (t - i)C_i. \quad (10.9)$$

Here  $Q(i, t)$  includes the occurrence label. During a pause the formula is applied separately at successive occurrences; as before, a pause at an integer belongs to the following period. For any two ordered occurrences in the same marked period, cancellation gives

$$Q(t, u) - g(u) + g(t) = (u - t)C_i. \quad (10.10)$$

Reversing a signed trip changes the sign of this identity. Splitting at integer levels shows that a prescribed window of length  $\ell \leq 1$  has absolute corrected cost at most  $\ell$  times the sum of  $|C_i|$  over its at most two periods. In particular, taking a supremum over attachment occurrences costs no factor for the number of lattice vertices in the window.

Writing  $P(t)$  for the calibration at the physical occurrence,

$$[g(t) - P(t)] - [g_i - P(i)] = D(i, t) - (t - i)C_i. \quad (10.11)$$

The right-hand side has expected absolute supremum at most  $C\beta$ , because  $0 \leq D(i, t) \leq D(i, i+1)$ . Together with the anchor increment estimates, this gives the simultaneous bounds needed for randomly selected attachments.

**Local extrema with variable endpoints.** All parameter windows used in a local optimization are closed and bounded. A bounded polygonal domain denotes its closed intersection with the metric graph, subdivided at the finitely many endpoints of point or segment intersections with its boundary. These conventions justify the extrema below even when an attachment lies inside an edge.

For fixed local inputs, a window contains finitely many labeled edge occurrences. Positive edge weights allow loops in a physical competitor to be deleted. Deletion preserves the later band and drawdown constraints, because it retains a subsequence of the traversal order. There are therefore finitely many combinatorial route types. For one type, let  $(a, b) \in [0, 1]^2$  be the fractions specifying its two endpoints. Membership in the closed parameter windows is a closed condition by continuity of the nominal maps. Membership in the subdivided domain, and the later band and drawdown conditions, are also closed: on each linear segment the coordinate extrema occur at its endpoints, leaving finitely many inequalities. Thus the feasible fractions form a compact set. Partial passage costs and the interpolated gauges are continuous in these fractions, including during a nominal pause. Each corrected objective consequently attains its extrema on every nonempty type.

The extrema and a selected optimizer are measurable from the same local inputs. To see this directly, enumerate the route types by a fixed order of their relative edge and occurrence codes. For one type its constraints can be written as  $F(a, b) = 0$ , where  $F \geq 0$  is continuous in  $(a, b)$  and measurable in the local record; write  $c(a, b)$  for its continuous cost. With the infimum of the empty set equal to  $+\infty$ , compactness gives

$$\min_{F=0} c = \lim_{j \rightarrow \infty} \inf \{c(q) : q \in \mathbb{Q}^2 \cap [0, 1]^2, F(q) < 1/j\}.$$

The right side is measurable. For an empty feasible set it is eventually  $+\infty$ . When some type is feasible, take the first minimizing type, then successively minimize the two endpoint coordinates on its compact set of minimizers. The same formula makes this selector measurable, and the relative order makes it commute with translations. If every type is empty, return a fixed no-path symbol. The same argument applied to  $-c$  handles suprema, and equality of two physical attachment points is one more closed constraint for switch costs. A nonnegative envelope over an empty family, including the negative part defining  $U'$ , is set to zero.

**Lemma 10.4** (Local envelope). *Fix  $m$  road labels and parameter windows of length at most  $L$  contained in a bounded macroscopic neighborhood of a deterministic cell. There is a nonnegative local random variable  $U$  with  $\mathbb{E}U \leq C_{m,L}\beta$  which bounds all of the following: absolute corrected signed costs within these windows; sums with a bounded number of switches between these labels at intersections; and the gauge differences for such switches at their common physical vertex. The support radius depends only on  $L$ , the neighborhood size, and the original road locality radius.*

*If a physical path is confined to a prescribed deterministic finite closed metric subgraph and has endpoints on two of the windows, its corrected cost, including signed adjustments to prescribed center anchors, is bounded below by  $-U'$  for a local nonnegative variable  $U'$  with  $\mathbb{E}U' \leq C_{m,L}\beta$ . Its support also includes the prescribed subgraph. This bound holds simultaneously for every such path and every choice of its endpoints.*

*Proof.* Define  $U$  as a fixed multiple of the sum of the suprema of the listed corrected costs and switch gauge differences, large enough for the allowed number of switches. It is local because the road windows and their gauges are local.

For its expectation, compare with an auxiliary sum over the finitely many relevant periods and anchors: include  $|C_i|$ , the defects  $D(i, i+1)$ , and the absolute increment discrepancies in (10.4). At a switch the two calibration values refer to the same physical vertex and cancel. Equations (10.10)

and (10.11) therefore bound  $U$  by a constant times this auxiliary sum. Its expectation is  $O(\beta)$ , regardless of how many microscopic occurrences lie in the windows.

For physical competitors, let  $U'$  be the negative part of the minimum endpoint-adjusted corrected cost in the prescribed subgraph. This is again a local measurable variable by the preceding endpoint-selection argument. Domination gives  $Q \geq \Delta P$  for every competitor. After endpoint corrections, only calibration-error increments and signed corrected anchor adjustments remain in the lower bound. The same auxiliary sum controls their suprema in expectation by  $O(\beta)$ .

The auxiliary estimates may be nonlocal;  $U$  and  $U'$  were defined from the local costs themselves. Thus the argument does not require local pointwise dominators for the absolute errors  $P - g$ .  $\square$

**Lemma 10.5** (Linear-scale tail bound). *Suppose the period excess bound satisfies  $\beta \leq C_s n^\alpha$  for some fixed  $\alpha < 1$ . Fix the neighborhood and finitely many labels of Lemma 10.4. After subtraction of one common deterministic reference-label mean, a local random variable  $V$  bounds the absolute gauges at all those occurrences and the absolute raw signed  $Q$ -costs of their bounded road adjustments. For any fixed  $\alpha' \in (\max\{\alpha, 1/2\}, 1)$ , Lemma 8.1 gives*

$$\mathbb{E}V \leq C_s n^{\alpha'}, \quad \mathbb{P}(V > \epsilon n) \leq C_s \epsilon^{-1} n^{\alpha'-1}.$$

*The constants may depend on the fixed neighborhood and label collection.*

*Proof.* We bound the expectations of the local suprema appearing in the statement, then take their sum as  $V$ . The passage-time deficits used to estimate these expectations need not be local.

Choose a fixed macroscopic box containing the road portions in question, the bounded comparison trips used below and every lattice edge meeting them. Let  $K_n^{\text{vert}}$  be the maximum of  $(\mu(n(y-x)) - T(nx, ny))_+$  over lattice vertices in that box, and let  $M_n$  be its maximum edge weight. Put

$$C_\mu = 2 \max_{i=1,2} \mu(e_i), \quad K_n = K_n^{\text{vert}} + 2M_n + C_\mu.$$

The lower concentration estimate, summed over polynomially many vertex pairs and then integrated, gives  $\mathbb{E}K_n^{\text{vert}} \leq Cn^{\alpha'}$ . The exponential tail and  $O(n^2)$  edges give  $\mathbb{E}M_n = O(\log(2+n))$ . Hence  $\mathbb{E}K_n \leq Cn^{\alpha'}$ .

The enlargement handles interior-edge endpoints as well. Given a road subinterval from  $x$  to  $y$ , attach partial edges to endpoints  $x', y'$  of the incident lattice edges. The resulting vertex-to-vertex path costs at most the subinterval cost plus  $2M_n$ , while

$$\mu(n(y' - x')) \geq \mu(n(y - x)) - C_\mu.$$

The definition of  $K_n^{\text{vert}}$  therefore shows that every forward road subinterval in the box satisfies  $Q \geq -K_n$ , because  $\mu \geq h$ . For a complete period  $q = Q(i, i+1)$  we therefore have  $\mathbb{E}|q| \leq \mathbb{E}q + 2\mathbb{E}K_n \leq \beta + 2\mathbb{E}K_n$ . If  $q = q_1 + q_2 + q_3$  is split at any two occurrences, then  $q_2 \geq -K_n$  and  $q_2 \leq q + 2K_n$ . Thus the supremum of absolute raw subinterval costs is bounded by  $|q| + 2K_n$ , including for intervals selected after inspecting the road. The same conclusion holds for a bounded signed trip by addition.

For the gauges, each summand in (10.6) is a projection of a potential difference. Lemma 10.1 approximates that difference by a bounded signed trip with  $L^1$  error  $O(\beta)$ . The projection norm and the fixed number of terms therefore give an  $O_s(n^{\alpha'})$  bound. Taking expectations of the same comparison bounds the differences of deterministic label means. Interpolation by (10.9) extends the bound to the windows. After the common deterministic centering, the sum of these local absolute suprema supplies  $V$ . Markov's inequality then gives the asserted tail.

For random phases, take the common centering constant to be the mean of a specified fixed-phase reference anchor, and attach its translated copy as an extra label near each site. The signed-trip comparison bounds the difference between any phase-conditioned anchor mean and this reference by  $C_s n^{\alpha'}$ , uniformly in the phase. The centered cocycle identity makes the reference mean independent of the site. Thus centering remains deterministic and introduces no shared random phase.  $\square$

#### 10.4 A common gauge in several exterior completions

The preceding construction uses a calibration from one complete environment. We now show that its gauge on a preserved incoming road can be approximated using only that road's own inputs. This gives a common endpoint correction when the exterior environment is completed in several different ways.

**Lemma 10.6** (A gauge from the preserved incoming road). *Fix deterministic tubes and a baseline road contained in one of them. Write  $\eta = (\eta_0, \eta_{\text{other}})$  for all preserved tube weights, where  $\eta_0$  consists exactly of the weights in the baseline's own tube and  $\eta_{\text{other}}$  contains the remaining preserved edges. Let  $\xi$  contain the baseline's private road seeds and a private completion of the entire complement of its own tube. Construct the baseline once, using the actual weights in its own tube and these private inputs everywhere else. The resulting road is a function of  $(\eta_0, \xi)$  alone.*

*In each world  $j$ , retain all prescribed tube weights and use an iid completion  $\zeta^j$  elsewhere. For every  $j$ , the private input  $\xi$  is independent of the joint tuple  $(\eta, \zeta^j)$ ; different worlds may share coordinates with one another. Assume that every world has the correct product weight law, uses the same measurable calibration map, and supplies the local comparison network above with period bound  $\beta$ . Assume also that the shared baseline is admissible in every world and has expected defect at most  $\beta$  on each marked period there. The latter condition follows from the random-cut period identity when the shared baseline rule has period excess at most  $\beta$ .*

*Let  $g_x^j$  be the gauge given by (10.6) at a marked baseline anchor. There is a local function  $g_x^{\text{in}}$  of  $(\eta_0, \xi)$  alone such that, in every world,*

$$\mathbb{E}|g_x^j - g_x^{\text{in}}| \leq C\beta. \quad (10.12)$$

*Its support radius and the constant  $C$  are uniform in the number of worlds. After interpolation along any bounded baseline interval, the same assertion holds with an expected supremum bound of order  $\beta$ . No identification of the calibrations in different worlds is asserted. In particular, incoming gauges for disjoint baseline tubes with independent private completions and seeds are independent.*

*Proof.* Use the same deterministic coarse cell partition in every world. Within each cell, group the common tube coordinates, common private coordinates, and additional coordinates of that world. These cells are independent. Until we establish that a projection is common to all worlds,  $\Pi_S$  denotes the finite-cell projection in the product space of the world under consideration.

We first replace each calibration difference in the gauge formula by a common baseline cost. Let  $x$  be the anchor's coarse index, and choose a marked anchor  $y$  on the same baseline, a fixed integer number  $L_0$  of periods earlier. Take  $L_0$  large enough that  $y$  lies outside every cutoff  $U_S$  contributing at  $x$  in (10.6), with an additional margin of  $4H + 20$ . The coordinate-conversion bound allows one choice of  $L_0$  for all  $0 < s < 1/4$ .

Let  $Q_{y,x}$  be the forward baseline cost from  $y$  to  $x$ . The traveled edges lie in the baseline's own preserved tube, and the path and its marked occurrences depend only on  $(\eta_0, \xi)$ . Thus  $Q_{y,x}$  is the

same random variable in every world. Adding the period defects along this fixed number of periods gives

$$\|Q_{y,x} - P^j(X_x) + P^j(X_y)\|_1 \leq C\beta.$$

Fix a contributing set  $S$ . Attach  $y$  to the comparison network locally near  $y$ , and connect that network anchor to the reference anchor at  $z_S$  by signed comparisons along the boundary of  $Q_R(k(S))$  and its exterior. Such a route can reach the square's boundary, follow it to the eastern side, and then move east to  $z_S$ , just as in the cutoff argument leading to (10.7). The initial attachment is separated from  $S$  by the margin at  $y$ . Along the remaining route, the local supports  $Q_H(u)$  miss  $S \subset Q_{2H}(k(S))$ , since  $R = 8H + 20$ . The route has bounded length:  $L_0$  is fixed,  $k(S) \in Q_R(x)$ , and  $z_S - k(S)$  is fixed by the gauge construction.

Every local comparison on this route therefore has zero  $\Pi_S$  projection. Telescoping their calibration differences, using their  $O(\beta)$  errors and the bounded projection norm, yields

$$\|\Pi_S(P^j(X_y) - P^j(X_{z_S,0}))\|_1 \leq C\beta.$$

Combining this estimate with the preceding baseline estimate shows that each summand of  $g_x^j$  differs by  $O(\beta)$  from  $\Pi_S Q_{y,x}$ .

It remains to check that these projected baseline costs, including the deterministic mean term, are common to all worlds. The joint marginal law of the world weights and the shared baseline is the same in every world: the baseline uses its own tube weights together with private inputs independent of that world's actual exterior. Since the calibration map is also the same, the mean  $m_x = \mathbb{E}P^j(X_x)$  is independent of  $j$ . Define

$$g_x^{\text{in}} = m_x + \sum_{\substack{S \in \mathcal{S} \\ x \in U_S}} \Pi_S Q_{y,x}.$$

For any cell subset, conditioning  $Q_{y,x}$  on its independent  $\eta_{\text{other}}$  or  $\zeta^j$  coordinates adds no information to its  $(\eta_0, \xi)$  coordinates. Consequently, every conditional expectation in  $\Pi_S Q_{y,x}$  is a function only of that subset's  $(\eta_0, \xi)$  coordinates. These conditional expectations, and hence the projections themselves, can be chosen as the same functions in every world. They have no dependence on weights in the other preserved tubes.

Only a fixed number of sets  $S$  contributes, all within the fixed support radius of (10.6). Summing the projection errors proves (10.12) and locality, with constants independent of the number of worlds. The displayed construction also proves the independence assertion: disjoint baseline tubes supply disjoint sets of product edge weights, their private input collections are independent, and  $m_x$  is deterministic.

Finally, interpolate the common anchor gauges using the common raw baseline cost. In (10.9), the raw  $Q$  terms cancel when the two interpolations are subtracted. For  $t \in [i, i+1]$ , their discrepancy is therefore the linear interpolation of the discrepancies at  $i$  and  $i+1$ . Its supremum is bounded by the sum of those two anchor discrepancies. A bounded interval meets only a bounded number of periods, which proves the stated expected supremum bound.  $\square$

*Remark 10.7* (Finite averaging of mesh offsets). Suppose the required row translations generate a finite group modulo the coarse mesh periods. Perform the construction for every translated coarse mesh, and average the resulting gauge functions. Each required translation permutes these finitely many functions, so their average has exact covariance under the row translations. All their supports lie in one fixed-radius ball. By the triangle inequality, averaging preserves every preceding  $L^1$  bound without a factor equal to the number of offsets. Applying the incoming-road construction separately for each offset and then averaging also preserves Lemma 10.6. The averaging is over deterministic offsets and introduces no shared random mesh phase.

## 11 Phase conventions and locality of the comparison fields

The gauges, links, and auxiliary comparison networks will all be realized on one probability space. We first specify how translations act on their private inputs, then choose the wall phases and the comparison meshes. These conventions give the column independence needed for the output road and the finite-template reduction needed for two-dimensional selected-path estimates.

**Spatial private inputs and wall phases.** Index a  $d$ -road by its structural role and its nominal origin modulo  $\mathbb{Z}d$ , and index each private input also by its spatial location. A unit longitudinal shift keeps the road-position class and shifts the spatial index; a transverse or fractional-phase shift acts on both. Auxiliary world starts and transition slots carry their own position indices and translate as well. Distinct private coordinates are independent, and all are independent of the edge weights. Their product law is invariant under these shifts, as required by Definition 9.1. Thus translated road copies can use the same physical weights and independent private completions; two distant portions of one road also use disjoint private coordinates when their prescribed input neighborhoods are disjoint.

Fix a rational row spacing  $\sigma$  such that every row increment in use, including  $s$ , is an integer multiple of  $\sigma$ . Let  $\mathcal{H} \subset \mathbb{R}/\mathbb{Z}$  be the finite subgroup generated by these increments. The canonical  $B$ -walls have nominal lines  $jA + \mathbb{R}B$ ,  $j \in \mathbb{Z}$ . Independently of the spatial inputs, give these walls independent uniform phases  $\phi_j \in \mathcal{H}$ , measured relative to their nominal origins  $jA$ . Their absolute phases in  $(v, B)$  coordinates are

$$\theta_j = js + \phi_j \pmod{1}.$$

All auxiliary  $A$ -roads have deterministic longitudinal phases relative to their nominal origins. Their private randomness follows the spatial indexing convention above.

Choose representatives  $\phi_j \in [0, 1)$ . A phase translates the road rule together with its marked cuts and private inputs. More explicitly, if  $\gamma_B(u)$  is the original  $B$ -road rule, the phased wall is

$$jA + \phi_j B + \gamma_B(\theta_{jA + \phi_j B \omega}, \text{translated private inputs}; u), \quad t = u + \phi_j.$$

Here  $t$  is the nominal parameter relative to  $jA$ . The marked cuts  $u = k$  have absolute heights  $js + \phi_j + k$ . We restrict  $n$  to sufficiently divisible integers so that every phase displacement is a lattice translation. At each fixed phase, translation preserves the product input law and the expected marked-period excess. Consequently the nonnegative period-defect estimates and all bounded-window comparisons remain uniform after the phases are fixed. Changing a phase representative by an integer simply invokes the road's unit-period covariance.

**Averaging the comparison gauges.** Fix one comparison radius  $H$  large enough for every structural local anchor attachment used below. In each offset mesh, register each labeled anchor once at the lexicographically first nearest mesh index to its deterministic nominal point, and reuse that registration wherever the label appears. This rule commutes with translations and has a uniform distance bound; longer comparisons telescope these local ones. Lemma 10.1 therefore permits this choice before the canonical gauges are defined. Keep  $H$ , the cutoff family, the reference anchors, and the offset group fixed when later labels are added.

For each  $a \in \mathcal{H}$ , take the unit  $(A, B)$  comparison mesh shifted by  $aB$ , with auxiliary  $A$ -roads on its horizontal lines. Apply Lemma 10.2 on this mesh. For each physical road label used in a bounded comparison—a canonical  $B$ -wall, an  $A$ -road, a corridor road, or a baseline introduced later—include the same physical anchor as an additional label in every mesh. Average its gauge

functions over  $a$ . The formula (10.6) for a label uses only that anchor and the mesh's reference anchors, so these labelwise definitions agree when two local comparisons overlap.

All labels taking part in a switch are averaged over the same offsets. In particular, their averaged gauge difference is the average of the within-mesh differences controlled by the local-envelope lemma. The private inputs of all translated meshes are indexed as above, so translations permute this finite family. This construction changes only endpoint potentials: the physical roads and their anchor occurrences are already fixed.

Adjoining a new road label in this fixed construction does not change an existing gauge formula. Any new primitive coordinates placed in the cells are jointly independent of the preexisting primitives; conditioning an old variable on those extra coordinates does not change its conditional expectation. Thus later baseline labels can be added without changing the canonical wall gauges that define the link field.

**Lemma 11.1** (Local phases and covariant gauges). *The averaged gauges retain the local bounds, means, interpolation identities, and common-tube comparisons of the preceding section. Their support radii and bounds do not acquire a factor  $|\mathcal{H}|$ . They are covariant under translation by  $A$ , and their laws are covariant under every row shift in  $\mathcal{H}$ .*

*For a link or gauge in a fixed neighborhood of wall  $j$ , all random phase dependence is confined to a fixed number of wall indices near  $j$ . Together with the independent spatial inputs, this gives independence of entire columns at sufficiently separated indices. The separation constant is independent of  $|\mathcal{H}|$  and  $\sigma$ .*

*Proof.* We first check phase dependence, including that of the conditional expectations used to construct a gauge. An anchor is determined by local weights, local private inputs, and the phases of the roads in a fixed neighborhood. The calibration itself depends on the weights alone. Each projected variable in (10.6) is the difference of calibration values at two anchors: the anchor carrying the gauge label and the reference anchor indexed by  $z_S$ . Both anchors lie within a fixed distance of the gauge's location. Indeed, for a gauge indexed by  $z$ , one has  $k(S) \in Q_R(z)$  and  $z_S = k(S) + (R + 4H + 10)e_1$ .

Regard the phases as fixed parameters when taking expectations over weights and private inputs. The variable being integrated depends on the phases only through the two anchors. Its conditional expectations therefore depend on the same finite list of phases. This applies to every projection and to the deterministic anchor mean in the gauge formula. The number of potentially involved  $B$ -wall indices is bounded by their spatial range, independently of the number of possible phase values. The spatial support of each projection is the one given by Lemma 10.2.

All mesh offsets lie in one bounded fundamental cell. Thus the supports of the gauges being averaged lie in one fixed larger neighborhood, uniformly in  $|\mathcal{H}|$  and  $\sigma$ . The triangle inequality for the normalized average preserves the  $L^1$  estimates without a factor  $|\mathcal{H}|$ . At a fixed physical anchor, each mesh gauge has the same mean, namely the expected calibration value at that anchor, so averaging preserves this mean. For a switch between two labels, the triangle inequality applied to the average of their within-mesh gauge differences preserves the switch estimate without a factor  $|\mathcal{H}|$ . Interpolation is affine in the anchor gauges and uses the same physical road cost on every mesh. It therefore commutes with averaging. The construction from the preserved incoming road can be averaged in the same way; its common-tube comparison and its dependence only on the baseline's own tube inputs and private field are preserved.

Translation by  $A$  advances the wall index by one and the absolute row height by  $s$ . Since  $\theta_j = js + \phi_j$ , its action on relative phases is the ordinary shift of the sequence  $(\phi_j)_{j \in \mathbb{Z}}$ . The road rules, spatial inputs, and gauge construction are covariant under this joint translation, giving covariance under  $A$ . Translating the decorated network by  $aB$ ,  $a \in \mathcal{H}$ , adds  $a$  modulo one to every

wall phase and permutes the offset meshes. Integer carries are absorbed by unit-period covariance. The independent uniform phase law is invariant under this addition, and the spatial product law is translation invariant. This proves the asserted covariance of the row laws.

Finally, the non-phase inputs for an entire column lie in a strip of fixed width about that column. Sufficiently separated columns have disjoint strips and disjoint finite sets of wall-phase indices. The auxiliary  $A$ -roads contribute only private coordinates at the spatial locations being inspected. Hence the corresponding column sigma-fields are generated by disjoint independent inputs. The strip width and the range of wall indices are independent of  $|\mathcal{H}|$  and  $\sigma$ , as required.  $\square$

**Finite templates for selected-path estimates.** A phase is shared along its entire wall, so the column independence above does not give two-dimensional finite dependence. For each fixed local test, however, Lemma 11.1 confines its phase dependence to finitely many wall indices, including both anchors in every gauge projection and the phases entering the anchor means. A template is an assignment of values in  $\mathcal{H}$  to this finite list. Section 13.6 extends these assignments periodically on sparse grids. Each template then gives an iid field, and their finite sum bounds the original nonnegative field pointwise. Different template fields need not be independent.

All translated offsets used here are rational and are fixed before  $n$ . Sufficient divisibility of  $n$  makes them lattice translations. The number of templates may depend on the fixed auxiliary meshes and on  $s$ , but it is independent of  $n$  and of the eventual denominators of the rational direction approximations. This auxiliary enumeration does not enter the input radius of the final road algorithm.

## 12 Uniform estimates for selected local observations

The local observations used later are chosen after inspecting the environment. We therefore need estimates for the largest reward over all admissible choices. There are two different tail estimates. Sparse-path counting controls fields with sufficiently light tails, including bounds obtained by choosing among separated candidates. For the tunnel rewards, the available tail is only of order  $t^{-2}$ ; their control instead uses the thin geometry of the selected diagram and a directed collection bound.

Selected-path estimates are related to the greedy lattice-animal and lattice-path bounds studied by Martin [28]. The critical quadratic tails here require the additional collection and geometric deformation arguments proved below.

The collection bound (12.7) is a hypothesis, proved for tunnels in Proposition 13.5. It does not follow from locality and a first-moment estimate. All constants below may depend on the fixed lattices, cones, dependence ranges, and observation templates, but are uniform in the index  $n$  of a family of laws. Constants denoted by  $C_F$  may also depend on the deformation parameter  $F$ ; constants denoted by  $C$  do not.

### 12.1 Sparse observations and light selected tails

Use the  $\ell^1$  norm on  $\mathbb{Z}^2$ , and let  $\mathcal{P}_L$  be the nearest-neighbor paths starting at the origin with at most  $L$  edges. Sums in this subsection count each visited site once. Allowing at most  $m_0$  occurrences per site multiplies the bounds by  $m_0$ .

**Lemma 12.1** (Sparse-path estimate). *Let  $(I_z)_{z \in \mathbb{Z}^2}$  be indicators with  $\mathbb{P}(I_z = 1) \leq p$ . Suppose  $\mathbb{Z}^2$  has a partition into  $q$  classes such that the variables within each class are mutually independent.*

For every integer  $L \geq 0$ ,

$$\mathbb{E} \max_{\pi \in \mathcal{P}_L} \sum_{z \in \pi} I_z \leq C_q \min\{L+1, (L+1)\sqrt{p}, (L+1)^2 p\}. \quad (12.1)$$

The same conclusion, with adjusted constants, holds for lists of distinct sites whose initial distance from the origin plus the sum of their successive distances is at most  $c(L+1)$ , for fixed  $c$ . It also holds on any fixed full-rank planar lattice. In particular, it applies to a field with a fixed range of dependence.

*Proof.* First suppose all indicators are independent. If a path visits  $\ell$  distinct marked sites, list them in their order of first encounter as  $z_1, \dots, z_\ell$ . With  $z_0 = 0$ ,  $\sum_{i=1}^\ell |z_i - z_{i-1}| \leq L$ . There are at most  $4(d+1)$  displacements of  $\ell^1$  length  $d$ . The number of possible lists is therefore at most

$$4^\ell \sum_{d_1 + \dots + d_\ell \leq L} \prod_{i=1}^\ell (d_i + 1) = 4^\ell \binom{L + 2\ell}{2\ell} \leq \left( \frac{C(L+1)}{\ell} \right)^{2\ell}, \quad 1 \leq \ell \leq L+1.$$

The identity follows from  $\sum_{d \geq 0} (d+1)x^d = (1-x)^{-2}$ , followed by summing coefficients through degree  $L$ . The count includes lists that cannot occur along a path, which only increases the bound.

Distinctness makes the probability that a specified list is marked at most  $p^\ell$ . If  $M_L$  is the maximum in the statement and  $a = C(L+1)\sqrt{p}$ , the union bound gives

$$\mathbb{P}(M_L \geq \ell) \leq \min\{1, (a/\ell)^{2\ell}\}. \quad (12.2)$$

For  $a \leq 1/2$ , summing over  $\ell \geq 1$  gives at most  $\sum_{\ell \geq 1} a^{2\ell} \leq 2a^2$ . For  $a > 1/2$ , split at  $\lceil 2a \rceil$ : the initial sum is  $O(a)$  and the remaining sum is geometric. Thus  $\mathbb{E} M_L \leq C \min\{a, a^2\}$ , after adjusting constants also for  $1/2 < a < 1$ . Combining this with  $M_L \leq L+1$  proves (12.1).

For a partition into independence classes, apply the same list count to sites in each class and sum the resulting  $q$  maxima. No independence between classes is needed. For example, when  $I_z$  depends on independent inputs in  $z + [-R, R]^2$ , residue classes modulo an integer greater than  $2R$  give the required partition. The same partition argument works when sigma-fields generated by sets at distance greater than a fixed range are independent.

Only the budget for successive distances entered the counting argument, so it also proves the assertion about lists. An invertible linear identification of a fixed full-rank lattice with  $\mathbb{Z}^2$  changes this budget by a fixed factor and proves the lattice version.  $\square$

**Corollary 12.2** (Integrated bound for selected tails). *Let  $(U_z)$  be nonnegative fields whose dependence range is uniform in a possible law index  $n$ , and suppose  $\sup_{n,z} \mathbb{P}(U_z > t) \leq p(t)$ . Then, for  $T \geq 0$ ,*

$$\mathbb{E} \sup_{\pi \in \mathcal{P}_L} \sum_{z \in \pi} U_z \mathbf{1}_{\{U_z > T\}} \leq C(L+1) \left[ T \sqrt{p(T)} + \int_T^\infty \sqrt{p(t)} dt \right]. \quad (12.3)$$

In particular, a uniform bound  $\mathbb{P}(U_z > t) \leq At^{-4}$  for  $t \geq 1$  gives  $C_A(L+1)/T$  for  $T \geq 1$ . The same bounds hold for the site lists and bounded multiplicities in Lemma 12.1.

*Proof.* For  $u \geq 0$ ,

$$u \mathbf{1}_{\{u > T\}} = T \mathbf{1}_{\{u > T\}} + \int_T^\infty \mathbf{1}_{\{u > t\}} dt.$$

Take the supremum over paths, move it inside the nonnegative integral as an upper bound, and apply Lemma 12.1 at every level. Tonelli's theorem gives (12.3), possibly with an infinite right-hand side. Integrating  $\sqrt{A} t^{-2}$  proves the fourth-power assertion.

The sharper minimum of  $(L+1)\sqrt{p(t)}$  and  $(L+1)^2p(t)$  in (12.1) can also be used inside the integral. In particular, the estimate has no additive constant independent of  $t$  to be integrated over an infinite interval.  $\square$

**Lemma 12.3** (Selection from separated candidates). *A fixed bounded region contains at most  $J$  candidate cells. For each cell  $c$ , let  $U_c \geq 0$  be measurable with respect to a specified subset of one family of independent inputs, with  $\mathbb{E}U_c \leq A$ . Suppose each admissible object provides a tuple of  $m$  cells whose input supports are pairwise disjoint. There is a local random variable  $V$  that bounds, simultaneously for every object, its smallest available  $U_c$ , and satisfies*

$$\mathbb{P}(V > t) \leq J^m \min\{1, (A/t)^m\}, \quad t > 0.$$

*For fixed  $J, m, A$ , the fourth moment of  $V$  is uniformly bounded when  $m > 4$ . The same assertion holds conditional on auxiliary data if the disjoint supports retain their product law and each conditional first moment is at most the same deterministic  $A$ .*

*Proof.* Before selecting an object, form

$$V = \max_{(c_1, \dots, c_m)} \min_{1 \leq i \leq m} U_{c_i},$$

where the maximum ranges over all tuples in the region with pairwise disjoint supports, including those that no admissible object supplies. Set  $V = 0$  if there are no such tuples. This is a local variable and gives the required simultaneous domination. For each fixed tuple, independence and Markov's inequality bound the probability that all  $m$  values exceed  $t$  by  $\min\{1, (A/t)^m\}$ . There are at most  $J^m$  tuples, so a union bound proves the tail estimate. Finally,

$$\mathbb{E}V^4 = 4 \int_0^\infty t^3 \mathbb{P}(V > t) dt < \infty \quad (m > 4).$$

The conditional proof is identical. The object itself may depend on the whole environment: independence is used only for each fixed tuple, before taking the finite maximum.  $\square$

## 12.2 Compressing a thin diagram by a lattice permutation

The critical-tail argument will compress one coordinate and stretch the other by reciprocal factors. The next lemma realizes this area-preserving deformation as a permutation of the lattice, so that transporting an iid field preserves its law exactly.

**Lemma 12.4** (Bounded-displacement matching). *Let  $\Lambda$  be a full-rank lattice in  $\mathbb{R}^2$  and let  $T$  be invertible with  $|\det T| = 1$ . There is a bijection  $\psi : \Lambda \rightarrow \Lambda$  and a finite  $b_T$  such that*

$$|\psi(z) - Tz| \leq b_T \quad (z \in \Lambda).$$

*For a bounded fundamental tile  $Q$  of  $\Lambda$ , one may take  $b_T \leq \sup_{q, q' \in Q} |Tq - q'|$ . For an affine map  $Tz + a$ , the same bound holds with  $|\psi(z) - (Tz + a)|$  on the left.*

*Proof.* Compare the tilings  $(T(z + Q))_{z \in \Lambda}$  and  $(w + Q)_{w \in \Lambda}$ . Join  $z$  to  $w$  when the corresponding tiles intersect in positive area. This bipartite graph is countable and locally finite. All tiles have the same area  $v_Q > 0$ . Any union of  $k$  tiles on either side has area  $kv_Q$  and is covered, up to a null set, by its neighboring tiles on the other side. It therefore has at least  $k$  neighbors. Both finite Hall conditions hold.

Here is the compactness step giving a perfect matching. By finite Hall's theorem [18, Theorem 1], any prescribed finite set of left vertices is covered by a finite matching  $M_1$ , and any prescribed finite set of right vertices by a finite matching  $M_2$ . The union of the two matchings consists of alternating paths and cycles, together with common edges. On each component one can choose an alternating matching covering all prescribed vertices. Internal vertices are covered by either choice. On an odd-length path, one choice covers both endpoints. On an even-length path, the endpoints lie on the same side, and the matching that covered the prescribed vertices on that side already covers every prescribed endpoint. Thus one matching covers both prescribed finite sets.

Exhaust the two vertex sets by finite sets and choose such a matching at each stage. A diagonal subsequence converges on every edge. Since each vertex has finite degree and is covered at all sufficiently large stages, exactly one of its incident edges belongs to the limit. The limit is a perfect matching.

Let  $\psi(z) = w$  for the matched pair. An intersection point has the form  $Tz + Tq = w + q'$ , with  $q, q' \in Q$ , giving the stated displacement bound. Translate the first tiling by  $a$  to obtain the affine statement, with the same error relative to  $Tz + a$ .  $\square$

We now specify the geometry and its uniformities. A cone here is closed, convex, pointed, and has nonempty interior. Fix a target cone  $\mathcal{K}$  strictly inside the positive quadrant, and fix linearly independent vectors  $u_1, u_2$  strictly inside  $\mathcal{K}$ , ordered so that  $U = (u_1, u_2)$  has positive determinant. Let  $E = (e_1, e_2)$  range over a fixed compact set of matrices with positive determinant. For each application,  $E$  and its lattice matching are deterministic. No supremum over an environment-dependent choice of  $E$  is taken. All constants in the thin-diagram estimates below are uniform over the allowed matrices  $E$ , with  $F$ -dependence indicated by subscripts.

**Lemma 12.5** (Thin diagrams and order errors). *Fix  $H, c > 0$  and a full-rank lattice  $\Lambda$ . Suppose  $z_1, \dots, z_k \in \Lambda$  satisfy the all-pairs condition*

$$E^{-1}(z_j - z_i) \geq -H(1, 1) \quad \text{coordinatewise for every } i < j, \quad (12.4)$$

*and lie in the parallelogram*

$$E([-H, cN + H] \times [-H, \delta N + H]). \quad (12.5)$$

*For  $F \geq 1$ , define*

$$T_F = \left( \frac{\det E}{\det U} \right)^{1/2} U \begin{pmatrix} F^{-1} & 0 \\ 0 & F \end{pmatrix} E^{-1}.$$

*Then  $\det T_F = 1$ . A matching  $\psi_F$  supplied by Lemma 12.4 sends these sites into a deterministic square of diameter at most*

$$C(N/F + F\delta N) + C_F. \quad (12.6)$$

*For every fixed  $r_0 > 0$ , there is a finite  $R_F$ , uniform in  $E$ , such that, if  $i < j$  and  $|z_j - z_i| > R_F$ , then  $\psi_F(z_j) - \psi_F(z_i) \in \mathcal{K}$  and both coordinates of this increment exceed  $r_0$ .*

*Proof.* The scalar prefactor makes the determinant one. In  $E$ -coordinates, the map contracts the first extent of the box by  $F$  and expands the second by  $F$ . The prefactor and  $U$  have uniform bounds, so the transformed diameter is at most  $C(N/F + F\delta N) + C_F$ . The fundamental-tile bound in Lemma 12.4 bounds the matching error uniformly in  $E$  for fixed  $F$ . Adding this error proves (12.6). Its  $N$ -dependent coefficient  $C$  is independent of  $F$ ; the deformation of the fixed margins and the matching error are included in  $C_F$ .

For  $v = z_j - z_i$ , condition (12.4) gives  $v = Ea + e$  with  $a \geq 0$  coordinatewise and  $|e| \leq CH$ . The vector  $T_F Ea$  is in the cone generated by  $u_1, u_2$ , strictly inside  $\mathcal{K}$ . Consequently its distance to the complement of  $\mathcal{K}$ , and each of its positive coordinates, are at least  $\rho_{\mathcal{K}}|T_F Ea|$  for a fixed  $\rho_{\mathcal{K}} > 0$ . For fixed  $F$ ,

$$|T_F Ea| \geq c_F(|v| - CH), \quad |T_F e| \leq C_F H,$$

with  $c_F > 0$ . Replacing  $T_F v$  by  $\psi_F(z_j) - \psi_F(z_i)$  adds at most twice the matching error. Choose  $R_F$  so that the cone margin and the two coordinate margins exceed these errors plus  $r_0$  whenever  $|v| > R_F$ . Compactness of the allowed matrices makes this choice uniform in  $E$ .  $\square$

The all-pairs condition is essential: a bounded error at each successive step may accumulate. The box condition follows, for example, when the same all-pairs condition also includes deterministic initial and terminal points with  $E$ -coordinates  $(0, 0)$  and  $(c'N, \delta'N)$ , where  $0 \leq c' \leq c$  and  $0 \leq \delta' \leq \delta$ . Neither condition bounds the number of visits to one site; that is a separate hypothesis below.

### 12.3 Critical tails in a thin diagram

Let  $(Y_z)_{z \in \Lambda}$  be a nonnegative iid field on a fixed full-rank lattice. For  $L \geq 1$  and  $x \in \mathbb{R}^2$ , let  $G(L, x)$  be the maximum of  $\sum_{z \in S} Y_z$  over chains in  $x + [-L, L]^2$  whose successive increments lie in  $\mathcal{K}$  and have both coordinates at least  $r_0$ . Include the empty chain and all singletons. Suppose the laws, possibly indexed by  $n$ , satisfy

$$\mathbb{E} G_n(L, x) \leq C_0 L + C_1, \quad L \geq 1, \quad x \in \mathbb{R}^2, \quad (12.7)$$

with the same constants for every  $n$ . This finite-volume bound will be proved for the charged tunnels in Proposition 13.5, before its use in Section 14.

**Lemma 12.6** (Weak second-moment bound). *Under (12.7),*

$$\sup_n \mathbb{P}(Y_z > t) \leq C \min\{1, t^{-2}\}, \quad t > 0.$$

*Proof.* Fix a law and  $t > 0$ , and set  $p = \mathbb{P}(Y_z > t)$ . There is nothing to prove if  $p = 0$ . Otherwise choose  $L$  comparable to  $p^{-1/2}$ , with lattice-dependent comparison constants, so that  $L \geq 1$  and  $[-L, L]^2$  contains at least  $p^{-1}$  lattice sites. Independence implies that some value in the square exceeds  $t$  with probability at least  $1 - e^{-1}$ . Since singletons are allowed in  $G$ ,

$$(1 - e^{-1})t \leq \mathbb{E} G_n(L, 0) \leq C_0 L + C_1 \leq C p^{-1/2}.$$

Rearrange and combine with  $p \leq 1$ . The constants do not depend on the law.  $\square$

This is a weak  $L^2$  estimate, not an assertion of a finite second moment. Substituting its  $t^{-2}$  tail into the square-root integral in (12.3) gives a divergent bound. The next argument uses the thin box to control the values that would cause this divergence.

**Lemma 12.7** (Negligible selected tails in a thin diagram). *Assume (12.7), and fix  $H, c, m_0$ . For  $N \geq 1$ , let  $\mathcal{S}(N, \delta)$  consist of finite site sequences satisfying (12.4) and (12.5), with at most  $m_0$  occurrences per site. Suppose also that their distinct sites can be visited by a path whose initial distance from the origin plus total length is at most  $c(N + 1)$ .*

*For every  $F \geq 1$  there is  $C_F < \infty$  such that, for  $T \geq 1$  and  $0 \leq \delta \leq F^{-2}$ ,*

$$\mathbb{E} \sup_{S \in \mathcal{S}(N, \delta)} \sum_{z \in S} Y_z \mathbf{1}_{\{Y_z > T\}} \leq \frac{CN}{F} + \frac{C_F(N + 1)}{T} + C_F. \quad (12.8)$$

Here  $C$  is independent of  $F, T, \delta, n$ . Consequently, for every  $\varepsilon > 0$  there are  $\delta_0 > 0$  and  $T_0 < \infty$  such that, for every sequence of laws and every  $N(n) \rightarrow \infty$ ,

$$\limsup_{n \rightarrow \infty} \frac{1}{N(n)} \mathbb{E} \sup_{S \in \mathcal{S}(N(n), \delta_0)} \sum_{z \in S} Y_z \mathbf{1}_{\{Y_z > T_0\}} \leq \varepsilon.$$

Finite sums of field types and independence sublattices satisfy the same conclusion if their number and all required constants are uniform.

*Proof.* Fix  $F$ , and take  $R_F$  from Lemma 12.5. We divide the selected values into nearby collisions, which have a lighter tail, and one separated set, to which the collection bound applies.

*Collision values.* Define the local field

$$D_z = Y_z \mathbf{1}_{\{\exists w \in \Lambda: 0 < |w-z| \leq R_F, Y_w \geq Y_z\}}.$$

If  $D_z > t$ , both  $Y_z$  and one distinct neighboring value exceed  $t$ . Independence and Lemma 12.6 give

$$\mathbb{P}(D_z > t) \leq \sum_{0 < |w-z| \leq R_F} \mathbb{P}(Y_z > t, Y_w > t) \leq C_F \min\{1, t^{-4}\}. \quad (12.9)$$

This argument includes laws with atoms. The dependence range of  $D$  is fixed once  $F$  is fixed. The path budget and Corollary 12.2 therefore bound the expected supremum of its selected tail sums by  $C_F(N+1)/T$ .

*Separated values.* For any admissible sequence, keep one occurrence of each site in its original order and pay a factor  $m_0$  for multiplicity. Among these sites with  $Y_z > T$ , process values in decreasing order, breaking ties by a fixed deterministic order. Retain a site if its distance from every previously retained site is greater than  $R_F$ . Every discarded site has a distinct retained neighbor with at least its value, so its value equals  $D_z$ . The retained sites are pairwise separated by more than  $R_F$ .

Put the retained sites back in their original order. The all-pairs order condition still holds, so their images under the deterministic bijection  $\psi_F$  form an admissible directed chain. They lie in a deterministic square with radius

$$L_F = \max\{1, C(N/F + F\delta N) + C_F\}.$$

The transported field  $\tilde{Y}_w = Y_{\psi_F^{-1}(w)}$  has exactly the original iid law. Thus the retained reward is at most the corresponding  $G_n(L_F, x_F)$  in this transported field, for a deterministic center  $x_F$ .

Crucially, the same square and the same field permutation work for every original admissible sequence. Hence this domination holds after taking the supremum over those sequences, even though the pruning and the retained chain depend on the observations. By (12.7), its expectation is at most

$$C(N/F + F\delta N) + C_F \leq CN/F + C_F \quad (\delta \leq F^{-2}).$$

Add the collision estimate and absorb the fixed factor  $m_0$  to obtain (12.8).

Choose  $F$  first so that  $C/F < \varepsilon/3$ , then choose  $T_0$  so that  $C_F/T_0 < \varepsilon/3$ , and set  $\delta_0 = F^{-2}$ . The remaining terms vanish after division by  $N(n) \rightarrow \infty$ , uniformly over the laws. For finitely many types, bound the supremum of a sum by the sum of the suprema. For a finite-range field, first split into finitely many sublattices on which each fixed template has an iid law, provided the collection hypothesis holds for every such law. Apply the argument separately and sum.  $\square$

Only one retained chain is charged to the collection bound. No packing factor depending on  $F$  multiplies  $N/F$ ; all costs of the increasing collision radius enter  $C_F/T$  or the additive  $C_F$ . This is why the order of choices  $F$ , then  $T$ , then a sufficiently large  $N$  is effective.

## 12.4 Concentration along slowly varying rows

**Lemma 12.8** (Uniform row concentration, including omissions). *For each  $n$ , let  $W_{i,j}^{(n)}$ ,  $i, j \in \mathbb{Z}$ , be real random variables with  $|W_{i,j}^{(n)}| \leq B$ . Suppose that for every deterministic sequence  $(j_i)$ , the variables  $(W_{i,j_i}^{(n)})$  in each residue class  $i \pmod{q}$  are mutually independent. The integer  $q \geq 1$  and  $B < \infty$  are fixed. One sufficient condition is mutual independence of the sigma-fields generated by entire columns in any set of columns separated by at least  $q$ .*

*Fix  $d < \infty$  and  $\varepsilon > 0$ . Consider rows with  $|j_1| \leq N^d$  and  $\sum_{i=1}^{N-1} |j_{i+1} - j_i| \leq \delta N$ , and omission sets  $O \subset \{1, \dots, N\}$  with  $|O| \leq \eta N$ . For sufficiently small fixed  $\delta, \eta > 0$ , depending only on  $\varepsilon, B, q, d$ , uniformly in  $n$ ,*

$$\mathbb{P} \left( \sup_{(j_i), O} \left| \sum_{i \notin O} (W_{i,j_i}^{(n)} - \mathbb{E} W_{i,j_i}^{(n)}) \right| > \varepsilon N \right) \rightarrow 0 \quad (N \rightarrow \infty). \quad (12.10)$$

*The convergence is exponential up to a polynomial factor. If all means equal  $m_n$ , the comparison sum is  $m_n(N - |O|)$ . The conclusion applies to indicators and to bounded sums over any fixed finite list of offsets or types that separately satisfy the hypotheses, with adjusted constants. A direct application to the sum requires independence at a spacing  $q$  appropriate to its combined input supports.*

*Proof.* The case  $B = 0$  is immediate. Otherwise fix a deterministic row and omission set. Split the centered sum into its  $q$  residue classes. Within each class the summands are independent, have support in intervals of length at most  $2B$ , and number at most  $N/q + 1$ . For the total sum to have absolute value greater than  $\varepsilon N$ , some class sum must have absolute value greater than  $\varepsilon N/q$ . Hoeffding's inequality [20, Theorem 2, Equation (2.6)] and a union bound therefore give, for  $N \geq q$ ,

$$2q \exp \left( -\frac{\varepsilon^2 N}{4qB^2} \right) \quad (12.11)$$

as an upper bound on this probability.

It remains to count the choices over which the supremum is taken. Put  $V = \lfloor \delta N \rfloor$ . For a fixed initial row, stars and bars gives  $\binom{N-1+V}{V}$  choices of the absolute increments with sum at most  $V$ . Their signs have at most  $2^V$  choices. The total number of rows is therefore at most

$$(2\lfloor N^d \rfloor + 1) 2^V \binom{N-1+V}{V} \leq (2\lfloor N^d \rfloor + 1) \exp\{N h_{\text{row}}(\delta)\},$$

where

$$h_{\text{row}}(\delta) = \delta \log 2 + (1 + \delta) \log(1 + \delta) - \delta \log \delta \rightarrow 0 \quad (\delta \downarrow 0).$$

For  $0 < \eta \leq 1/2$ , there are at most  $(N+1) \exp\{N h_{\text{bin}}(\eta)\}$  omission sets, where  $h_{\text{bin}}(\eta) = -\eta \log \eta - (1-\eta) \log(1-\eta)$ . Choose  $\delta, \eta$  so that the two entropy terms sum to less than  $\varepsilon^2/(8qB^2)$ . Multiplying the counts by (12.11) gives a polynomial factor times  $2q \exp\{-\varepsilon^2 N/(8qB^2)\}$ , uniformly in  $n$ .

If rows are specified only at nonomitted columns, assume the same bound on their first specified row and on their successive jumps. Extend to the full interval by holding that first row on the initial gap and, at each later gap, holding the old row until the next specified column. Hold the last row on the final gap. This does not increase the total variation and gives a row counted above. Indicators have  $B = 1$ . For a fixed finite sum, apply the result to each summand with a divided tolerance, then use the triangle inequality and a finite union bound. Alternatively, the sum itself has a fixed bound and can be treated directly when its combined supports satisfy the column independence condition.  $\square$

The supremum in (12.10) is taken before any row or omission set is selected from the environment. Its hypotheses must also be checked at the level where it is used. After conditioning on a shared phase, the same proof is available if the bound and independence spacing remain uniform, but the centering is then by conditional means. Replacing those means by unconditional ones requires a separate equality or error estimate. Alternatively, deterministic phase templates or a local phase construction may directly give the stated unconditional independence and row invariance. Averaging functions over finitely many mesh conventions is permitted when their input supports still give the required independence spacing.

## 12.5 A common dyadic truncation level

The final observation chooses a truncation from a fixed finite list. A first-moment bound is enough for this choice, although it does not give uniform integrability of the original family.

**Lemma 12.9** (Good dyadic levels). *Let  $U_1^{(n)}, \dots, U_k^{(n)}$  be nonnegative variables with*

$$\sup_n \sum_{a=1}^k \mathbb{E} U_a^{(n)} \leq A.$$

*Fix  $M_0 > 0$  and an integer  $J \geq 1$ . For each  $n$ , there is  $M_n \in \{M_0, 2M_0, \dots, 2^{J-1}M_0\}$  such that*

$$M_n \sum_{a=1}^k \mathbb{P}(U_a^{(n)} > M_n) \leq \frac{2A}{J}. \quad (12.12)$$

*Thus this quantity can be made arbitrarily small within a fixed finite range of levels, uniformly over the laws. The level may depend on the law, but is deterministic for that law and is not chosen from the realization of the field.*

*Proof.* For every  $u \geq 0$ , a geometric sum gives

$$\sum_{j=0}^{J-1} 2^j M_0 \mathbf{1}_{\{u > 2^j M_0\}} \leq 2u.$$

Take expectations and sum over  $a$ . At least one of the  $J$  resulting nonnegative summands is no larger than their average, which is at most  $2A/J$ .  $\square$

For finitely many thresholds  $c_a M$  with fixed  $c_a > 0$ , apply the lemma to  $U_a/c_a$ . The conclusion controls the tail probability multiplied by its threshold; it makes no assertion that  $\mathbb{E}[U_a^{(n)} \mathbf{1}_{\{U_a^{(n)} > M\}}]$  is uniformly small. When a concentration or comparison estimate is needed at  $M_n$ , establish it simultaneously for the finite candidate list, or use its largest bound  $2^{J-1}M_0$  in Lemma 12.8. All concentration tolerances can then be fixed before  $n$  varies.

## 13 Selected access and charged tunnels

The probabilistic estimate needed later concerns savings selected along an optimizing path. A first-moment bound for a fixed local saving does not control such a selection. We obtain the stronger estimate by joining any directed chain of local improvements into one physical path and comparing its cost with the calibration.

There are three steps. First, local envelopes control both corrected access costs and the gauges at cuts, uniformly over the possible cut occurrences. Second, a conditional access charge converts a local negative cost into a tunnel saving that can be realized by a physical path. Finally, a sparse corridor network joins every directed chain of charged tunnels at an expected cost proportional to its span. This proves Proposition 13.5, which supplies the collection hypothesis of Lemma 12.7.

Throughout, an observation domain for a fixed test is a deterministic union of independent input cells. All maxima needed to choose a path, a cut, or a test are taken before that choice is made. Conditional expectations below refer to the fixed tests, and retain local supports.

### 13.1 The local input used below

Fix an admissible pair of input-road rules with common period bound

$$0 < \beta \leq C_s n^\alpha, \quad \alpha < 1.$$

The constants  $C_s$  and  $\alpha$  are fixed as  $n$  and the rules vary. Propositions 9.4 and 9.5 place the near-minimizing roads used in the contraction in this regime.

Road tracking and radius-ten locality are measured in the fixed  $(v, w)$  maximum norm. We use coordinates in the ordered basis  $(A, B)$  below; for  $s < 1/4$  this change of coordinates has operator norm at most two. Comparison with Euclidean coordinates may depend on the fixed flat, which is harmless here. Write  $\xi = (\xi_1, \xi_2)$  for these coordinates and put  $\ell_1 = h(A) > 0$ ,  $\ell_2 = h(B) > 0$ . An  $A$ -road has nominal equation  $\xi_2 = j$ , and a  $B$ -road has nominal equation  $\xi_1 = i$ . Their tracking width in these coordinates is at most  $\rho = 2r < 1/100$ .

Assign each lattice edge to the half-open unit coordinate square containing its midpoint. In each square, collect the assigned edge weights and independent private seeds into one random element. The resulting cell inputs

$$(\eta_c : c \in \mathbb{Z}^2)$$

are independent. Auxiliary road phases are fixed for the moment; they are treated explicitly in Section 13.6. Every expectation and conditional expectation below integrates the spatial cell inputs with those phase values held fixed. The bounds are uniform over the fixed admissible phase assignments. Enlarging a support by a fixed number of unit cells incorporates the dependence neighborhoods of both road rules and gauges.

We use Lemmas 10.4 and 10.5 for a fixed finite list of road labels near a cut and bounded parameter windows on those roads. For each unit cell  $c$ , they give nonnegative local envelopes  $U_c, V_c$ , supported on a set  $\mathcal{N}(c)$  of at most  $m_0$  cells within distance  $R_0$  of  $c$ , with

$$\mathbb{E}U_c \leq C\beta, \quad \mathbb{E}V_c \leq a_n, \quad a_n = C_1 n^{1-\vartheta}, \quad \vartheta > 0. \quad (13.1)$$

All constants are uniform in  $n$  and the admissible input-road rule. The two envelopes serve different purposes. The corrected-cost envelope  $U_c$  bounds the absolute signed cost of travel within the listed windows and of a change of road label at a physical intersection. Finite sums of operations use the corresponding sums of envelopes. The gauge envelope  $V_c$  bounds the absolute local gauges, after their common deterministic centering, and the absolute raw signed  $Q_n$  costs of the same short operations. Both envelopes are functions of the local inputs. Their construction requires no local pointwise bound for the calibration discrepancy  $P_n - g$ .

The interpolation convention makes these bounds uniform over all attachment occurrences, including repeated visits to one vertex. If  $C_j$  is the corrected cost between successive unit-period anchors, then the corrected signed cost between parameter values  $u, t$  in a bounded interval satisfies

$$|C_W(u, t)| \leq \sum_{j: [j, j+1] \cap [u \wedge t, u \vee t] \neq \emptyset} |C_j|. \quad (13.2)$$

Here each road uses its own parameter, with marked levels at the integers; the same formula in a shifted wall parameter uses the correspondingly shifted period endpoints. The number of terms is bounded by the parameter length plus two. For switches, the local-gauge lemma supplies the analogous supremum estimate over the finitely many crossing windows. Its proof can use calibration defects, but the envelopes in (13.1) themselves are functions of the local inputs. Enlarging  $U_c, V_c$  by a fixed finite sum will be understood whenever a cut allows both road families or several prescribed center labels.

The envelope constants may depend on the fixed flat, the auxiliary row mesh, and the fixed lists of local labels. Their support radii are chosen before the macroscopic piece size  $K$  and the number of candidate shells. The observation domains used below prove estimates for these local variables; they are not domains on which the final road rule is run.

### 13.2 Observation domains and conditional charges

A cut cell  $c$  specifies a finite list of road labels and a closed bounded parameter window on each road. Endpoints are labeled by path *occurrences*, so repeated visits to a lattice vertex remain ordered. For each listed road, prescribe one reference center: its first occurrence at the absolute nominal longitudinal coordinate of the center of  $c$ , using the  $\xi_1$  coordinate for an  $A$ -road and  $\xi_2$  for a  $B$ -road. This level is fixed by the cell and road label. The first-occurrence convention is local and commutes with integer cell translations, and tracking keeps its parameter at bounded distance from every cut occurrence in  $c$ . Choose the windows before forming the envelopes so that they contain these reference centers and all finitely many allowed row centers. The reference choice adds one center convention per road label; its half-integer absolute parameter and the fixed finite wall phase list give only finitely many relative anchor phase classes.

A local test type consists of two cut cells  $c_-, c_+$ , their road labels and one prescribed center convention at each endpoint, a reference index  $z \in \mathbb{Z}^2$ , and a deterministic closed polygon  $D_0$ . It must satisfy

$$\begin{aligned} D_0 &\subset [-CK, CK]^2 + z, \\ D_0 &\subset \{\xi : h(c_- - z) - b_0 \leq h(\xi - z) \leq h(c_+ - z) + b_0\}, \end{aligned} \tag{13.3}$$

where a cut cell in an argument of  $h$  means its center, and  $b_0$  is fixed. A path of diameter  $O(K)$  with backward  $h$ -excursion at most one fits in an inner polygon obtained by intersecting a bounding square with the slab of relative levels  $h(c_- - z) - b_{\text{in}}$  and  $h(c_+ - z) + b_{\text{in}}$ , where the fixed padding  $b_{\text{in}}$  includes the unit excursion and the cut-cell diameters. Round the square coordinates  $\xi - z$  and these two relative slab levels outward to a fixed rational mesh of spacing  $\delta_{\text{mesh}} > 0$ , and call the resulting closed polygon  $D_0 = z + D_t$ . It contains the inner polygon and satisfies (13.3) with  $b_0 = b_{\text{in}} + \delta_{\text{mesh}}$ , after a fixed enlargement of  $C$ . The rounded lower and upper slab levels remain at bounded distance from the corresponding cut windows. The bounded relative cut-cell offsets leave finitely many  $D_t$ , so these are exact translated types even when  $h(A)$  or  $h(B)$  is irrational. Physical access below uses this final padding  $b_0$ .

For each relative type, choose a deterministic union  $\mathcal{D}_t$  of whole input cells before specifying its local phase pattern. Choose it large enough to determine the weights on  $D_t$  and the roads, labels, and gauges in its cut windows for every admissible local phase pattern and every input realization. Set  $\mathcal{D} = \mathcal{D}_{z,t} = z + D_t$ . The uniform locality bounds place this stencil in a fixed-width enlargement of  $D_0$ . Set

$$\mathcal{F}_{\mathcal{D}} = \sigma(\eta_c : c \in \mathcal{D}).$$

An access envelope may overlap this observation. The next lemma controls its conditional mean simultaneously for every possible overlap, without letting the size of  $D_0$  enter the bound.

**Lemma 13.1** (A local bound for every observation). *There is a local random variable  $W_c$ , with the same support as  $U_c$ , such that*

$$\mathbb{E}[U_c \mid \mathcal{F}_{\mathcal{D}}] \leq W_c, \quad \mathbb{E}W_c \leq 2^{m_0} C\beta. \quad (13.4)$$

*The inequalities hold simultaneously for every deterministic union-cell observation  $\mathcal{D}$ .*

*Proof.* For  $I \subseteq \mathcal{N}(c)$  put  $\mathcal{F}_I = \sigma(\eta_d : d \in I)$ , and define

$$W_c = \sum_{I \subseteq \mathcal{N}(c)} \mathbb{E}[U_c \mid \mathcal{F}_I].$$

Independence of the input cells and measurability of  $U_c$  give

$$\mathbb{E}[U_c \mid \mathcal{F}_{\mathcal{D}}] = \mathbb{E}[U_c \mid \mathcal{F}_{\mathcal{D} \cap \mathcal{N}(c)}].$$

This is one of the nonnegative summands defining  $W_c$ . There are at most  $2^{m_0}$  summands, each with expectation  $\mathbb{E}U_c$ . Since only the finitely many intersections with  $\mathcal{N}(c)$  matter, versions can be chosen for which all the inequalities hold simultaneously.  $\square$

Only the subsets of  $\mathcal{N}(c)$  enter the construction of  $W_c$ . Thus neither its support radius nor its mean depends on the number of possible  $K$ -sized observation domains. We can consequently choose the separation of candidate cuts before choosing  $K$ .

A later selection of a test or label uses these simultaneous bounds. For a chosen deterministic type, its charge is the conditional mean computed for that type. We do not condition again on the event that the type was chosen; this distinction will allow arbitrary selection from the finite test family.

### 13.3 Signed center labels and physical forward access

Center labels let adjacent pieces use a common additive cost. The resulting center adjustments can have either sign. Before applying calibration domination, we therefore realize their sum by an actual path with forward road attachments.

For an oriented road  $W$ , denote by  $C_W(u, t)$  its additive signed corrected cost, with the gauge of  $W$  at both endpoints. Thus  $C_W(u, t) = -C_W(t, u)$  as bookkeeping, even when the reverse physical trip has positive cost. If a physical internal path  $\pi$  goes from the occurrence  $x$  on  $W_-$  to the occurrence  $y$  on  $W_+$ , define

$$C_\pi(x, y) = Q_n(\pi) - g_{W_+}(y) + g_{W_-}(x).$$

Let  $x_0, y_0$  be the chosen center labels on these walls. The internal cost measured at center labels is

$$H(\pi) = C_{W_-}(x_0, x) + C_\pi(x, y) + C_{W_+}(y, y_0). \quad (13.5)$$

**Lemma 13.2** (Realization of center-label costs). *There are exterior intersection occurrences  $a$  on  $W_-$  and  $b$  on  $W_+$ , at bounded parameter distance from their cut windows, such that  $a$  precedes every admissible  $x, x_0$ ,  $b$  follows every admissible  $y, y_0$ , and the input supports at  $a, b$  lie beyond the corresponding observed  $h$ -faces. For every admissible internal path, the physical forward concatenation  $a \rightarrow x, \pi, y \rightarrow b$  has corrected cost*

$$H(\pi) + C_{W_-}(a, x_0) + C_{W_+}(y_0, b). \quad (13.6)$$

*After including changes to exterior network labels, the absolute extra cost is bounded by  $U_{c_-} + U_{c_+}$ , enlarging the fixed local envelopes if necessary.*

*Proof.* The nominal  $h$ -coordinate increases along either road at a fixed positive rate, and tracking changes it by a bounded additive amount. The slab condition therefore lets us pass the full observation halo by moving a fixed number of periods before the incoming window or after the outgoing window. The relevant transverse-road index is chosen from the deterministic test type. For example, on an incoming  $B$ -wall of nominal first coordinate  $i$ , take the  $A$ -road of index

$$j_a = \left\lfloor \frac{h(c_-) - M_0 - \ell_1 i}{\ell_2} \right\rfloor,$$

where the fixed  $M_0$  exceeds the final slab padding  $b_0$ , its input halo, and the support radius of a connector by a strict margin. Then use the canonical intersection of those two prescribed roads. On an incoming  $A$ -wall interchange the coordinates; for exits use the analogous ceiling above the upper face. Tracking changes the  $h$ -level by only  $O(\rho)$ , absorbed by the margin. The chosen parameter differs by a bounded amount from the cut window. Thus the index choice uses no unrevealed input, and all possible attachment occurrences remain in fixed local windows.

The first and last road portions in the asserted concatenation are physical forward portions. Additivity of signed corrected costs gives

$$\begin{aligned} C_{W_-}(a, x) &= C_{W_-}(a, x_0) + C_{W_-}(x_0, x), \\ C_{W_+}(y, b) &= C_{W_+}(y, y_0) + C_{W_+}(y_0, b). \end{aligned}$$

Adding  $C_\pi(x, y)$  proves (13.6), whatever the relative orders of  $x, x_0$  and of  $y, y_0$ . In particular, neither signed center arm has to be traversed backward. A label change at an exterior intersection adds the corresponding difference of the two local gauges. All extra signed travels and switches lie in the bounded windows of Section 13.1. Their absolute costs are therefore bounded by  $U_{c_-} + U_{c_+}$  after a fixed enlargement of those envelopes.  $\square$

### 13.4 Cuts selected uniformly over optimizing paths

We next select cuts for which both the corrected access charge and the raw gauge are small. The same selection must work for a path chosen after the whole environment is known. We achieve this by maximizing over a finite list of separated candidate tuples.

The union of the two unit-spaced road families divides the plane into components of uniformly bounded diameter. Indeed a  $B$ -road lies in  $|\xi_1 - i| \leq \rho$  and is proper in its other coordinate. The planar crossing property prevents a component of its complement from crossing this entire strip. Applying this to the neighboring  $B$ -roads bounds the first coordinate of every component; the  $A$ -roads similarly bound the second. Equivalently, inside a large rectangle a left-to-right crossing intersects every top-to-bottom crossing. This argument also applies when roads have self-intersections. Their nominal parameter makes the relevant bounded crossing portions finite.

For this cut construction, distances are measured by the maximum norm in the  $(A, B)$  coordinates. Given an initial mark on a parametrized path, place each successive  $K$ -exit mark at the first subsequent point at distance  $K$  from the preceding mark, stopping if no such point remains.

**Lemma 13.3** (Uniform selected cuts). *Fix a polynomial-box exponent  $q$  and an integer  $m > 4$ . The integer  $m$  can be increased so that  $m\vartheta > q$ . There are fixed radii  $H_m$  and  $K > 20H_m$ , a finite-range field  $(E_z)$ , and cuts of any continuous nearest-neighbor path at its successive  $K$ -exit marks, apart from bounded end portions, with the following properties.*

1. *Each selected cut is a road occurrence within  $K/10$  of the corresponding exit mark. Consecutive selected cuts have displacement between  $K/2$  and  $2K$ , and the intervening path piece has diameter  $O(K)$ .*

2. If  $c$  is its cut cell and  $z$  the unit cell containing its exit mark, then

$$W_c/\beta \leq E_z, \quad V_c \leq a_n E_z. \quad (13.7)$$

All finitely many allowed labels and center conventions at that occurrence are included in these inequalities.

3. Uniformly in the input-road rule and  $n$ ,

$$\mathbb{P}(E_z > t) \leq C_K \min(1, t^{-m}), \quad t > 0. \quad (13.8)$$

In particular  $\sup_{n,z} \mathbb{E} E_z^4 < \infty$ .

4. For any fixed  $\varepsilon > 0$ , the probability that one of these cuts in a collection of at most  $O(n^q)$  possible mark cells has  $V_c > \varepsilon n$  tends to zero.

The construction is valid simultaneously for all paths, including a path chosen after the entire environment has been observed. At an exit mark, call the portion after its last preceding hit of the boundary of its  $H_m$ -ball the terminal arc. Suppose a fine occurrence partition has segment traces of diameter at most  $\delta_f$  in the present max norm, and every fine interval meeting that arc chronologically, including at an endpoint, has displacement in

$$\mathcal{G}_\kappa = \{z : z_1 \geq 0, |z_2| \leq \kappa z_1\}, \quad 0 \leq \kappa < 1, \quad H_m > \frac{2(1+\kappa)}{1-\kappa} \delta_f. \quad (13.9)$$

Then all candidates for that mark may be taken on  $B$ -walls.

*Proof.* Normalize the two costs together:

$$S_c = W_c/\beta + V_c/a_n.$$

By (13.1) and Lemma 13.1,  $\mathbb{E} S_c \leq C$  uniformly. Increase the fixed support radius  $R_0$  to include the cell diameter.

*Separated candidates.* Choose  $m$  nested annuli about a potential exit mark, with annulus widths larger than the diameter bound for a component of the road complement, and with gaps larger than  $4R_0$ . Also choose an integer  $J > 2R_0 + 1 + 2\rho$ . Take  $H_m$  large enough that all annuli fit inside its ball and  $H_m > (m-1)J + 2\rho + 1$ , then take  $K > 20H_m$ . On the path portion preceding an exit mark, use the last passage from the outer boundary of this ball to the mark. Its successive crossings of the annuli meet the road network; choose one occurrence at such a meeting in each annulus. If the mark is not at a cell center, allow every unit cell meeting the annulus enlarged by two cell diameters. The larger gaps ensure disjoint input supports for any candidate tuple having one cell in each enlarged annulus. The finite number of such tuples depends on  $K, m, R_0$ , but not on  $n$ .

*One bound for every path.* For the mark cell  $z$ , let  $\mathcal{T}_z$  contain every possible annular tuple and every horizontal tuple described below, allowing all center offsets within  $z$ , and set

$$E_z = \max_{(c_1, \dots, c_m) \in \mathcal{T}_z} \min_{1 \leq j \leq m} S_{c_j}.$$

A finite list of discretized annular and horizontal patterns suffices, because the allowed candidate sets are subsets of a fixed finite list of unit cells. For each fixed tuple, independence, Markov's inequality, and the uniform mean bound give

$$\mathbb{P}(S_{c_1} > t, \dots, S_{c_m} > t) \leq (C/t)^m.$$

Taking the union over  $\mathcal{T}_z$  proves (13.8). The actual path supplies a tuple in this list. Choose the candidate minimizing  $S_c$ , breaking ties by the deterministic cell and occurrence order. This gives (13.7) for every path on the same event, regardless of how that path was selected.

*Geometry and gauges.* The selected cut lies within  $H_m < K/10$  of its mark. A preliminary exit has displacement exactly  $K$  from the preceding mark, in the continuous max norm, so the triangle inequality gives the asserted displacement bounds. Each portion between preliminary marks stays in its  $K$ -ball. The two adjacent short terminal portions give the  $O(K)$  diameter bound. The selected cut occurs after the preceding preliminary mark, since that mark lies outside the small terminal ball. Thus the cuts preserve path order.

Finally,

$$\mathbb{P}(a_n E_z > \varepsilon n) \leq C_K \left( \frac{a_n}{\varepsilon n} \right)^m \leq C_{K,\varepsilon} n^{-m\vartheta}.$$

A union bound over  $O(n^q)$  mark cells proves the fourth assertion. The single objective  $S_c$  is essential here: it selects both bounds at once, so the same product estimate controls the charge and the gauge.

*Horizontal terminal portions.* Let  $[t_-, t_+]$  be the terminal occurrence-time interval and let  $P$  linearly interpolate the fine cuts on their time intervals. The diameter bound gives  $|P(t) - \gamma(t)|_\infty \leq \delta_f$  in the present coordinates. For  $t_- \leq a \leq b \leq t_+$ , the increment  $P(b) - P(a)$  is a sum of nonnegative multiples of the good fine increments. Hence, for  $\Delta = \gamma(b) - \gamma(a)$  in these coordinates,

$$\Delta_1 \geq -2\delta_f, \quad |\Delta_2| \leq \kappa\Delta_1 + 2(1 + \kappa)\delta_f. \quad (13.10)$$

This includes both partial fine segments at the ends. At  $a = t_-, b = t_+$  we have  $|\Delta|_\infty = H_m$ . The strict inequality in (13.9) excludes  $\Delta_1 = -H_m$  and  $|\Delta_2| = H_m$ , so  $\Delta_1 = H_m$ . Thus the last boundary hit is on the left face. By lastness and continuity, all later points of the terminal arc are in the open ball.

Put  $p_1 = \xi_1(\gamma(t_-))$  and choose the wall levels

$$i_j = \lfloor p_1 + \rho \rfloor + 1 + (j - 1)J, \quad 1 \leq j \leq m.$$

Our choice of  $H_m$  gives  $p_1 < i_j - \rho$  and  $\xi_1(\gamma(t_+)) > i_j + \rho$ . A proper  $B$ -road in each such vertical strip separates these two sides: apply the planar rectangle-crossing property to a bounded vertical road portion and a left-to-right subarc. Select the first meeting along the terminal arc. It is a vertex occurrence, since an interior meeting of two square-lattice paths lies on a common edge whose entering vertex would already be a meeting; the initial arc point is outside the wall. At repeated road visits choose the first eligible vertex occurrence in the bounded window determined by the cut cell. The first meetings with  $i_1, \dots, i_m$  occur in that order, because reaching a later wall requires crossing every earlier wall. All of them lie strictly inside the terminal ball.

The center of a cell meeting  $W_{i_j}$  has first coordinate within  $\rho + 1/2$  of  $i_j$ . Distinct candidate centers are therefore separated by at least  $J - 1 - 2\rho > 2R_0$ , so their input supports are disjoint. For a fixed mark cell all candidates lie in a fixed finite  $(H_m + 1)$ -neighborhood. The horizontal tuples were included in  $\mathcal{T}_z$  before the common objective was maximized, so the same product estimate applies without conditioning on the path or on its terminal test. The selected point stays on the terminal arc, preserving the preceding order and displacement bounds.  $\square$

### 13.5 The charged tunnel field

Fix one local test type and its observation  $\mathcal{F}_D$ . The internal improvement is the minimum of the center-label cost over physical paths in  $D_0$  with the specified endpoint windows and labels. Writing

$H(\pi)$  as in (13.5), define

$$-u = \beta^{-1} \min_{\pi} H(\pi).$$

If no admissible path exists, set  $u = -\infty$ . Otherwise the closed-window endpoint-selection argument preceding Lemma 10.4 gives an attained minimum and a measurable local optimizer. It applies to the finitely many labeled edge occurrences and loop-erased route types in  $D_0$ , while retaining the continuum of possible fractions on each endpoint edge.

The improvement must pay for the two forward attachments of Lemma 13.2. Charge their conditional expected cost:

$$a_{\mathcal{D}} = \beta^{-1} \mathbb{E}[U_{c_-} + U_{c_+} \mid \mathcal{F}_{\mathcal{D}}], \quad Y = (u - a_{\mathcal{D}})_+, \quad (13.11)$$

with  $Y = 0$  when no admissible internal path exists. Both  $a_{\mathcal{D}}$  and  $Y$  are functions of the observed cells. Thus  $Y$  is a local saving after access has been paid for. Its sign comes from the additive center-label cost; its realization will use physical forward attachments.

More explicitly, for a fixed local phase pattern  $\phi$ , the charge is the kernel

$$a_{\mathcal{D}}^{\phi}(\eta) = \beta^{-1} \mathbb{E}_{\eta}[U_{c_-}^{\phi} + U_{c_+}^{\phi} \mid \eta_{\mathcal{D}}],$$

where  $\mathbb{E}_{\eta}$  integrates spatial inputs only. We suppress  $\phi$  while it is fixed. For the original random phase field  $\Phi$ , evaluate these same kernels and set  $Y(\eta, \Phi) = (u^{\Phi}(\eta) - a_{\mathcal{D}}^{\Phi}(\eta))_+$ . For each relative type and phase pattern, define a product-integral version of this kernel at the reference site by integrating the unobserved spatial coordinates against their product law. Define its copies by translation with  $z$ , and use these same kernels for the original and template fields. This fixes the versions covariantly.

The physical-competitor envelope of Lemma 10.4, with this domain and these center labels, gives  $Y \leq u_+ \leq U'/\beta$  and hence  $\mathbb{E}Y \leq C$ . In particular, the maximum reward over any finite array is integrable, being bounded by the sum of its finitely many  $Y$  variables.

**Lemma 13.4** (Tunnel accounting). *If a path piece with the specified type has normalized center-label cost  $c$ , then*

$$c^- \leq Y + W_{c_-}/\beta + W_{c_+}/\beta. \quad (13.12)$$

*If  $Y > 0$ , the internal minimizer has a physical forward realization between its exterior intersection labels whose conditional expected normalized corrected cost is at most  $-Y$ .*

*Proof.* Minimality gives  $c \geq -u$ , and  $a_{\mathcal{D}} \geq 0$  gives

$$c^- \leq u_+ \leq (u - a_{\mathcal{D}})_+ + a_{\mathcal{D}}.$$

Lemma 13.1 bounds  $a_{\mathcal{D}}$  by the two  $W/\beta$  terms in (13.12).

On  $\{Y > 0\}$ , Lemma 13.2 realizes the internal minimizer with normalized corrected cost at most  $-u + (U_{c_-} + U_{c_+})/\beta$ . Taking its conditional mean yields  $-u + a_{\mathcal{D}} = -Y$ . The envelopes already bound every admissible attachment occurrence, so this estimate holds for the selected minimizer.  $\square$

At selected cuts, (13.12) reads

$$c^- \leq Y + E_{z_-} + E_{z_+}.$$

The error has the light tail (13.8). To assign one error to the piece center, sum the finitely many  $E_z$  in its  $O(K)$  neighborhood; this preserves locality and the tail exponent.

The remaining task is to control selected tunnel rewards  $Y$ . A piece may choose its type after inspection: we bound the resulting sums by summing the estimates for the finite type list. This needs no independence between types at one site.

### 13.6 Compatible deterministic phase templates

To use the abstract thin-diagram estimate, we need an iid copy of each local tunnel law on a sparse lattice. Shared wall phases are fixed first; we will then dominate the original random-phase field by a finite sum of such template fields.

Fix the rational mesh, the road periods, and the finitely many row and anchor phase classes used by a local test. Choose an even integer  $S$  that is a multiple of every resulting period and is much larger than the diameter of  $\mathcal{D}$  plus every dependence radius. In  $(A, B)$  coordinates place copies of the test at

$$p_{ij} = S(i, j), \quad (i, j) \in \mathbb{Z}^2.$$

If two translated copies refer to the same infinite  $A$ -road, their longitudinal translations differ by an integer multiple of  $S$ ; their phase prescriptions therefore agree. The same is true of shared  $B$ -roads. The finitely many transverse offset classes used by the test can be repeated periodically. Thus one fixed local phase template has a consistent extension to the whole road network. Give all remaining roads fixed phases with the same periodicity.

Once the phases are deterministic, the copied observation sets have disjoint cell supports, including all private seeds. Translation covariance gives a common law, and independence of the cells makes  $(Y_{ij})$  iid. Shared phases along one wall have introduced no additional randomness into this assertion.

For the original construction, enumerate the finitely many local phase templates, together with the needed coarse residue and row classes, and apply the collection estimate to each. The local moment bounds hold uniformly for every fixed admissible phase assignment. The enumeration and  $S$  may depend on the fixed mesh, type, and road periods, but are fixed before  $n$  varies.

The domination is pointwise in one common coupling. A template records every wall phase used by the test and its access envelopes, as well as the phases of both anchors in every projection in (10.6). This includes the reference anchors  $z_S$ . Lemma 11.1 bounds the resulting finite list uniformly. For each sparse residue class and each assignment  $t$  to the list, repeat  $t$  periodically and evaluate the field  $Y_z^t$  using the original weights and the same road-indexed spatial private inputs. Only the phase arguments are overridden. At each site  $z$ , one assignment is its original local phase pattern; locality then gives  $Y_z = Y_z^t$  for that assignment, and hence

$$Y_z \leq \sum_t Y_z^t.$$

For fixed  $t$  the sparse field is iid; the different template fields need not be independent. The same coupling applies to local error fields. The conditional expectations defining gauges and access charges integrate weights and private coordinates only. Thus the full local phase list determines their kernels, including projected terms and anchor means, and the matching template uses exactly the original charge.

### 13.7 Corridors avoiding every observed halo

We now build the routes used in the collection estimate. Their ordinary connectors must avoid every observed input, including observations at sites that the chosen chain does not use. This ensures that choosing the chain does not bias the expected connector costs.

Increase  $S$  by an integer factor so that the observation at  $p$  and the supports of its two charged access envelopes lie in

$$p + [-S/10, S/10]^2.$$

The physical trace and full input support of each ordinary road segment, and the full input support of each intersection cost, lie within distance  $R_*$  of the corresponding nominal segment or intersection. Choose  $S > 100R_*$ . Reserve the  $A$ -roads on  $\xi_2 = (j + \frac{1}{2})S$  and the  $B$ -roads on  $\xi_1 = (i + \frac{1}{2})S$ . Their entire input-support strips miss every observation square. Adding these sparse road copies does not change any already specified road. At each crossing choose a canonical intersection occurrence. Since tracking is much smaller than the spacing, the intersections with consecutive transverse roads occur in the forward order. A monotone lattice path on this corridor grid is therefore an actual forward concatenation of roads.

The southwest and northeast portals of the cell centered at  $p_{ij}$  are the intersections near

$$p_{ij} - (S/2, S/2), \quad p_{ij} + (S/2, S/2),$$

respectively. Connect its southwest portal to the exterior entry  $a$  from Lemma 13.2 as follows. If the last incoming road is a  $B$ -wall, approach its transverse  $A$ -road along the left corridor boundary and then along that  $A$ -road to  $a$ . If the last incoming road is an  $A$ -wall, interchange the two coordinates and use the bottom boundary. The final transverse portion is below the lower observed  $h$ -face:  $a$  was chosen beyond that face by more than the dependence radius, and every earlier point of a forward  $A$ - or  $B$ -segment has no larger nominal  $h$ , including during pauses. All preceding boundary portions are outside the observation square. The analogous construction from  $b$  to the northeast portal uses the upper observed  $h$ -face and the top or right boundary.

The nominal segments of the ordinary connector portions assigned to this cell stay in  $p + [-S/2, S/2]^2$  and have total length  $O(S)$ . Their physical traces and full input supports lie in its  $R_*$ -enlargement. Apart from the two short charged attachments in (13.11), their input supports avoid the cell's observation. They also avoid every other observation: the  $R_*$ -enlargement misses the inner observation squares because  $S > 100R_*$ . Shift a corridor or extend a short access by a fixed extra period whenever necessary to preserve a full input buffer. The available margins are fixed fractions of  $S$ , so these changes preserve the forward ordering and the  $O(S)$  length bound.

Figure 2 separates the ordinary route portions, whose input supports avoid every observation, from the two charged attachments at each selected cell.

The construction works for every allowed incoming and outgoing label. After conditioning on all observations, the selected labels and nominal itineraries are fixed, whereas their actual road geometry and intersection occurrences remain random. For each itinerary, use the envelopes over its full prescribed crossing windows. Its uncharged support avoids all observations, so those envelopes retain their unconditional means. The mean bound is uniform over the finite label list and therefore also applies to the observation-dependent choice.

**Proposition 13.5** (Uniform finite-volume collection bound). *For the iid tunnel field associated with any fixed local type and phase template there are constants  $C, C_0 < \infty$ , independent of  $n$  and the input-road rule, such that the following holds. Let  $G_n(L)$  be the maximum of  $\sum_{p \in \Gamma} Y_p$  over finite distinct chains in an arbitrary deterministic translated coordinate square of side at most  $L$ , whose consecutive differences have both coordinates at least  $2S$ . The empty chain and singletons are allowed. Then*

$$\mathbb{E}G_n(L) \leq CL + C_0. \tag{13.13}$$

*The same bound therefore holds if consecutive differences are additionally required to lie in any fixed inner cone of the positive quadrant.*

*Proof.* Choose the chain and realize its route. Observe the inputs of every test in the sparse array. In the given finite square, choose a reward-maximizing chain, resolve ties lexicographically, and omit zero-reward sites. This is a function of the observations. Fix an external southwest and an

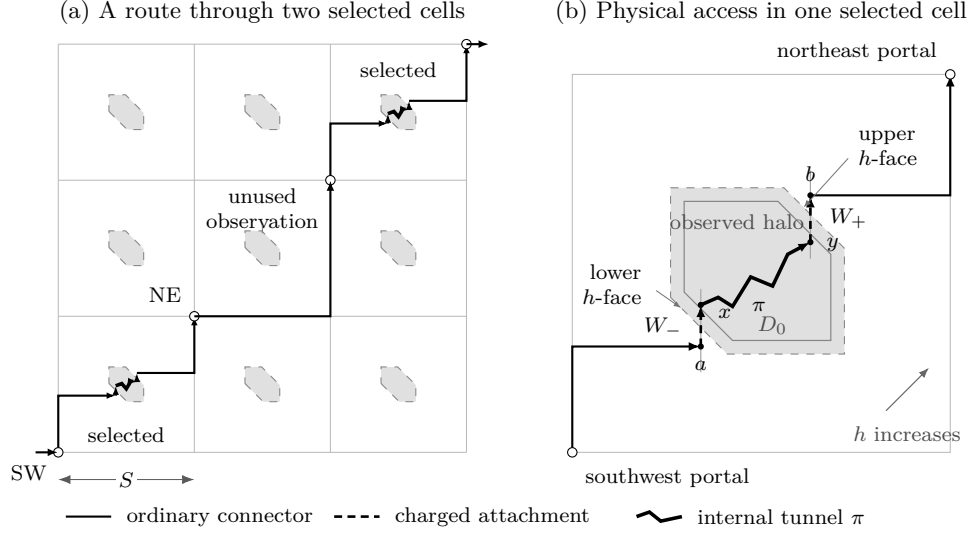


Figure 2: Physical routing through selected cells, schematically and not to scale. Gray regions represent observed input halos, including observations at unused cells. The ordinary connectors have input supports disjoint from every observed halo. The two dashed attachments at a selected tunnel are also physical forward portions; their supports may meet only that cell's observation. In the enlarged cell, the inner outline is  $D_0$ , the exterior entry  $a$  is below the lower observed  $h$ -face and the exterior exit  $b$  is above the upper face. The transverse connectors can therefore reach them without meeting the observation. The internal path is  $\pi : x \rightarrow y$ . Only physical route portions are drawn: the signed center-label terms in  $H(\pi)$  are bookkeeping and do not prescribe additional traversals.

external northeast corridor intersection, outside the square by  $3S$  in both coordinates, with the same prescribed intersection-label convention. These endpoints depend on the square, not on the observed chain.

Route monotonically on the corridor grid from the start to the southwest portal of the first selected cell, use that tunnel and its private access, then continue from its northeast portal to the next southwest portal. The assumed  $2S$  gaps make both coordinates of each latter displacement nonnegative, with a full corridor spacing to spare. Continue similarly to the fixed end. Every portion is a physical forward road portion or a physical internal tunnel path. Thus the result is an actual path.

Its total ordinary-corridor length is  $O(L + S)$ , by telescoping the two positive coordinate increments. A chain has at most  $C(L/S + 1)$  sites. The private uncharged connector lengths are  $O(S)$  per site, so their total length is also  $O(L + S)$ . The number of road switches has the same bound when measured in unit road spacings. The two short charged attachments at each site are excluded from this count.

All selected route costs are integrable. A used tunnel has  $Y > 0$  and therefore has negative center-label cost. Rearranging that cost identity bounds its positive physical passage cost by a bounded  $nh$  displacement and the integrable local gauge and raw-arm envelopes. Charged attachments and ordinary finite road portions are integrable by the same window estimates. The chain is finite, and its maximum reward is integrable by the bound following (13.11). Thus the conditional costs and the expectation rearrangements below are finite.

*Take conditional expected costs.* For every prescribed ordinary connector, its full input support

misses *all* observations. Its inputs therefore retain their product law under the conditioning. Subdivide the connector into unit road periods and apply the local envelopes to those periods and its switches. Its expected normalized corrected cost is at most a constant times its nominal length plus its number of switches. Summing uses only linearity of conditional expectation; connectors may share unobserved inputs.

The support of a charged access envelope meets no observation other than its own test. Independence gives

$$\mathbb{E}[U_{c-} + U_{c+} \mid \text{all observations}] = \mathbb{E}[U_{c-} + U_{c+} \mid \mathcal{F}_{\mathcal{D}_p}].$$

Thus each selected tunnel together with its charged attachments has conditional expected normalized corrected cost at most  $-Y_p$ , by Lemma 13.4. Consequently the completed path  $\Pi$  satisfies

$$\mathbb{E} \left[ \frac{Q_n(\Pi) - \Delta g(\Pi)}{\beta} \mid \text{all observations} \right] \leq C(L + S) - G_n(L). \quad (13.14)$$

*Apply the calibration at the fixed endpoint windows.* Corrected costs use only differences of gauges. For this endpoint calculation, undo any common deterministic centering and use the original mean-matching gauge representatives; the corrected cost of  $\Pi$  is unchanged. The constructed path has endpoint labels prescribed by translated copies of the exterior anchor convention. Calibration domination gives, pathwise,

$$Q_n(\Pi) - \Delta g(\Pi) \geq (P_n - g)(x_{\text{end}}) - (P_n - g)(x_{\text{start}}).$$

Although an endpoint intersection is random, it lies in a prescribed bounded road window. Put  $R_W = P_n - g_W$  on that exterior road and choose a fixed marked anchor  $k$  before the entire window of possible endpoint occurrences  $t$ . Equation (10.3) gives  $\mathbb{E}R_W(k) = 0$ , and the defect and interpolation identities give

$$R_W(t) - R_W(k) = -D_W(k, t) + C_W(k, t).$$

The interval is covered by a fixed number of complete road periods. Additivity and nonnegativity bound its defect by the defects of those periods; Lemma 10.4 controls its corrected cost. Thus the expected supremum of the absolute right side is at most  $C\beta$ . In particular,

$$|\mathbb{E}(P_n - g)(x_{\text{start}})| + |\mathbb{E}(P_n - g)(x_{\text{end}})| \leq C\beta.$$

The estimate covers every possible intersection occurrence and allowed endpoint label. Consequently the expected normalized corrected cost of  $\Pi$  is at least  $-C$ . Taking expectations in (13.14) yields  $\mathbb{E}G_n(L) \leq C(L + S) + C$ , and the fixed  $S$  is absorbed into  $C_0$ . The bound holds at each finite  $L$ , uniformly over the tunnel law; no law-dependent limiting passage is needed.  $\square$

The constants in Proposition 13.5 may depend on the fixed type, mesh, and sparse spacing, and are uniform over all  $n$  and road rules satisfying the local input assumptions. Taking their maximum over the finite type list gives one bound for that list. For rational  $v, w$  in fixed sufficiently small neighborhoods of the flat endpoints, the coordinate comparisons and the positive rates  $h(A), h(B)$  have uniform bounds. The domains, support buffers, and sparse spacing can therefore be fixed throughout those neighborhoods. In the application in Section 14, this test family is fixed before the final thin-diagram cone is narrowed. Rational direction denominators affect only the divisibility required of  $n$ .

Together with the phase-template domination, the proposition supplies the iid collection bounds required by Lemma 12.7. The observation domains, candidate shells, and corridors enter this distributional proof; they do not enlarge the input support of the final road-improvement rule.

## 14 Horizontal links extracted from a long geodesic

The aim is to bound the capped mean cost of a local horizontal crossing. A long geodesic toward an endpoint of the putative flat has negligible raw excess. We extract many crossings from it, control the cost lost at exceptional pieces by the charged-tunnel estimates, and use row concentration to compare the surviving sum with a single-link capped mean. The last step requires two truncation estimates, proved separately below. Only extremality of the endpoint is used in this extraction.

Costs remain in lattice units, while positions belong to  $\mathcal{G}_n = n^{-1}\mathbb{Z}^2$ . Put  $d = 1 - s^2$ , and use coordinates  $(u, y)$  in the basis  $(v, B)$ ; thus  $A = (d, s)$  and  $B = (0, 1)$ . Tracking and radius-10 locality are still measured in the fixed  $(v, w)$  maximum norm. For  $s < 1/4$ , changes to either  $(v, B)$  or  $(A, B)$  coordinates and back cost a factor at most 2. Thus a  $B$ -wall stays within  $2r$  of its nominal line in the present coordinates. Lattice-edge rounding errors are at most  $2K_{\text{lat}}/n$  in this norm and are absorbed by taking  $n$  large.

Constants may depend on the fixed flat. Those subscripted by  $s$  or  $K$ , and the large constants used only in the proof, may also depend on the indicated auxiliary parameters. The dependence radius of a link is fixed independently of all truncation and coarse-graining choices.

### 14.1 The signed link and its translation conventions

Let  $W_j$  be the  $B$ -road with nominal line  $\{(jd, js + t) : t \in \mathbb{R}\}$ . We keep its ordered occurrence labels, so two visits to one lattice vertex remain distinct. The nominal parameter is continuous and nondecreasing, and a fixed first-occurrence convention selects its marked cuts. This convention also selects a unique cut at a pause, where the occurrences retain their order. The road is locally finite.

Write  $R_j(t, t')$  for the signed raw  $Q_n$ -cost between the indicated occurrences at  $t$  and  $t'$ . Reversing the arguments negates this bookkeeping quantity; a physical reverse traversal has its own positive passage cost. With the interpolated local gauge  $g_j$ , put

$$C_j(t, t') = R_j(t, t') - g_j(t') + g_j(t).$$

Signed costs are additive in occurrence labels. Interpolation gives

$$C_j(t, t') = (t' - t)C_j(\kappa, \kappa + 1), \quad \kappa = \phi_j + m, \quad m \in \mathbb{Z},$$

for occurrences in the marked wall period  $[\kappa, \kappa + 1]$ , with the relative wall phase  $\phi_j$  from Section 11. This includes pauses. Longer intervals are split at the actual marked levels.

Fix a rational row spacing  $\sigma < s/1000$ , chosen so that  $s$  and all later row offsets are integral multiples of  $\sigma$ . Write  $y_k = k\sigma$ ,  $b = s/50$  and  $b' = s/20$ . Let  $c_{j,k}$  be the center occurrence of  $W_j$  with nominal height  $y_k$ , that is with wall parameter  $t = y_k - js$ , selected by the first-occurrence convention. An admissible crossing from  $W_j$  to  $W_{j+1}$  at row  $k$  is a finite physical path  $\gamma$ , with endpoint occurrences  $x \in W_j$  and  $x' \in W_{j+1}$ , such that

1. every point has  $|y - y_k| \leq b$  and  $jd - b' \leq u \leq (j + 1)d + b'$ ;
2. for any two points in their traversal order, their horizontal coordinates satisfy  $u_2 \geq u_1 - b'$ .

The second condition bounds every horizontal drawdown along the crossing, including one that occurs after an earlier visit to the terminal wall. At finite  $n$ , round the band boundaries outward to the nearest lattice boundaries; the  $O(1/n)$  error fits inside the strict margins used below. Equivalently, fix rational padded boundaries and require sufficient divisibility of  $n$ . Measure the crossing

at the two row centers by setting

$$L_{j,k}(\gamma) = C_j(c_{j,k}, x) + Q_n(\gamma) - g_{j+1}(x') + g_j(x) + C_{j+1}(x', c_{j+1,k}), \quad (14.1)$$

$$X_{j,k} = \min_{\gamma} L_{j,k}(\gamma). \quad (14.2)$$

The closed-window endpoint-selection argument preceding Lemma 10.4 gives an attained measurable local minimum. The band and all-pairs drawdown conditions are closed and survive loop deletion. The class is nonempty for large  $n$ : follow the horizontal center line by a lattice path and retain the portion between appropriate consecutive crossings of the two proper vertical roads. The planar rectangle-crossing property ensures these intersections even when the roads have self-intersections.

When a link is used in a new road, its minimizer may be chosen with vertex endpoints. For a loop-erased minimizer starting at an interior point  $x$  of a wall edge, let  $v$  be the first adjacent vertex reached by the crossing. Delete the partial prefix from  $x$  to  $v$  and move the wall attachment to the corresponding occurrence of  $v$ . The change in the center-adjusted cost is

$$R_j(x, v) - Q_n(x \rightarrow v) = \begin{cases} 0, & \text{if the partial traversal follows the wall,} \\ -2\tau(x \rightarrow v), & \text{if it reverses the wall.} \end{cases}$$

The right endpoint has the same property after deleting its final partial edge. The two walls have positive separation, so for large  $n$  these deletions are disjoint. The remaining subpath retains the band and drawdown conditions. Its cost cannot exceed the minimum and therefore equals it. Choose the first such vertex-ended minimizer in the relative order above. This supplies a lattice path at every link splice, while the definition still compares all allowed interior attachments.

The minimum has a uniform bound on its negative first moment: Lemma 10.4 gives

$$\mathbb{E}(X_{j,k})^- \leq C\beta. \quad (14.3)$$

To see why taking a minimum causes no loss, apply calibration domination to each physical  $\gamma$ . After taking the difference between the two center labels, the endpoint calibration–gauge discrepancies have  $L^1$  suprema at most  $C\beta$ . The two interpolated center arms obey the same bound. Their suprema provide one lower bound valid for all competitors before minimization. The calibration enters this estimate; the definition of  $X_{j,k}$  uses only local inputs. The constant  $C$  is independent of small  $s$ , the row mesh, and the input rule, because only two labels and bounded parameter windows are compared, and finite averaging of gauges preserves these bounds.

The same first-moment proof applies with every phase entering the link and its gauge projections fixed. The calibration depends only on the edges, while the local-envelope proof treats phases as parameters and uses period, interpolation, and projection bounds uniform over their admissible assignments. Thus a link  $X^t$  evaluated under any fixed local phase template of Section 13.6 satisfies

$$\mathbb{E}[(X^t)^-] \leq C\beta. \quad (14.4)$$

This fixed-phase form will be used for the sparse negative-link flags.

Two properties of the phase construction will turn selected sums into a statement about one link. Use Lemma 11.1 with  $\mathcal{H}$  containing  $s$ ,  $\sigma$ , and all required row offsets. Translation by  $dv = A - sB$  advances a column at fixed height; translation by  $\sigma B$  advances a row. Covariance of the phases and averaged gauges therefore gives every  $X_{j,k}$  the same law, denoted by  $X$ . The lemma also gives a fixed integer  $q$  for which entire-column sigma-fields with indices in one residue class modulo  $q$  are mutually independent. These are precisely the stationarity and independence used in Lemma 12.8. For a finite family of crossing templates, apply this construction and the extraction separately to each template.

## 14.2 The geometric extraction

Two geometric facts are needed. Near-minimal norm length forces most fine increments toward the extreme endpoint. At the coarser scale, deleting neighborhoods of bad pieces gives the all-pairs order required by the tunnel-tail estimate. Write

$$\mathcal{C} = \{z : \mu(z) = h(z)\}$$

for the cone on the flat, and choose pointed closed cones with

$$\mathcal{C} \setminus \{0\} \subset \text{int } \mathcal{C}_1, \quad \mathcal{C}_1 \setminus \{0\} \subset \text{int } \mathcal{C}_2.$$

They can be arbitrarily close to  $\mathcal{C}$ . On every such cone sufficiently close to  $\mathcal{C}$ ,  $h(z) \geq c|z|$ , with  $c > 0$  independent of the enlargement.

**Lemma 14.1** (A uniform modulus at an extreme point). *Let  $a$  be an extreme point of the norm ball, let  $S_\mu = \{z : \mu(z) = 1\}$ , and let  $U$  be a relatively open neighborhood of  $a$  in  $S_\mu$ . For  $t \geq 0$ , put*

$$\omega_U(t) = \sup \left\{ \nu(S_\mu \setminus U) : \nu \in \mathcal{P}(S_\mu), \left| \int z d\nu(z) - a \right| \leq t \right\}.$$

*Then  $\omega_U(t) \rightarrow 0$  as  $t \downarrow 0$ . If  $N > 0$ ,  $0 \leq d_0 < 1/2$ , and finitely many increments satisfy*

$$\left| \sum_i z_i - Na \right| \leq d_0 N, \quad \left| \sum_i \mu(z_i) - N \right| \leq d_0 N,$$

*then, with  $C_a = 2(1 + |a|)$ ,*

$$\sum_{i: z_i/\mu(z_i) \notin U} \mu(z_i) \leq (1 + d_0)N \omega_U(C_a d_0). \quad (14.5)$$

*Zero increments are omitted. In particular, endpoint error  $o(N)$  and total norm length  $N + o(N)$  imply  $o(N)$  norm mass outside  $U$ .*

*Proof.* Put  $F = S_\mu \setminus U$ . If the first assertion failed, there would be  $t_j \downarrow 0$  and probability measures  $\nu_j$  with barycenters within  $t_j$  of  $a$  and  $\nu_j(F) \geq e > 0$ . A weakly convergent subsequence exists on the compact sphere. Its limit  $\nu$  has barycenter  $a$ , and the closed-set inequality gives  $\nu(F) \geq \limsup_j \nu_j(F) \geq e$ .

The convex hull of  $F$  is compact and omits  $a$ . Indeed, Carathéodory's theorem would otherwise express  $a$  as a finite nontrivial convex combination of points different from  $a$ , contrary to extremality. If  $\nu(F) = 1$ , its barycenter lies in that convex hull. If  $0 < \nu(F) < 1$ , the conditional barycenters on  $F$  and its complement lie in the norm ball and express  $a$  as a nontrivial convex combination. Extremality makes both equal to  $a$ , whereas the first lies in the convex hull of  $F$ . Both cases are impossible, proving  $\omega_U(t) \rightarrow 0$ .

For the second assertion put  $S = \sum_i \mu(z_i) > 0$  and give  $z_i/\mu(z_i)$  mass  $\mu(z_i)/S$ . Its barycenter is  $(\sum_i z_i)/S$ , at distance at most

$$\frac{d_0 + d_0|a|}{1 - d_0} \leq C_a d_0$$

from  $a$ . The mass on  $F$ , multiplied by  $S \leq (1 + d_0)N$ , is the left side of (14.5).  $\square$

Deleting only the bad indices would not control an interval that crosses several of them. The next lemma removes a density halo: every interval touching a retained index then contains enough good increments to enforce forward order. The raw penalty on bad indices pays for all removed effective gauge differences.

**Lemma 14.2** (Removal of bad intervals). *Let  $x_0, \dots, x_m$  be successive cuts of a path. All distances in this lemma use the  $(A, B)$  maximum norm. Suppose  $K/2 \leq |x_{i+1} - x_i| \leq CK$ , and associate to each starting cut  $x_i$  a center  $z_i$  within a fixed distance  $H_0$ , independent of the cone enlargement. Declare an index bad when its increment is outside  $\mathcal{C}_1$ , or its physical subpath has a backward  $h$ -excursion of size at least one. Suppose, for  $c_* > 0$  and  $\epsilon_n^{\text{raw}} \rightarrow 0$ , its raw costs satisfy*

$$Q_i \geq -\epsilon_n^{\text{raw}} n \quad \text{for all } i, \quad Q_i \geq c_* n \quad \text{for bad } i.$$

*Let effective cut labels satisfy  $|\tilde{g}_i - g_{\text{ref}}| \leq \lambda n$  for one common constant  $g_{\text{ref}}$ . For sufficiently small  $\lambda > 0$  there is a set  $R$  of removed indices containing all bad indices, of cardinality at most a fixed multiple of their number, for which*

1.  $\sum_{i \in R} (Q_i - \tilde{g}_{i+1} + \tilde{g}_i) \geq 0$ ;
2. retained centers  $z_i$  are in  $\mathcal{C}_2$ -order up to a bounded additive error, for every pair of retained indices;
3. the same order includes the initial and terminal points, up to a bounded additive error;
4. repetitions of any one center have bounded multiplicity, independently of how small the cone enlargement is.

*Proof.* Let  $B_0$  be the bad set. For  $\theta > 0$ , put

$$R = \{i : \text{some index interval } I \ni i \text{ has } |I \cap B_0| > \theta |I|\}.$$

A one-dimensional maximal-interval bound gives  $|R| \leq 3|B_0|/\theta$ : select disjoint witness intervals in decreasing order of length. Their triple enlargements cover all discarded witness intervals, and their lengths sum to at most  $|B_0|/\theta$ . Every interval containing a retained index therefore has bad density at most  $\theta$ .

Let  $\ell_1, \ell_2$  be the coordinate functionals of  $\mathcal{C}_2$ . Compactness and the strict cone inclusion give  $\ell_j(x_{i+1} - x_i) \geq \kappa_0 K$  at good indices, whereas at bad indices they are at least  $-CK$ . Choose  $\theta < \kappa_0/(2(C + \kappa_0))$ . Every interval containing a retained index then satisfies

$$\ell_j \left( \sum_{i \in I} (x_{i+1} - x_i) \right) \geq \frac{1}{2} \kappa_0 K |I|.$$

Apply this to the interval between two retained cut indices, then absorb the bounded errors in choosing their centers. To include the initial endpoint and a retained starting-cut representative  $x_i$ , use the prefix through increment  $i$  and subtract that increment, whose size is at most  $CK$ . The suffix from  $x_i$  already contains the retained index. Center rounding adds another bounded error. The coordinate matrices range in a fixed compact set, so these errors remain uniform as the cone enlargement shrinks.

For multiplicity, apply the same density argument to  $h$ . After decreasing  $\theta$  if necessary, it gives  $h(x_j - x_i) \geq cK(j - i)$  on the relevant intervals, where  $c$  is independent of the cone enlargement. Only boundedly many retained centers can therefore lie in any fixed-radius ball.

For the cost assertion, bound each removed gauge difference separately:

$$\begin{aligned} \sum_{i \in R} (Q_i - \tilde{g}_{i+1} + \tilde{g}_i) &\geq n \{c_* |B_0| - (\epsilon_n^{\text{raw}} + 2\lambda) |R|\} \\ &\geq n \left\{ c_* - \frac{3(\epsilon_n^{\text{raw}} + 2\lambda)}{\theta} \right\} |B_0|. \end{aligned}$$

Choose  $\lambda < c_*\theta/24$  and then  $n$  large. The bracket is at least  $c_*/2$ , retaining a positive penalty per bad index. The set  $R$  depends only on the bad indices, so this conclusion holds for every assignment of effective gauges satisfying the stated bound.  $\square$

**Costs at shared cuts.** For an allowed road label  $W$ , a cut occurrence  $x$ , and a permitted datum  $\bar{x}$  on the same wall, define

$$\tilde{g}_W(x; \bar{x}) = g_W(\bar{x}) + R_W(\bar{x}, x). \quad (14.6)$$

The datum is a prescribed row center or reference center in the bounded windows of the access construction. At a shared cut  $x_i$ , both incident pieces will use one datum  $\bar{x}_i$ . Writing  $\tilde{g}_i = \tilde{g}_{W_i}(x_i; \bar{x}_i)$ , define

$$\begin{aligned} c_i^{\text{cost}} &= \beta^{-1} \{C(\bar{x}_i, x_i) + Q_n(\gamma_i) - g(x_{i+1}) + g(x_i) + C(x_{i+1}, \bar{x}_{i+1})\} \\ &= \beta^{-1} \{Q_n(\gamma_i) - \tilde{g}_{i+1} + \tilde{g}_i\}. \end{aligned} \quad (14.7)$$

The second equality is additivity of signed wall costs. The cut selection takes its suprema over every allowed label, cut occurrence, and future datum before any row is chosen. Lemma 10.5 therefore bounds all the effective gauges in (14.6) relative to one deterministic reference mean for the law and scale.

### 14.3 Uniform extraction

We state the output shared by the two capped estimates. For the final fixed rational directions, let  $\mathfrak{R}_n$  be the class of admissible input-road laws with

$$c_0 \leq \beta \leq C_s n^\alpha, \quad \alpha < 1,$$

where these constants are fixed as the law and  $n$  vary. The preceding local, selected-cut, and collection estimates apply uniformly to this whole class.

The order of choices is as follows. Fix  $s, \sigma$ , the local access construction, and  $\chi = 0.99$ . Choose an integer  $L > 99$ , put  $N = n^L$ , and take  $m = m_* > 4$  in Lemma 13.3 with  $m_* \vartheta > 2L + 10$ . Retain its radii  $H_m$  and  $K > 20H_m$ ;  $K$  is the actual preliminary exit radius. Fix the finite tunnel types and their collection constants in one larger cone. The capped proofs then choose  $F, T$ , their caps or finite cap menu, and all entropy tolerances. For their requested thin width  $\delta$  and row tolerance  $\eta$ , choose the final cones, the halo parameters  $\theta, \lambda$ , the fine cone and radius, and the rational direction errors. Finally impose divisibility and take  $n$  large. The constants multiplying  $\eta$  below are fixed before  $\eta$ ; later fine and cone choices enter the error sequence. The finite phase lists are independent of the eventual direction denominators, as explained in Section 11. These proof parameters never enlarge the input support of an individual link.

**Proposition 14.3** (Uniform extraction from a long geodesic). *For every sufficiently small fixed  $\delta, \eta > 0$ , the final geometric choices above can be made so that there are deterministic  $\epsilon_n, \pi_n \rightarrow 0$  and fixed constants  $H, C, C_K, c > 0$  and integers  $q_K, m_0$  with the following property. For each  $\mathfrak{r} \in \mathfrak{R}_n$  there is an event  $\Omega_{n, \mathfrak{r}}$  such that*

$$\sup_{\mathfrak{r} \in \mathfrak{R}_n} \mathbb{P}_{\mathfrak{r}}(\Omega_{n, \mathfrak{r}}^c) \leq \pi_n. \quad (14.8)$$

Let  $x_n^*$  be a deterministic nearest vertex of  $\mathcal{G}_n$  to  $Na$ , and put

$$j_n = \left\lfloor u(x_n^*)/d + \frac{1}{2} \right\rfloor, \quad \mathcal{J}_n = \{0, 1, \dots, j_n - 1\}.$$

For large  $n$ ,  $cN \leq j_n \leq CN$ . On  $\Omega_{n, \mathfrak{r}}$ , a geodesic from 0 to  $x_n^*$  has consecutive selected pieces  $\gamma_i : x_i \rightarrow x_{i+1}$ ,  $0 \leq i < m_N$ , after two bounded terminal fragments are removed, and the following objects and bounds.

1. There are at most  $CN/K+C$  pieces, each of diameter  $O(K)$ , with consecutive cut displacement between  $K/2$  and  $2K$  in the  $(A, B)$  maximum norm. The shared data in (14.7) have  $|\tilde{g}_i - g_{\text{ref}}| \leq \lambda n$ . A removed set  $R$  contains every bad piece of Lemma 14.2 and satisfies

$$|R| \leq \epsilon_n N, \quad \sum_{i \in R} c_i^{\text{cost}} \geq 0, \quad \sum_{i \notin R} c_i^{\text{cost}} \leq \epsilon_n N. \quad (14.9)$$

2. The retained pieces have types  $t_i$  from the fixed access family and centers  $z_i$  on its fixed finite family of full-rank center lattices, after bounded rounding. Let  $E = [e_1 \ e_2]$ , where the  $e_j$  are the boundary generators of  $\mathcal{C}_2$ , normalized by  $h(e_j) = 1$  and ordered so that  $e_1 \rightarrow a$  and  $e_2 \rightarrow b$  as  $\mathcal{C}_2$  shrinks to  $\mathcal{C}$ . In chronological order the centers satisfy

$$E^{-1}(z_j - z_i) \geq -H(1, 1) \quad (i < j), \quad z_i \in E([-H, CN + H] \times [-H, \delta N + H]). \quad (14.10)$$

The inequality is coordinatewise. Each rounded site occurs at most  $m_0$  times, and the initial distance from the origin plus total successive center distance is at most  $C(N + 1)$ . These bounds persist on every type and sparse-residue subsequence. The matrices  $E$  range in a fixed compact family, and  $H, C, m_0$  remain bounded when the final cone is narrowed.

3. An irregular retained set  $\mathcal{I}$  has  $|\mathcal{I}| \leq (C_K \eta + \epsilon_n)N$ . Each  $i \notin R \cup \mathcal{I}$  supplies at most  $q_K$  admissible links  $\mathcal{L}_i$ , with

$$c_i^{\text{cost}} \geq \sum_{(j,k) \in \mathcal{L}_i} X_{j,k}/\beta. \quad (14.11)$$

In the chronological list  $\mathcal{L} = \bigcup_{i \notin R \cup \mathcal{I}} \mathcal{L}_i$ , columns are distinct, increasing, and belong to  $\mathcal{J}_n$ . For sufficiently small fixed  $\eta$ ,  $|\mathcal{L}| \geq cN$  for all large  $n$ , after decreasing  $c$  once. If  $k_1, \dots, k_\ell$  are the selected rows, then

$$|\mathcal{J}_n \setminus \{j : (j, k) \in \mathcal{L}\}| \leq (C_K \eta + \epsilon_n)N, \quad |k_1| \leq j_n^2, \quad \sum_{\nu < \ell} |k_{\nu+1} - k_\nu| \leq (C_K \eta + \epsilon_n)N. \quad (14.12)$$

Rows can be held across omitted columns without increasing their variation.

4. Every retained piece obeys the charged-tunnel bound

$$(c_i^{\text{cost}})^- \leq Y_{z_i, t_i} + E_{z_i}. \quad (14.13)$$

Here  $Y$  is the local saving of Section 13.5, and  $E$  is a fixed local sum at the piece center that dominates both selected endpoint errors for every allowed piece. The finite phase-template fields for  $Y$  satisfy Proposition 13.5; those for  $E$  have fixed range and tail  $C_K \min(1, t^{-m_*})$ .

Set every selected list and count, including  $m_N, R, \mathcal{I}, \mathcal{L}$ , to the empty or zero value off  $\Omega_{n,r}$ . For finitely many link templates, apply the proposition separately, intersect their events, and take the maximum of their error sequences and the sum of their failure sequences.

*Proof. An edge-only event and fine directions.* Choose the nearest endpoint and the geodesic by fixed tie conventions. The metric and cheap-path estimates give an event  $M_n$  depending only on the edge weights, with  $\rho_n = \mathbb{P}_{\text{edge}}(M_n^c) \rightarrow 0$ , on which

$$|Q_n(\gamma)| \leq q_n N, \quad |\gamma|_{\text{edges}} \leq CNn, \quad \gamma \subset [-CN, CN]^2, \quad q_n = C \frac{(Nn)^\chi + 1}{N} \rightarrow 0. \quad (14.14)$$

Here  $L > 99$  makes  $(Nn)^{0.99} = o(N)$ ; endpoint rounding contributes only  $O(1)$  to  $Q_n$ . Write  $T_n(x, y) = T(nx, ny)$  for scaled metric-graph endpoints. On the same event, every such pair in a fixed enlargement of this box and at the bounded scaled distances used below satisfies

$$T_n(x, y) \geq n\mu(y - x) - e_n, \quad e_n = C_K n^\chi = o(n).$$

The graph-endpoint assertion includes the maximum incident-edge bound from Lemma 8.1. Thus it holds simultaneously before any cuts are chosen.

The fine choices in the stated parameter order can be made as follows. Choose the rational approximation to  $a$ , a neighborhood  $U$ , and  $\varepsilon_f > 0$  so that every increment directed in  $U$  satisfies

$$u(z) \geq 0, \quad |y(z)| \leq \varepsilon_f u(z) \leq \frac{1}{2} \eta \sigma \mu(z).$$

Indeed  $u(a) > 0$ ,  $y(a) \rightarrow 0$  as  $v \rightarrow a$ , and  $u(z) \leq C\mu(z)$  by fixed norm comparison. Take  $\varepsilon_f$  small enough and choose the fixed radius  $\varrho$  so that

$$\kappa = s + d\varepsilon_f \leq \frac{1}{2}, \quad 12\varrho < H_m, \quad \varrho < \sigma/100, \quad 2(1+s)\varrho < b/10, \quad 2r + 4d\varrho < b'. \quad (14.15)$$

These conditions are compatible because  $s < 1/4$  and  $2r < b'$ .

Use the fine exit partition of radius  $\varrho$  in the  $(A, B)$  maximum norm. Each fine trace has diameter at most  $2\varrho$ , including the final incomplete trace. The edge budget bounds the number of fine segments by  $CN/\varrho + 1$ . For their increments  $z_i$ , put  $S_n = \sum_i \mu(z_i)$ . Summing the metric lower bounds gives

$$nS_n \leq nN + C + q_n N + (CN/\varrho + 1)e_n,$$

while the triangle inequality and endpoint rounding give  $S_n \geq N - C/n$  and  $|\sum_i z_i - Na| \leq C/n$ . Thus both errors in Lemma 14.1, divided by  $N$ , are at most

$$d_n = C \left( \frac{1}{nN} + \frac{q_n}{n} + \frac{e_n}{n\varrho} + \frac{e_n}{nN} \right) \rightarrow 0.$$

Define

$$\epsilon_n^{\text{fine}} = (1 + d_n)\omega_U(C_a d_n) \rightarrow 0.$$

The total norm mass outside  $U$  is at most  $\epsilon_n^{\text{fine}} N$ . Every full fine increment has norm length at least  $c\varrho$ , so there are at most  $C\epsilon_n^{\text{fine}} N/\varrho + 1$  bad increments, including a possible bad incomplete one. A zero final segment is omitted. The good transverse variation is at most  $\frac{1}{2}\eta\sigma S_n \leq \eta\sigma N$  for large  $n$ , and the bad variation is at most  $C\epsilon_n^{\text{fine}} N$ . All these bounds hold on the same edge event for every road law.

*Cuts and the uniform good event.* Place the actual preliminary  $K$ -exit marks in the  $(A, B)$  maximum norm. At each mark test every fine interval whose occurrence-time interval meets the terminal arc, including at either endpoint. The coordinate identities

$$u = d\xi_1, \quad y = s\xi_1 + \xi_2$$

put every good fine displacement in  $\mathcal{G}_\kappa$  of (13.9). With  $\delta_f = 2\varrho$ ,  $\kappa \leq 1/2$  and  $12\varrho < H_m$  imply its exact strict condition

$$H_m > \frac{2(1+\kappa)}{1-\kappa} \delta_f.$$

If every fine interval meeting the terminal arc is good, use the  $B$ -wall candidates of Lemma 13.3; otherwise use its annulus candidates. Both tuple families and all future datum choices were included before this test in the single objective  $W_c/\beta + V_c/a_n$ .

With  $a_n = C_1 n^{1-\vartheta}$ , that lemma supplies

$$\mathbb{P}_{\mathbf{r}}(E_z > t) \leq C_K \min(1, t^{-m_*}), \quad W_{\text{cut}}/\beta \leq E_z, \quad V_{\text{cut}} \leq a_n E_z. \quad (14.16)$$

The inequalities cover every allowed effective gauge at the chosen cut. Let  $\mathcal{Z}_n$  be the deterministic set of possible mark cells in the box of (14.14), with its fixed boundary enlargement. It has at most  $Cn^{2L+10}$  cells. Define

$$\Omega_{n,\mathbf{r}} = M_n \cap \left\{ \max_{z \in \mathcal{Z}_n} a_n E_z \leq \lambda n \right\}.$$

The union bound in the selected-cuts lemma gives

$$\mathbb{P}_{\mathbf{r}}(\Omega_{n,\mathbf{r}}^c) \leq \rho_n + C_{\lambda,K} n^{2L+10-m_*\vartheta} =: \pi_n \longrightarrow 0$$

uniformly in  $\mathbf{r}$ . No independence between the metric event and the cut fields is needed. A selected cut lies within  $H_m$  of its mark. Since successive preliminary displacements equal  $K$ , selected displacements lie between  $K - 2H_m$  and  $K + 2H_m$ , in particular between  $K/2$  and  $2K$ . The intervening pieces and the two terminal fragments have diameter  $O(K)$ .

*The halo and its cost.* Each terminal fragment has raw cost at least  $-e_n$ , so the selected pieces have total raw cost at most  $q_n N + 2e_n$ . Every such piece has raw cost at least  $-e_n$ . For an off- $\mathcal{C}_1$  displacement in the fixed annulus, compactness gives a positive lower bound for  $\mu(z) - h(z)$ . For a unit backward  $h$ -excursion, split before, during, and after the excursion;  $\mu \geq |h|$  gives raw cost at least  $2n - 3e_n$ . For some fixed  $c_* > 0$ , bad pieces therefore cost at least  $c_* n$  for large  $n$ . Let  $B_0$  be their index set. The edge budget gives at most  $CN/K + C$  pieces, whence

$$\frac{|B_0|}{N} \leq C \left( \frac{q_n}{n} + \frac{e_n}{nK} + \frac{e_n}{nN} \right).$$

Use the fixed  $\lambda < c_*\theta/24$  from the halo lemma and  $\epsilon_n^{\text{raw}} = e_n/n$ . The halo set  $R$  depends only on  $B_0$ , and  $|R| \leq 3|B_0|/\theta$ . On  $\Omega_{n,\mathbf{r}}$  the gauge bound holds simultaneously for every later datum, so the removed cost is nonnegative for the eventual shared choices. They telescope on the full consecutive list, giving

$$\sum_{i \notin R} c_i^{\text{cost}} \leq \sum_i c_i^{\text{cost}} \leq c_0^{-1}(q_n N + 2e_n + 2\lambda n).$$

The halo's endpoint order puts the retained centers in (14.10): the second  $E$ -coordinate of  $a$  tends to zero as  $\mathcal{C}_2$  shrinks to  $\mathcal{C}$ . The endpoint and center errors stay bounded because  $E$  ranges in a compact family. Each preliminary exit uses at least  $cKn$  edges, so there are  $O(N/K)$  centers. Their successive-distance budget, including the initial  $O(K)$  distance, is  $O(N)$ . The halo lemma also gives the stated multiplicity bound.

*Rows and regular links.* Track the fine polygon by hysteresis: keep its current row until it leaves the central sub-band of half-width  $b/4$ , then choose the nearest row. Since  $\sigma < b/10$ , each change except possibly the first consumes at least  $b/10$  of transverse variation. The row-index variation is bounded by a constant times that variation divided by  $\sigma$ , plus a constant times the number of changes. It is at most

$$C_K \eta N + C_{\text{fixed}} \epsilon_n^{\text{fine}} N + O(1).$$

Let  $\mathcal{I}_0$  contain exactly the pieces whose closed occurrence-time intervals meet a row change or a bad fine interval, or which are incident to a failed terminal test. Define this set on the full selected list, regardless of membership in  $R$ . A row-change time lies in at most two closed coarse intervals. A fine trace has diameter at most  $2\varrho$ . Its closed occurrence-time interval can meet at most two closed coarse intervals, since otherwise it contains two consecutive cuts separated by  $K/2$ . Each

terminal arc occurs after its preceding preliminary mark. A fine interval meeting three terminal arcs would contain two consecutive preliminary marks, separated by  $K$ . Each failed test is incident to at most two pieces and is charged to a bad fine interval meeting its terminal arc. Thus

$$|\mathcal{I}_0| \leq C_K \eta N + C_{\text{fixed}} \epsilon_n^{\text{fine}} N + O(1).$$

On a piece outside  $\mathcal{I}_0$ , every fine interval meeting it chronologically is good. The polygon calculation in (13.10), now applied to any submotion of this piece with  $\delta_f = 2\rho$ , gives horizontal drawdown at most  $4d\rho$ . The path is within transverse distance  $2(1+s)\rho$  of its fine polygon. The hysteresis and (14.15) therefore keep the whole piece in one full row band, with drawdown less than  $b'$ . Its two successful terminal tests supply  $B$ -wall vertex occurrences. Write  $k_i$  for this constant hysteresis row. Two adjacent pieces outside  $\mathcal{I}_0$  have the same row: their closed time intervals share a cut, and a change there would put the incident pieces in  $\mathcal{I}_0$ .

The first  $(A, B)$  coordinate increases across this piece. Its maximum norm displacement is at least  $K - 2H_m$ . If the negative first coordinate realized that norm, its size would be at most  $4\rho$  by (13.10); if the second coordinate realized it, its size would be at most  $4(1+\kappa)\rho/(1-\kappa) \leq 12\rho$ . Both contradict  $12\rho < H_m < K/20$ . Hence the first-coordinate increase is at least  $K - 2H_m$ . Since  $\rho = 2r$  is the wall tracking width in these coordinates, the terminal wall index exceeds the initial one by at least  $K - 2H_m - 2\rho > 0$ .

Split the piece at successive first meetings of the next  $B$ -wall. At each intermediate meeting, use one occurrence of that wall for both adjoining subpaths. Planarity forces a meeting before passage beyond that wall's far padded side, and the drawdown bound for every submotion supplies the backward allowance. End the final subpath at the original terminal cut, even if it met that wall earlier. Its overshoot is at most  $2r + 4d\rho < b'$ , so the entire trailing portion is retained. These are admissible links in (14.2); the piece diameter bounds their number by a fixed  $q_K$ . Write  $q_i \leq q_K$  for this number.

Figure 3 shows the terminal-wall candidates and the first-meeting decomposition of a regular piece, including its final subpath after an earlier visit to the terminal wall.

*Distinct columns.* At every original cut put

$$v_h = \left\lfloor \xi_1(x_h) + \frac{1}{2} \right\rfloor, \quad 0 \leq h \leq m_N.$$

At a cut on  $W_j$ , tracking gives  $|\xi_1(x_h) - j| \leq \rho + O(1/n) < 1/2$  for large  $n$ , so  $v_h = j$ . Form one integer walk by expanding each piece outside  $\mathcal{I}_0 \cup R$  into its  $q_i$  positive unit wall steps from  $v_i$  to  $v_{i+1}$ , and replacing each piece in  $\mathcal{I}_0 \cup R$  by the jump between these endpoints. Include the two terminal jumps  $0 \rightarrow v_0$  and  $v_{m_N} \rightarrow j_n$ . The walk begins at 0, ends at  $j_n$ , and is fixed before any whole piece is rejected by the following scan. Each jump has magnitude  $O(K)$ , including the terminal jumps. If  $q_{\text{all}}$  is the number of unit steps and  $P, D$  are the sums of the positive and negative jump magnitudes, then

$$q_{\text{all}} + P - D = j_n, \quad P + D \leq C_K(|\mathcal{I}_0| + |R| + 1).$$

Scan this fixed walk once, updating its running maximum at every visited vertex. Call a unit step  $j \rightarrow j+1$  a candidate if its endpoint is a new strict maximum and is at most  $j_n$ , and let  $q_{\text{rec}}$  be the number of candidates. For each integer level  $\ell \in \{1, \dots, j_n\}$ , the first step whose endpoint is at least  $\ell$  is either a candidate unit step ending at  $\ell$  or a positive jump. A positive jump is charged at most its magnitude. Hence

$$j_n - q_{\text{rec}} \leq P, \quad q_{\text{all}} - q_{\text{rec}} = D - P + (j_n - q_{\text{rec}}) \leq D.$$

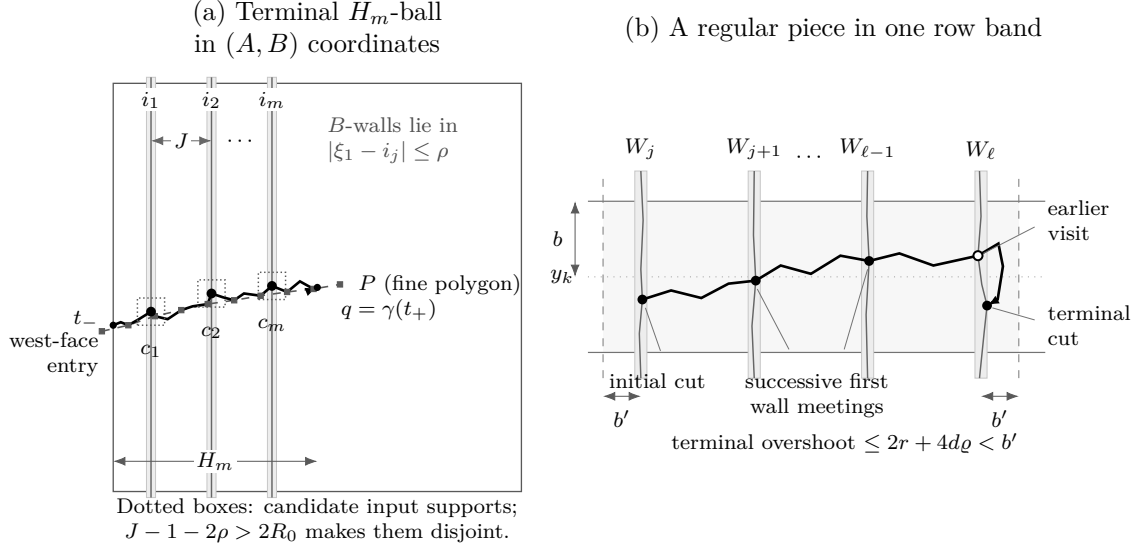


Figure 3: Horizontal selected cuts and their use in extracting links (schematic). In (a), the solid terminal arc starts at its last boundary hit  $\gamma(t_-)$  and ends at the mark  $q = \gamma(t_+)$ . The dashed fine polygon  $P$ , whose fine cuts are open squares, may use partial fine intervals at either end. The good-increment and radius hypotheses force entry through the west face. The filled dots are ordered first vertex meetings with the  $B$ -walls in the separated strips; intermediate candidates are omitted. Their cut cells  $c_j$  have the disjoint candidate input supports represented by the dotted boxes. In (b), a regular piece ending on  $W_\ell$  stays in  $|y - y_k| \leq b$  and is split at successive first meetings with intermediate walls, most of which are omitted from the drawing. The hollow point is an earlier visit to the terminal wall, not the retained terminal cut. The all-pairs drawdown bound and wall tracking give terminal overshoot at most  $2r + 4d\varrho < b'$ , so the whole final subpath satisfies the link window. The dashed vertical boundaries show the outermost  $b'$  margins; each unit link has the same margins.

This counts every rejected unit step, including record steps beyond  $j_n$ . Let  $\mathcal{C}_{\text{rec}}$  be the pieces outside  $\mathcal{I}_0 \cup R$  containing at least one rejected step, and set

$$\mathcal{I} = (\mathcal{I}_0 \cup \mathcal{C}_{\text{rec}}) \setminus R, \quad \mathcal{G} = \{0, \dots, m_N - 1\} \setminus (R \cup \mathcal{I}).$$

Distinct pieces in  $\mathcal{C}_{\text{rec}}$  contain distinct rejected steps, so  $|\mathcal{C}_{\text{rec}}| \leq D$ . Keep all links of the pieces in  $\mathcal{G}$ , without rescanning the walk. They are candidate steps, so their columns are distinct, increasing, and in  $\mathcal{J}_n$ . Discarding a whole piece loses at most  $q_K$  candidates, and therefore

$$|\mathcal{I}| \leq |\mathcal{I}_0| + D, \quad |\mathcal{J}_n \setminus \{j : (j, k) \in \mathcal{L}\}| \leq P + q_K D.$$

The bounds on  $P, D$  prove the stated irregular and omission bounds. Since  $u(a)/d$  is bounded above and below,  $j_n$  is comparable to  $N$ ; for sufficiently small  $\eta$ , the omission bound also gives  $|\mathcal{L}| \geq cN$  for large  $n$ .

The selected rows form a chronological subsequence of the hysteresis rows, so deleting terms cannot increase variation. Hold the old row across each omitted gap, as in Lemma 12.8. The box in (14.14) bounds the first selected row by  $CN/\sigma \leq j_n^2$  for large  $n$ . This proves (14.12).

*Shared data and the tunnel charge.* After these masks are fixed, assign one datum to every original cut  $x_h$ ,  $0 \leq h \leq m_N$ , retaining its selected road label and occurrence. If at least one

incident original piece belongs to  $\mathcal{G}$ , use that piece's row center on the cut's  $B$ -wall. When both incident pieces belong to  $\mathcal{G}$ , their rows agree by the argument for closed time intervals above, and they use the same selected wall occurrence, so they request the same center. If neither incident piece belongs to  $\mathcal{G}$ , use the prescribed reference center for the cut cell and road label. At  $h = 0, m_N$ , only the one existing incident piece is considered. Use this datum for both incident pieces even if one belongs to  $R$  or  $\mathcal{I}$ ; all incidence here is incidence in the original list.

The masks  $R, \mathcal{I}_0, \mathcal{C}_{\text{rec}}$  did not use the eventual data, and the simultaneous  $V_{\text{cut}}$  bound covers all the prescribed choices. Thus these are the shared data used in the telescope over the full list and the halo estimate above, which holds for every allowed datum choice. Every piece in  $\mathcal{G}$  receives its own row center at both endpoints. Any retained nonregular piece accepting a neighboring center is already in  $\mathcal{I}$ , while any removed incident piece is in  $R$ ; the datum choice requires no further irregular or omission charge. For  $i \in \mathcal{G}$ , let  $\gamma_{i,\nu}$  be its consecutive physical subpaths from the decomposition at first meetings. Intermediate center arms cancel, including across the retained terminal subpath, and give

$$\beta c_i^{\text{cost}} = \sum_{\nu=0}^{q_i-1} L_{v_i+\nu, k_i}(\gamma_{i,\nu}) \geq \sum_{\nu=0}^{q_i-1} X_{v_i+\nu, k_i}.$$

This proves (14.11).

For each retained piece, choose the access type with its cut cells, road labels, the two prescribed center conventions just used, and its rounded padded domain. This type belongs to the fixed finite family, and  $H(\gamma_i) = \beta c_i^{\text{cost}}$  in (13.5). For the tunnel comparison, choose an exterior wall occurrence  $a_i$  before both  $\bar{x}_i, x_i$ , and  $b_i$  after both  $x_{i+1}, \bar{x}_{i+1}$ , as in Lemma 13.2. Additivity gives

$$C(a_i, \bar{x}_i) + \beta c_i^{\text{cost}} + C(\bar{x}_{i+1}, b_i) = C(a_i, x_i) + Q_n(\gamma_i) - g(x_{i+1}) + g(x_i) + C(x_{i+1}, b_i).$$

The right side is the corrected cost of the physical forward concatenation in Lemma 13.2. Those attachments enter the access charge; a negative signed arm is never treated as a negative physical traversal. Every retained piece lies in its padded  $h$ -slab because it has no unit backward  $h$ -excursion. Lemma 13.4 therefore gives (14.13). At each coarse center, sum all cut error fields in its fixed  $O(K)$  neighborhood and over the finite cut-type list. This fixed local field dominates the two selected endpoint errors for every allowed piece, keeps a fixed range, and preserves the tail exponent. The finite type and phase-template lists are those fixed in Section 13.6.

Finally, one common deterministic error sequence is

$$\epsilon_n = C_{\text{fixed}} \left( q_n + \frac{e_n}{n} + \frac{n}{N} + \epsilon_n^{\text{fine}} + \frac{1}{N} \right) \rightarrow 0.$$

The fixed constant absorbs  $1/\varrho, 1/\sigma, 1/\theta, 1/c_0$  and the bounded endpoint adjustments. The preceding cost, halo, irregular, omission, and row estimates then have exactly the stated bounds. Together with the uniform  $\pi_n$  above, this proves the proposition.  $\square$

**Tail loss and capping.** Let  $\mathcal{S}_N$  be all finite lists on those center lattices, with types from the fixed family, satisfying (14.10) and its path and multiplicity bounds. Define  $H_N$  on the whole probability space by

$$H_N = \sup_{S \in \mathcal{S}_N} \sum_{(z,t) \in S} \left[ Y_{z,t} \mathbf{1}_{\{Y_{z,t} > T\}} + E_z \mathbf{1}_{\{E_z > T\}} \right].$$

The supremum includes the fixed type list. It is not set to zero outside the extraction event. For a finite template domination  $V \leq \sum_{\nu=1}^{r_0} V^\nu$ , use the tail comparison

$$V \mathbf{1}_{\{V > T\}} \leq r_0 \sum_{\nu=1}^{r_0} V^\nu \mathbf{1}_{\{V^\nu > T/r_0\}}.$$

On  $\{V > T\}$ , the largest summand exceeds  $T/r_0$  and the sum is at most  $r_0$  times that largest summand. Apply this to the  $Y$  and error fields, then sum over the fixed templates and sparse residues. If  $1 \leq T < r_0$ , split each summand at 1: values at most 1 contribute at most  $C(N+1)$  by the path and multiplicity bounds, and values above 1 use the tail bounds at threshold 1. Since  $r_0$  is fixed and  $T < r_0$ , these terms are absorbed by  $C_F(N+1)/T + C_F$  after increasing  $C_F$ . Lemma 12.7 and Corollary 12.2 give, for  $F, T \geq 1$  and uniformly in  $\mathfrak{r}$ ,

$$\mathbb{E}_{\mathfrak{r}} H_N \leq \frac{CN}{F} + \frac{C_F(N+1)}{T} + C_F, \quad \delta \leq F^{-2}.$$

The error fields have exponent  $m_* > 4$ , so their integrated bound is included in the  $C_F/T$  term. Thus for every  $\varepsilon > 0$  one can first choose  $F$ , then  $T$ , and then restrict the final cone so that

$$\limsup_{n \rightarrow \infty} \sup_{\mathfrak{r} \in \mathfrak{R}_n} \mathbb{E}_{\mathfrak{r}} H_N / N \leq \varepsilon. \quad (14.17)$$

The test family and its collection constants remain those fixed in the larger cone. This is an expectation bound, separate from  $\Omega_{n,\mathfrak{r}}$ .

For  $J, M > 0$ , write  $f_{J,M}(x) = \max(-J, \min(x, M))$ . On the extraction event let  $\mathcal{E}_J$  contain every irregular retained piece and every regular piece containing a link below  $-J\beta$ ; set  $\mathcal{E}_J = \emptyset$  off that event. On a nonexceptional regular piece, replacing links by  $f_{J,M}$  only decreases the right side of (14.11). On an exceptional regular piece the capped sum is at most  $q_K M$ , while (14.13) controls the negative cost of every exceptional retained piece. Splitting its two charges at  $T$  gives, on  $\Omega_{n,\mathfrak{r}}$ ,

$$\sum_{i \notin R} c_i^{\text{cost}} \geq \sum_{(j,k) \in \mathcal{L}} f_{J,M}(X_{j,k}/\beta) - (q_K M + 2T)|\mathcal{E}_J| - H_N. \quad (14.18)$$

The list  $\mathcal{L}$  includes links in exceptional regular pieces. Only the upper cap enters the loss coefficient.

**Sparse negative-link flags.** The asymmetric estimate needs a two-dimensional sparse-path bound for  $\mathcal{E}_J$ . A phase is shared along a wall, so column independence alone does not give that bound. For each coarse center  $z$ , let  $\mathcal{A}_z$  be the fixed finite list of potential links in its  $O(K)$  box, including its row and offset classes, and let

$$\Phi_z^J = \mathbf{1}_{\{\exists \ell \in \mathcal{A}_z: X_\ell < -J\beta\}}.$$

Enumerate the local phase templates for the union of these links, including both anchors and anchor means in every gauge projection. For a fixed template  $t$ , let  $\Phi_z^{J,t}$  be the same indicator with every link evaluated under that template. At each site the matching template reproduces every original link in  $\mathcal{A}_z$  exactly, so

$$\Phi_z^J \leq \sum_t \Phi_z^{J,t}.$$

Because one template agrees exactly, every indicator uses the same threshold  $J$ . The original first-moment bound gives  $\mathbb{P}(X < -J\beta) \leq C_s/J$ . The fixed-phase first-moment bound (14.4), followed by Markov and a union bound over  $\mathcal{A}_z$ , gives

$$\mathbb{P}_{\mathfrak{r}}(\Phi_z^{J,t} = 1) \leq C_K/J, \quad J \geq 1,$$

uniformly in the law and template.

On the extraction event, every exceptional regular piece is counted by  $\Phi_z^J$  at its center. With selected counts empty off that event,

$$\mathbf{1}_{\Omega_{n,\mathfrak{r}}} |\mathcal{E}_J| \leq (C_K \eta + \epsilon_n)N + \sup_{S \in \mathcal{S}_N} \sum_{(z,\tau) \in S} \sum_t \Phi_z^{J,t}.$$

When fixing the core geometry, take the sparse spacing in Section 13.6 large enough for the supports of  $\mathcal{A}_z$ , and split the center lattice into its finitely many residues. For each residue and fixed template, the copied indicators have disjoint product input supports; the different template fields need not be independent. Every selected subsequence retains the path budget and bounded multiplicity. Lemma 12.1, summed over these fixed residues and templates, therefore gives

$$\frac{\mathbb{E}_{\mathbf{r}}[\mathbf{1}_{\Omega_{n,\mathbf{r}}}|\mathcal{E}_J|]}{N} \leq C_K\eta + \epsilon_n + C_K(1 + N^{-1})J^{-1/2}. \quad (14.19)$$

The irregular term uses the good-event bound in the extraction proposition; the other term bounds the supremum over all allowed center lists before one is selected. The definition off the good event makes all selected counts zero there.

#### 14.4 The two truncation estimates

Capping makes link costs bounded, so row concentration applies. The first estimate allows the lower cap to be chosen after a prescribed upper cap. The second chooses equal caps from a fixed finite menu and also controls their tail probabilities; this stronger form will be needed for road replacement.

**Proposition 14.4** (Asymmetric capped-link estimate). *For each fixed  $0 < M < \infty$  and  $\zeta > 0$ , there are  $1 \leq J < \infty$ , fixed geometric tolerances and  $n_0$  such that, for every admissible  $n \geq n_0$ ,*

$$\mathbb{E}f_{J,M}(X/\beta) \leq \zeta. \quad (14.20)$$

*The tolerances can be imposed by taking the rational  $v, w$  sufficiently close to the endpoints of the flat. The choices are uniform over input road laws satisfying the stated local estimates and the sublinear upper bound for  $\beta$ .*

*Proof.* Fix an error  $\varepsilon > 0$ , to be chosen relative to  $\zeta$ . Choose  $F, T$  in (14.17), reserving the corresponding upper bound for the final  $\delta$ . By (14.19), choose  $J$  after  $T, M$  so that  $(q_K M + 2T)C_K J^{-1/2} < \varepsilon$ , and then choose  $\eta$  so that its contribution has the same bound.

Apply Lemma 12.8 to  $f_{J,M}(X_{j,k}/\beta)$ , using the original column independence from Lemma 11.1. Its fixed bound is  $\max(J, M)$ . The interval length  $j_n$  is comparable to  $N$ ; hence (14.12) meets the lemma's row and omission tolerances after decreasing  $\eta$  and taking  $n$  large. Uniformly over the road law, with probability tending to one,

$$\left| \sum_{(j,k) \in \mathcal{L}} f_{J,M}(X_{j,k}/\beta) - |\mathcal{L}|\mathbb{E}f_{J,M}(X/\beta) \right| \leq \varepsilon N$$

on the extraction event. The supremum in the row lemma is taken before the rows and omissions are selected.

If (14.20) failed uniformly, choose an offending sequence of scales and road laws. On the geometric and row events the capped sum is at least  $c\zeta N - \varepsilon N$ , while (14.9) bounds the left side of (14.18) by  $\epsilon_n N$ . Since  $\mathcal{E}_J$  is empty off the good event, (14.19) and (14.17) bound the expected nonnegative losses in (14.18) by  $C\varepsilon N$  for all large  $n$ . By Markov's inequality, they are at most  $4C\varepsilon N$  with probability at least  $3/4$ . This event intersects the geometric and row events. Choosing  $\varepsilon < c\zeta/(10C + 10)$  contradicts (14.18). Every threshold and geometric tolerance was fixed before  $n$ , proving the asserted uniformity.  $\square$

The same estimate also bounds the untruncated positive-tail probability. Since  $f_{J,M}(X/\beta)$  is at least  $M$  on  $\{X > M\beta\}$  and at least  $-(X/\beta)^-$  elsewhere,

$$\mathbb{P}(X > M\beta) \leq (C_s + \zeta)/M. \quad (14.21)$$

This uses no assumption that the positive tails are uniformly integrable.

**Proposition 14.5** (Matched caps in a finite dyadic range). *Let  $X^{(1)}, \dots, X^{(r)}$  be finitely many required row-link templates, and let  $U_1, \dots, U_p$  be additional nonnegative local variables with a uniform bound on the sum of their first moments. For every  $\zeta > 0$  and every prescribed lower bound  $M_{\text{prescribed}}$ , there is a fixed finite dyadic set  $\mathcal{M} = \{M_0, 2M_0, \dots, 2^{H-1}M_0\}$ , with  $M_0 \geq M_{\text{prescribed}}$ , and fixed geometric tolerances, such that for every sufficiently large  $n$  some deterministic law-dependent  $M \in \mathcal{M}$  satisfies, simultaneously,*

$$\mathbb{E}f_{M,M}(X^{(a)}/\beta) \leq \zeta \quad (1 \leq a \leq r), \quad (14.22)$$

$$M \left[ \sum_{a=1}^r \mathbb{P}((X^{(a)}/\beta)^- > M) + \sum_{b=1}^p \mathbb{P}(U_b > M) \right] \leq \zeta. \quad (14.23)$$

In particular  $U_1$  may be the positive part of the normalized primary increment used in the road improvement.

*Proof.* Fix an error  $\varepsilon > 0$ . Choose  $F, T$  as in (14.17), uniformly over the finite type family, and reserve the corresponding upper bound for the final  $\delta$ . Choose  $M_0 \geq M_{\text{prescribed}}$  so large that  $2TC_s/M_0 < \varepsilon$ . Let  $A$  bound the sum of the first moments of  $(X^{(a)}/\beta)^-$  and the  $U_b$ . Choose  $H$  so large that  $2A/H < \varepsilon$ . Lemma 12.9 ensures that for each law some level in the fixed menu satisfies

$$M \sum_a \mathbb{P}(X^{(a)} < -M\beta) + M \sum_b \mathbb{P}(U_b > M) \leq \varepsilon. \quad (14.24)$$

We fix all concentration and geometric tolerances for the whole menu before selecting this law-dependent level.

For equal caps, the square-root probability in (14.19) would be multiplied by the cap. Instead use the original row-link fields and their column independence. For every  $M \in \mathcal{M}$  and every link template, apply Lemma 12.8 separately to  $\mathbf{1}_{\{X_{j,k}^{(a)} < -M\beta\}}$  and to the capped costs. Take a finite union of these conclusions, using  $M_{\text{max}} = 2^{H-1}M_0$  as a common bound. The resulting event has probability tending to one uniformly over the law, and controls all candidate levels before one is selected.

On this event and the corresponding extraction events, the selected negative-link indicators have frequency at most their marginal probability plus a fixed error  $\nu$ . Columns are distinct, so each exceptional regular piece is counted by at least one such indicator. For any candidate  $M$ , the regular exceptions in (14.18) with  $J = M$  therefore cost at most

$$C_K(M + T)N \left[ \sum_a \mathbb{P}(X^{(a)} < -M\beta) + \nu \right].$$

The irregular contribution is at most  $(q_K M + 2T)(C_K \eta + \epsilon_n)N$ . After fixing  $M_{\text{max}}$ , choose  $\nu$ , the row and omission tolerances, and  $\eta$  so that their contributions are at most  $\varepsilon N$ . The term containing  $\epsilon_n$  then has the same bound for large  $n$ . Select a level satisfying (14.24). Its  $M$ -weighted term is  $O(\varepsilon N)$ ; its  $T$ -weighted term is small by the same bound or by  $\mathbb{P}(X < -M\beta) \leq C_s/M_0$ .

The capped row sums at this level are compared with their means by the same simultaneous concentration event. The extraction gives  $|\mathcal{L}| \geq cN$  and retained cost at most  $\epsilon_n N$  for each template. Markov applied to the finite sum of their  $H_N$ 's supplies a loss event of fixed positive probability, which intersects all the events whose probabilities tend to one. The contradiction in the asymmetric proof, with  $\epsilon$  sufficiently small relative to  $\zeta$ , therefore proves (14.22) for every template. Equation (14.24) gives (14.23). The same cap is used for both conclusions; no expectation bound on a discarded heavy negative tail is required.  $\square$

## 15 Improving a road while preserving locality

The link estimates supply inexpensive crossings after clipping, whereas an admissible road needs a bound on its full expected cost. We bridge this gap in two stages. First, compare each crossing with the corresponding piece of the input road. Then replace clusters of large positive costs by pieces of an independent spare road. The replacement tests use a fixed number of neighboring slots. After a fixed enlargement of the spatial scale, the resulting path belongs to the original road class.

We construct the improved  $A$ -road; interchanging the two families gives the  $B$ -road. Positions are in  $\mathcal{G}_n$  and costs remain in their original units. Write  $(x, y)$  for coordinates in the basis  $(v, B)$ ; the horizontal coordinate  $x$  was denoted by  $u$  in Section 14. Put

$$a_0 = d = 1 - s^2, \quad A = (a_0, s), \quad B = (0, 1)$$

and take  $0 < s < 1/10$ . The tracking and locality norm is still the maximum norm in  $(v, w)$  coordinates. The change  $(x_v, x_w) \mapsto (x_v - sx_w, x_w)$  and its inverse have norm at most  $1 + s \leq 1.1$ ; this factor is included in every band clearance below. Scaled lattice edges have length at most  $2K_{\text{lat}}/n$  in these coordinates, so their rounding errors fit the stated strict clearances once  $n$  is large. Constants are uniform over the fixed neighborhoods of the flat endpoints. Euclidean metric estimates may use constants depending on the hypothetical flat.

### 15.1 Two inputs to the construction

Fix an input pair whose mean period excesses are bounded by  $\beta$ , where

$$c_0 \leq \beta \leq C_s n^{1-c}.$$

We use the preceding results in the following two forms. The gauge input (G) combines Lemmas 10.2, 10.4, 10.6, and 11.1, with the interpolation identity (10.10). The link input (L) combines Propositions 14.4 and 14.5 with (14.3). Stating their support bounds here will make the eventual rescaling independent of all auxiliary proof parameters.

(G) In an iid joint world, all specified road anchors have integrable local gauges, with the deterministic means of their label types included. Corrected increments along a bounded number of road periods, and changes of labels at the specified intersections, have  $L^1$  bounds  $C\beta$ . Comparisons with increments of the calibration have the same bound. These comparisons hold for the supremum over attachment occurrences in the indicated bounded parameter intervals. Along each unit road period the gauge is interpolated so that its corrected signed cost over a nominal interval of length  $u$  is  $u$  times that period's corrected increment. Consequently, over an interval of length at most  $c_1 s$ , situated in a prescribed bounded number of periods, its absolute corrected cost is bounded by  $c_1 s$  times a local variable of mean  $C\beta$ .

A baseline generated from its tube and private completion admits a compatible gauge depending only on these common inputs. In each joint world containing that baseline the difference

from its world gauge, including the supremum over bounded attachment intervals, has  $L^1$  norm at most  $C\beta$ . Every computed gauge, corrected road increment, and gauge-to-gauge seam record has a fixed support radius  $R_g$ , independent of  $s, n, \beta$ , probability tolerances, phase-group size, and the chosen input road rule. Comparisons involving the calibration itself are analytic  $L^1$  bounds; those calibration errors are not asserted to be local.

- (L) Between the walls  $x = ja_0$  and  $x = (j+1)a_0$ , let  $X$  be the minimum signed link cost, corrected at the two wall center levels, in the horizontal band of half-width  $s/50$ . The admissible crossings have horizontal drawdown at most  $s/20$ , and the signed endpoint arms are along the labeled walls. The link is a function of the inputs in a fixed neighborhood of this band. Its negative part satisfies  $\mathbb{E}(X/\beta)^- \leq C_X$ . For every fixed upper cap  $K$  and  $\eta > 0$ , its asymmetric clipped mean is at most  $\eta$  after a sufficiently large fixed lower cap is chosen. All direction tolerances and lower thresholds for  $n$  are uniform over the input road rules satisfying the displayed common bounds, and over the finitely many row and phase versions used below. The matched-cap conclusion is also available. Given  $M_0 < \infty$ ,  $\eta > 0$ , and finitely many additional nonnegative local variables with uniformly bounded first moments, there is a fixed finite dyadic set  $\mathcal{M} \subset [M_0, \infty)$  such that at each sufficiently large admissible  $n$  a deterministic  $M \in \mathcal{M}$  satisfies

$$\mathbb{E} \text{clip}_M(X/\beta) \leq \eta, \quad M \sum_{\ell} \mathbb{P}(U_{\ell} > M) \leq \eta. \quad (15.1)$$

Here  $\text{clip}_M(z) = f_{M,M}(z) = \max\{-M, \min\{z, M\}\}$ . The geometry and threshold for  $n$  are chosen uniformly over this finite set. This is Proposition 14.5, not a consequence merely of asymmetric clipping.

The support assertions in (G) and (L) refer to all the weights and private random inputs used to calculate a variable, including the inputs to its gauges. Finite averaging of gauge functions over mesh offsets preserves both the estimates and a fixed support radius: all offsets lie in one bounded fundamental cell, irrespective of the number of offsets.

## 15.2 Independent baselines and completed environments

Only the weights on a path's traveled edges need to be the actual weights. We may therefore complete the environment privately outside a deterministic band, while retaining the correct iid law for every local calculation. We arrange the completions so that the entire primary field is independent of the spare baseline, and so that transitions can use both baselines with compatible gauges.

Put  $z(x, y) = y - sx/a_0$ . We use the deterministic primary band

$$\mathcal{T}_p = \{|z| \leq 3s\}$$

and the narrow spare-baseline tube around  $z = 10s$ . The input tracking bound  $r = s/1000$  places the primary and spare baselines in disjoint subsets of these bands. Edges meeting their boundaries can be assigned by a fixed convention; for sufficiently large  $n$  the  $O(1/n)$  lattice padding leaves the two preserved sets disjoint.

Generate each baseline once from the actual weights in its narrow tube, its own independent private exterior, and its private road seeds. Its physical path stays in that tube. Whenever a completed environment uses this baseline, insert that same path with those same private inputs. Choose these baseline private arrays jointly independent of the actual edge field and of all world completion and auxiliary arrays. Denote the compatible baseline gauges from (G) by  $H_p$  and  $H_s$ .

The primary world uses actual weights in  $\mathcal{T}_p$  and fresh iid weights outside  $\mathcal{T}_p$ . Its auxiliary road seeds and exterior completion are independent of all spare-baseline inputs. The primary baseline is inserted using its already fixed private completion. Although that completion is not the primary world’s exterior, it is an admissible independent randomization for a path whose physical edges all have the primary world’s weights. Keep the canonical wall network and wall gauges of (L) unchanged, and attach the common baseline as an extra path label using (G). In particular its gauge does not replace any gauge defining  $X$ . Do the same with both common baselines in every transition world. The link variables therefore retain exactly their laws in (L), while (G) supplies the new endpoint-label comparisons. The spare world is constructed in the same way around the spare tube. Thus the entire primary field constructed below is independent of the entire spare baseline and its compatible gauge.

For each prospective transition make a joint world using actual weights in

$$\mathcal{T}_t = \{-3s \leq z \leq 13s\}$$

and a fresh iid exterior. Insert both common baselines, with their original private completions. Use fresh auxiliary road seeds. Distinct transition worlds may share the actual weights in their preserved bands; their exterior completions are independent. We never equate calibrations from two different worlds. Comparisons at their seams use (G).

Apply (G) separately to each baseline with only its own narrow tube prescribed: take  $\eta_0$  to be that tube’s actual weights and take  $\eta_{\text{other}}$  empty. In each world  $j$ , let  $\zeta^j$  be the entire weight array on the complement of this tube, using the actual weights where that world retains them and its fresh exterior weights elsewhere. The partition is deterministic, so each hybrid complement has the iid exponential law and is independent of  $\eta_0$ . The baseline’s private field is jointly independent of  $(\eta_0, \zeta^j)$ . The complements in different worlds may share actual coordinates. Applying the lemma first to the primary baseline in its primary and transition worlds, and then to the spare baseline in its spare and transition worlds, gives the common gauges  $H_p, H_s$  used above.

| World      | Actual weights retained | Inserted baselines | Compatible gauges |
|------------|-------------------------|--------------------|-------------------|
| Primary    | $\mathcal{T}_p$         | primary            | $H_p$             |
| Spare      | narrow spare tube       | spare              | $H_s$             |
| Transition | $\mathcal{T}_t$         | primary and spare  | $H_p, H_s$        |

Table 1: The completed environments and the inputs they share. Each world uses fresh iid weights outside its retained region. Every inserted baseline keeps its fixed private exterior and road seeds. A compatible gauge is compared with that baseline’s gauge in the indicated world; wall-to-baseline switches are handled within that world.

The following phase convention makes the slot construction stationary. If  $\mathcal{H}$  is a finite subgroup of  $\mathbb{R}/\mathbb{Z}$  containing  $s$  and all the row-mesh increments used in the construction, let  $\phi_j$  be iid uniform in  $\mathcal{H}$ . Give wall  $j$  the relative phase  $\phi_j$  in the rule of Section 11; its absolute marked-height residue is

$$\theta_j = \phi_j + js \pmod{1}. \quad (15.2)$$

The translation by  $A$  takes wall  $j$  to wall  $j + 1$  and adds  $s$  to its vertical coordinate. Consequently ordinary translation of the iid sequence  $(\phi_j)$  gives exactly the required translation of the wall family, including its marked cuts. A uniform wall phase is needed only on the bounded wall portions queried by this construction; its seed is stored at the corresponding primary slot. Different transition worlds receive their own such seeds. No random phase is shared by all slots. Gauge functions, rather than

a single random mesh choice, are averaged over mesh offsets. The number of phases can depend on the previously fixed rational mesh and  $s$ ; their physical storage locations and the union of their support sets remain in a fixed neighborhood.

For the copy with nominal origin zero, group the globally indexed primitives of Section 11 into slabs indexed by  $j = \lfloor x/a_0 \rfloor$ , resolving boundary ties deterministically. The coordinates retain their structural roles and road or world origins; translated copies carry translated origins and slabs. Each slab contains countably many independent auxiliary coordinates, with separate coordinates for the baselines, the primary completion, and each translated transition type, and with the same marginal law in every translated slab for each role. The global indexing supplies the measure-preserving action of every lattice shift; the slab grouping here establishes locality and separation. All quantities used for slot  $i$  are measurable with respect to the slab inputs with  $|j - i| \leq R_*$ , for a fixed integer  $R_*$ . Only bounded transverse subsets of these slabs are queried. The retained actual primary and spare edges are disjoint, and their auxiliary coordinates are disjoint. These facts establish unconditional independence of the two entire fields, and independence of a slot variable from a remote transition test. They are stronger than conditional finite dependence given phases.

### 15.3 The primary choice in each slot

A slot has two physical candidates with the same endpoints: the baseline piece and a horizontal crossing reached by forward wall travel. Their common endpoint convention lets us choose the cheaper candidate before estimating its cost.

Every minimizing link used in this section is the vertex-ended representative constructed after (14.2).

Choose the primary baseline's common-vertex intersection  $c_i$  with wall  $i$  by the local occurrence rule of Lemma 10.1. All such intersections have baseline parameter  $i + O(r)$  and wall height  $is + O(r)$ , by the tracking bounds. The permissible parameter intervals for different walls are disjoint and ordered. The choice is therefore local and commutes with translation by  $A$ . Ties between multiple occurrences are resolved in their order along the baseline. Let  $g_i$  be the primary world's wall gauge at  $c_i$ .

The two competing paths from  $c_i$  to  $c_{i+1}$  are the baseline subpath and the midpoint link, whose row is  $(i + 1/2)s$ , with its attachments along the two walls. Every left attachment is forward because

$$is + O(r) < (i + 1/2)s - s/50 - O(r),$$

and every right attachment is forward for the analogous reason. The link is followed in its crossing direction. Thus both candidates are physical paths with positive edge traversal costs. The signed arms in the definition of  $X_i$  are only bookkeeping: they cancel inside these forward attachments.

Let  $B_i$  and  $L_i$  be the normalized corrected costs of these two paths, both in the endpoint convention  $g_i, g_{i+1}$ . By the linear interpolation in (G),

$$L_i = X_i/\beta + E_i, \quad \mathbb{E}|E_i| \leq C_1 s. \quad (15.3)$$

Indeed the algebraic difference consists exactly of the two signed wall movements from the baseline intersections to the midpoint center levels. Each has nominal length  $O(s)$  in a bounded number of wall periods. This estimate precedes, and does not condition on, any later path selection.

Choose the cheaper path and put  $Z_i = \min(B_i, L_i)$ . This is one common endpoint convention, with wall gauge  $g_i$  shared by its two neighboring slots. It is essential to retain these gauges here: replacing them slot by slot by  $H_p$  would add errors of order  $\beta$ , whereas (15.3) is of order  $s\beta$ . The compatible baseline gauge is used only at the rare seams below.

Both tails of  $Z_i$  have a uniform first-moment bound. For the negative tail, calibration domination leaves only the gauge error between  $c_i$  and  $c_{i+1}$ ; the attachment supremum in (G) bounds its expectation by  $C\beta$  before the candidate is selected. For the positive tail, the baseline competitor has defect bounded by a fixed number of complete period defects, because its endpoints lie in fixed parameter windows. The endpoint gauge comparison costs a further  $C\beta$ . After normalization,

$$\mathbb{E}|Z_i| \leq C_0. \quad (15.4)$$

The complete-period estimate applies even though the endpoints were chosen from the baseline and wall environment.

Since clipping is monotone and 1-Lipschitz,

$$\mathbb{E} \text{clip}_M(Z_i) \leq \mathbb{E} \text{clip}_M(X_i/\beta) + C_1 s. \quad (15.5)$$

Apply the matched-cap estimate with the additional variable  $Z_i^+$ . For any prescribed  $M_0$  and  $\eta$ , a cap in a fixed finite dyadic range therefore gives

$$\mathbb{E} \text{clip}_M(Z_i) \leq C_1 s + \eta, \quad \text{MP}(Z_i > M) \leq \eta. \quad (15.6)$$

The cap depends on the input law and the scale, but is deterministic before the environment is sampled. This dependence adds no spatial input to the rule.

All physical pieces just constructed lie in  $|z| \leq s$ , after increasing the lower threshold for  $n$ . Thus their actual costs in the primary world are their actual costs in the original environment. Their full calculation, including the wall gauges, is local in the sense established in Section 15.2.

## 15.4 Forward transitions to the spare road

A replacement must leave the primary baseline and return to it using physical forward travel. We reserve a fixed number of slots for each transition. Its geometry is fixed; only the numerical success threshold will depend on the desired failure probability.

Fix the transition length  $T_* = 64$ . For each integer  $k$ , construct a candidate from  $c_k$  to the marked spare cut at nominal parameter  $k + T_*$ . Follow the primary baseline to its common-vertex intersection with the joint world's wall  $k + 1$ , chosen by the same local rule, cross successive walls through midpoint bands, and after wall  $k + T_* - 1$  follow the spare baseline forward to that marked spare cut. For the intervening link between walls  $j$  and  $j + 1$ , use row

$$m_j = (j + 1/2)s + 10s \frac{j + 1/2 - k}{T_*}, \quad k + 1 \leq j \leq k + T_* - 2. \quad (15.7)$$

For the transition from the spare cut at  $k$  to  $c_{k+T_*}$ , use instead

$$m_j = (j + 1/2)s + 10s \left( 1 - \frac{j + 1/2 - k}{T_*} \right). \quad (15.8)$$

All the rows belong to a fixed finite rational mesh. At interior walls the row increment is  $s(1 \pm 10/T_*)$ . At the first and last attachment, the smaller available clearance is

$$\left( \frac{1}{2} - \frac{15}{T_*} \right) s = \frac{17}{64} s > s/4.$$

This exceeds twice the band half-width and the  $O(r)$  tracking error. Hence every wall attachment in both transitions is forward. The initial and terminal baseline pieces are also forward, since the

walls at  $k + 1$  and  $k + T_* - 1$  have disjoint ordered baseline intersection intervals. Every edge of these candidates lies in  $\mathcal{T}_t$ .

We use incoming records indexed by their primary starting cut:  $\Gamma_j^-$  starts at  $c_j$  and ends at spare cut  $j + T_*$ . Outgoing records are indexed by the last primary slot they replace:  $\Gamma_j^+$  starts at spare cut  $j + 1 - T_*$  and ends at  $c_{j+1}$ . Let their normalized total corrected costs be  $C_j^-$  and  $C_j^+$ , using  $g_j$  or  $g_{j+1}$  at the primary end and  $H_s$  at the spare end. In calculating them, insert the primary compatible gauge  $H_p$ , the joint world's baseline and wall gauges, and the spare compatible gauge  $H_s$  at the seams. All intermediate potentials telescope. Every gauge change is included in  $C_j^\pm$ .

Here is the seam estimate at a random primary intersection. Let  $g_{\text{base}}^p$  and  $g_{\text{base}}^t$  denote the gauges of the common primary baseline in the primary and transition worlds. Let  $I_i$  be all baseline occurrences with nominal parameter in  $i + [-Cr, Cr]$ , so  $c_i \in I_i$ . Then

$$|g_i - g_{\text{base}}^t(c_i)| \leq |g_i - g_{\text{base}}^p(c_i)| + \sup_{c \in I_i} |g_{\text{base}}^p(c) - H_p(c)| + \sup_{c \in I_i} |H_p(c) - g_{\text{base}}^t(c)|.$$

The switch envelope bounds the first term in  $L^1$  by  $C\beta$ ; Lemma 10.6 bounds the other two. These are bounds under the full coupling, obtained from the respective world marginals and uniform over the occurrence window. They require no conditioning on the selected  $c_i$ . The other seams are treated in the same way.

For each  $\delta > 0$  there is a finite  $K_\delta$  such that the local tests

$$U_j^\pm = \mathbf{1}_{\{|C_j^\pm| \leq K_\delta\}} \quad \text{satisfy} \quad \mathbb{P}(U_j^\pm = 0) \leq \delta. \quad (15.9)$$

To prove this, first obtain a uniform upper-tail bound for each link. For any fixed  $K$ , asymmetric clipping and the negative first moment in (L) give

$$K \mathbb{P}(X > K\beta) \leq \mathbb{E} f_{J,K}(X/\beta) + \mathbb{E}(X/\beta)^- \leq \eta + C_X.$$

There are  $T_* - 2$  links, so a union bound makes all their normalized costs at most a sufficiently large fixed  $K$  with probability at least  $1 - \delta/4$ . The remaining forward baseline portions, interpolated wall movements, and all the gauge seams are bounded in absolute value by a fixed sum of local variables having mean  $C\beta$ . Markov's inequality bounds this sum by a finite multiple of  $\beta$  with probability at least  $1 - \delta/4$ . This controls the upper tail of  $C_j^\pm$ . Its negative part has bounded mean: in the joint world the candidate is a physical path, hence its defect is nonnegative, and (G) controls its two endpoint corrections, including the comparison to  $g_j$  or  $g_{j+1}$ . Markov's inequality controls the lower tail. Increasing the numerical cap gives (15.9). The support and the geometry of a test do not increase when its numerical cost cap increases.

Let  $V_i$  be the absolute normalized corrected cost of the spare baseline from its marked cut  $i$  to cut  $i + 1$ , using  $H_s$ . Then

$$V_i \geq 0, \quad \mathbb{E} V_i \leq C_V, \quad (V_i)_{i \in \mathbb{Z}} \text{ is independent of } (Z_i)_{i \in \mathbb{Z}}. \quad (15.10)$$

The transition tests may depend on both fields. Later we will bound a successful replacement by the presence of a nearby primary flag, so the spare-cost estimate will use the independence in (15.10).

Choose once and for all an integer  $R \geq T_*$  large enough that, whenever  $|i - j| > R$ ,  $Z_i$  is independent separately of  $Z_j$ ,  $U_j^-$ , and  $U_j^+$ . Such an  $R$  follows from the explicit iid input supports. It is independent of  $s$ ,  $\delta$ ,  $K_\delta$ , and the clipping caps. The whole joint collection is stationary under translation by  $A$ .

## 15.5 A finite-range replacement lemma

The remaining probabilistic step uses only the stationary fields just constructed and their stated independence. We isolate it to make clear how a first-moment bound suffices for the rare positive tail.

Write  $F_i = \mathbf{1}_{\{Z_i > M\}}$ ,  $p = \mathbb{E}F_0$ , and  $W_i = \text{clip}_M(Z_i)$ . Fix

$$B_* = R + 1, \quad G_* = 4B_*, \quad L_* = 8B_*, \quad H_* = 10B_* + 1. \quad (15.11)$$

**Lemma 15.1** (Local replacement of large positive costs). *For the stationary local fields and compatible physical pieces constructed above, assume (15.4), (15.9)–(15.10), and the stated independence at distances greater than  $R$ . There is a stationary finite-range rule selecting disjoint replacement intervals. It concatenates primary pieces, successful transitions, and spare pieces. Charge each retained primary cost to its own slot and each replacement's total normalized corrected cost to its leftmost flag. The resulting integrable local charge  $Z_i^{\text{out}}$  satisfies*

$$\mathbb{E}Z_0^{\text{out}} \leq \mathbb{E}W_0 + H_*Mp + C_0\{2G_*p + 2(L_* + 1)\delta\} + (H_*C_V + 2K_\delta)p. \quad (15.12)$$

The range of the rule depends on  $R$  and  $T_*$ , independently of  $M$ ,  $\delta$ , and  $K_\delta$ .

*Proof. The local rule.* Consecutive flagged indices belong to the same cluster when their difference is at most  $G_*$ . A cluster with extremes  $a \leq b$  is accepted exactly when  $b - a \leq L_*$  and

$$U_{a-B_*}^- = U_{b+B_*}^+ = 1.$$

For such a cluster replace the primary slots shown in Figure 4:

$$I(a, b) = [a - B_*, b + B_*] \cap \mathbb{Z}. \quad (15.13)$$

Use the incoming transition, then the spare baseline, then the outgoing transition. The spare portion runs from parameter  $a - B_* + T_*$  to parameter  $b + B_* + 1 - T_*$ . This interval has nonnegative length because  $B_* > T_*$ . The endpoints of the whole replacement are precisely  $c_{a-B_*}$  and  $c_{b+B_*+1}$ . Its normalized corrected cost  $C(a, b)$  is in their primary gauge convention and satisfies

$$|C(a, b)| \leq 2K_\delta + \sum_{i \in I(a, b)} V_i. \quad (15.14)$$

This follows by exact telescoping through the two transitions and the spare gauge. Different accepted intervals are disjoint because distinct clusters have gap greater than  $G_* > 2B_*$ . All long or infinite clusters are left unchanged.

The latter instruction has an exact finite implementation. A proposed leftmost flag  $a$  and rightmost flag  $b \in [a, a + L_*]$  form a complete short cluster if and only if they are flagged, all consecutive flag gaps in  $[a, b]$  are at most  $G_*$ , and there is no flag in  $[a - G_*, a - 1]$  or  $[b + 1, b + G_*]$ . These tests read only  $[a - G_*, a + L_* + G_*]$ . They can never accept a portion of a longer or infinite cluster. Let  $O_i$  indicate replacement of slot  $i$ . A cluster covering  $i$  has

$$a \in [i - L_* - B_*, i + B_*],$$

so  $O_i$  is determined by flags within distance  $13B_*$  of  $i$  and transition tests within distance  $10B_*$  of  $i$ . It follows in particular that

$$O_i \leq \sum_{a=i-L_*-B_*}^{i+B_*} F_a, \quad \mathbb{E}O_i \leq H_*p. \quad (15.15)$$

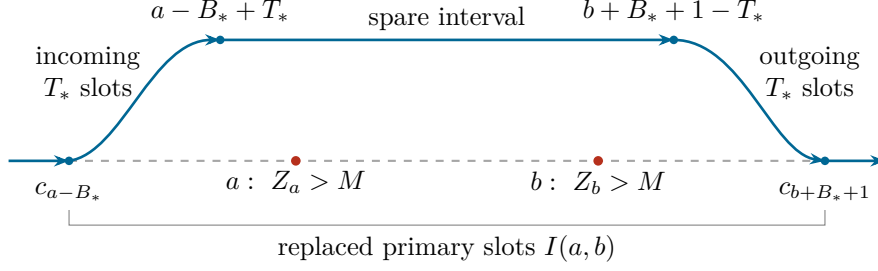


Figure 4: An accepted replacement, shown in nominal-parameter coordinates. The marked primary slots are the extreme flags  $a, b$  of the accepted cluster; they may coincide. The flags use the matched cap  $M$ , while the arrows show the actual uncapped physical route. The diagram is schematic and does not represent a sampled FPP path; each transition and the intervening spare piece is traversed forward.

*Uncovered positive costs.* If a flagged slot is left uncovered because its cluster is long or infinite, it contains a flag at distance strictly greater than  $R$ . Following its chain from  $i$  until the first such flag gives an index  $j$  with

$$R < |j - i| \leq R + G_*.$$

There are  $2G_*$  possible witnesses. If the cluster is short but not accepted, one of the possible incoming tests indexed in  $[i - L_* - B_*, i - B_*]$ , or outgoing tests indexed in  $[i + B_*, i + L_* + B_*]$ , fails. Each of these at most  $2(L_* + 1)$  tests is independent of  $Z_i$ , because its index is more than  $R$  away. Applying independence to these deterministic possible witnesses, before any random choice of an extreme, gives

$$\mathbb{E}[Z_i^+ F_i (1 - O_i)] \leq C_0 \{2G_* p + 2(L_* + 1)\delta\}. \quad (15.16)$$

For example each remote-flag term is  $\mathbb{E}[Z_i^+ F_i F_j] = \mathbb{E}[Z_i^+ F_i] p \leq C_0 p$ ; the failed-test terms are analogous. Thus the expectation of an uncovered large cost is bounded directly, even if rare values carry a substantial part of the first moment.

*Retained and discarded primary costs.* The retained costs satisfy

$$(1 - O_i) Z_i \leq W_i + M O_i + Z_i^+ F_i (1 - O_i). \quad (15.17)$$

Indeed on a covered slot the right side is at least  $-M + M = 0$ ; on an uncovered slot only the portion of  $Z_i$  above  $M$  can increase its clipped value. In particular discarding negative primary costs loses at most  $M$  times the covered density, whether or not those costs are correlated with nearby flags.

*Replacement costs.* By (15.15) and independence of the entire spare and primary fields,

$$\mathbb{E}[V_i O_i] \leq H_* C_V p. \quad (15.18)$$

We have bounded the success indicator by the event of a nearby primary flag; success itself need not be independent of  $V_i$ .

Accepted clusters have density at most  $p$ , so their two successful transitions contribute at most  $2K_\delta p$  to the mean upper bound. Charge the remaining bound in (15.14) to the leftmost flag of its cluster. By stationarity, its mean per slot is bounded by  $\mathbb{E}[V_0 O_0]$ ; unused spare slots in that display are simply overcharged. Summing these bounds with (15.16)–(15.18) proves (15.12). Each charge and each reassignment involves a uniformly bounded number of slots. Hence all charges are integrable, and the expectation calculation requires only first moments.  $\square$

## 15.6 Physical parameterization and endpoint accounting

The replacement lemma bounds stationary charges assigned to whole pieces. Three further facts are needed to obtain an admissible road: the physical concatenation must track the nominal line, its actual period cuts must have the same mean excess as these charges, and its full input support must remain bounded. We verify them in this order.

**Tracking.** Each crossing has horizontal drawdown at most  $s/20$ . The wall and baseline attachments add only  $O(r)$  to this drawdown; successive walls have horizontal separation  $a_0$ . On each unchanged primary slot, or each transition, take the running maximum of the horizontal coordinate and apply the affine adjustment sending its endpoint values to the prescribed endpoint parameters. This is continuous and nondecreasing; constant portions are allowed. The adjustment moves the horizontal nominal position by  $O(s)$ , since the horizontal endpoint errors and the drawdown are  $O(s)$ . On spare portions use the original spare nominal parameter. These parameterizations agree at every concatenation and cover the real line.

Throughout a primary slot the vertical displacement from the nominal line is  $O(s)$ . Throughout a transition it lies between its two baseline offsets, up to  $O(s)$ , by (15.7)–(15.8). Spare pieces have offset  $10s$ . Thus in the  $(v, w)$  tracking norm the entire output has tracking error at most  $C_{\text{tr}}s$ , with  $C_{\text{tr}}$  independent of the cluster and probability parameters. Local edge repetitions and nominal pauses retain their occurrence labels; no negative signed traversal is ever substituted for physical motion. Every baseline–wall and wall–link splice is at a lattice vertex. A marked spare cut inside an edge only subdivides consecutive forward travel along that same baseline occurrence. Merging those subdivisions therefore leaves a two-sided lattice walk with locally finite edge occurrences, as required by Definition 9.1.

**Actual period cuts and integrability.** To convert the charge bound (15.12) into a mean actual excess, assign the primary gauges to the designated primary concatenation endpoints, and the compatible spare gauges to spare endpoints. On each primary or transition piece interpolate its two endpoint gauges linearly in the fraction of its positive physical cost already traversed; on spare pieces use the spare interpolation. These assignments agree at concatenations, are local, and commute with translation by  $A$ . In particular they assign a gauge at every actual first occurrence of an integer nominal level, including when that occurrence precedes the designated endpoint during a nominal pause. The assigned gauges are integrable at all these cuts. Integrability on a transition follows from its positive physical costs. On success, the total physical passage cost is at most

$$K_\delta\beta + |g_{\text{first}}| + |g_{\text{last}}| + Cn,$$

where the appropriate endpoint gauge may be  $H_s$ . The right side is integrable by (G), and any physical subpiece costs at most the whole physical path. A selected primary candidate has physical cost no larger than its baseline competitor: their endpoints and corrections are identical. That competitor lies in a fixed number of integrable input periods. On primary and transition pieces, linear interpolation bounds each assigned gauge by the absolute endpoint gauges. On spare pieces, the original interpolation and (G) give integrability over bounded parameter windows. Partial primary pieces, successful transitions, and spare periods are therefore integrable. This integrability is required at each fixed  $n$ ; the bound need not be uniform in  $n$ .

For an interval of  $N$  slots, summing the local upper charges used in (15.12) bounds the corrected cost of all complete primary and replacement pieces. Passing to the actual first-occurrence endpoint cuts changes this bound only within the bounded number of primary pieces and replacement intervals meeting the two boundaries. The primary boundary pieces must be included even when no

replacement is accepted: a terminal wall arm may follow the first occurrence of the terminal integer. Each boundary remainder is bounded by a fixed finite sum of the integrable costs and gauges just described. Its expected absolute value is bounded independently of  $N$ . After division by  $N$  this boundary error vanishes. The endpoint gauge difference has zero mean: the complete output rule, including the choice of endpoint label, is stationary under  $A$ , so the gauges at translated cuts have the same integrable distribution. Thus (15.12) indeed bounds the expected actual  $Q_n$  per old slot, divided by  $\beta$ . This uses stationarity of the chosen labels and does not assume that the mean of the calibration at an arbitrary random point is zero.

**Input locality.** The finite cluster test and the input supports in Section 15.2 imply a fixed output dependence radius  $R_{\text{out}}$  in old spatial units. Numerical caps affect only comparisons of already calculated local quantities. The large auxiliary boxes used to prove (L), and the size of its finite cap set, are not inputs to this path construction. Neither  $R_{\text{out}}$  nor  $C_{\text{tr}}$  depends on  $s$ , the directional tolerances, the cap, the seam failure probability, or the input road rule.

## 15.7 Choice of constants and contraction of the infimum

The construction can now be closed in the class used to define  $b(n)$ . The order of choices is essential: choose the rescaling from structural support and tracking bounds first, then the small inward tilt, and only then the probability and cap tolerances.

**Proposition 15.2** (Contraction in the original road class). *For suitable fixed rational  $v, w$  near the flat endpoints and  $s > 0$ , there are an integer  $D \geq 1$  and  $\rho < 1$  such that*

$$b(Dn) \leq \rho b(n)$$

*for every sufficiently large admissible  $n$ . The infimum on both sides uses the road class of Definition 9.1, with costs in their original units.*

*Proof.* The structural constants  $C_{\text{tr}}, R_{\text{out}}, C_1$  are uniform in sufficiently small  $s$  and the nearby rational directions. First choose an integer

$$D \geq \max\{1000C_{\text{tr}}, R_{\text{out}}/10, 1\}. \quad (15.19)$$

Next fix rational  $0 < s < 1/10$  and  $\epsilon > 0$  so small that  $DC_1s < 1/4$  and  $D\epsilon < 1/4$ . The geometry and the constants  $R, B_*, G_*, L_*, H_*, C_0, C_V$  have already been fixed uniformly. If  $C_0 = 0$ , the primary costs vanish almost surely and no replacements are necessary. Otherwise choose

$$\delta \leq \min\left\{\frac{1}{2}, \frac{\epsilon}{6(L_* + 1)C_0}\right\},$$

then the corresponding transition cap  $K_\delta$ . Set

$$A_* = 2G_*C_0 + H_*C_V + 2K_\delta, \quad M_0 \geq \max\{1, 3C_0A_*/\epsilon\}, \quad \eta \leq \frac{\epsilon}{3(H_* + 1)}.$$

Choose the finite matched-cap range from (L) with this  $M_0, \eta$ , and then fix the geometric comparison tolerances and the rational directions so that it applies uniformly over that range. At every sufficiently large admissible  $n$ , choose its deterministic good cap. Equation (15.6) gives  $Mp \leq \eta$ , while  $p \leq C_0/M_0$  by Markov's inequality. In (15.12) the failed-test error, the terms  $A_*p$ , and the terms  $(H_* + 1)\eta$  are each at most  $\epsilon/3$ . We obtain an actual stationary local road with

$$\mathbb{E}Q_n(\text{one old output slot}) \leq (C_1s + \epsilon)\beta. \quad (15.20)$$

Regard  $D$  old slots as one new period and use space units  $Dn$ . The tracking error is then at most  $s/1000$ , the input radius at most 10, and the rule commutes with translation by  $DA$  in old units, that is by  $A$  in new units. Reindexing the private product field by this rescaling preserves its shift-invariant law: a new lattice shift corresponds to the old lattice shift multiplied by  $D$ . Thus it is in the original road class at scale  $Dn$ . The same construction in the other direction gives a pair with common mean bound

$$D(C_1s + \epsilon)\beta. \quad (15.21)$$

All the rational meshes have fixed denominators; multiplication of an admissible  $n$  by the integer  $D$  preserves admissibility.

Let  $b(n)$  be the infimum of the common mean bound over that road class. Invoke the uniform input estimates with the fixed enlarged prefactor  $2C_s$ ; this changes only the threshold for  $n$ . That threshold is therefore uniform over members with  $\beta \leq 2C_sn^{1-c}$ . Since  $b(n) \leq C_sn^{1-c}$ , for each sufficiently small  $t > 0$  one can choose a pair with  $\beta \leq b(n) + t \leq 2C_sn^{1-c}$ . Apply (15.21) and then let  $t \downarrow 0$ ; existence of a minimizing road is unnecessary. This proves

$$b(Dn) \leq D(C_1s + \epsilon)b(n). \quad (15.22)$$

All directions, meshes, and tolerances are fixed in the stated order, and the choices give  $\rho = D(C_1s + \epsilon) < 1/2$ . This proves the assertion for all admissible  $n$  above the common threshold.  $\square$

## 16 Excluding flat faces

*Proof of Theorem 1.1.* Suppose  $\partial\mathcal{B}$  contains a nontrivial segment, and fix the maximal supporting face with extreme endpoints  $a, b$  described in Section 8. The same admissible road class has the bounds

$$0 < c_0 \leq b(n) \leq C_sn^{3/4}$$

from Propositions 9.5 and 9.4.

Proposition 15.2 supplies fixed rational directions,  $s > 0$ , an integer  $D \geq 1$ , and  $\rho < 1$  such that  $b(Dn) \leq \rho b(n)$  for every sufficiently large admissible  $n$ . Choose an admissible  $n_0$  above the common threshold. Multiplication by  $D$  preserves admissibility, so iteration gives

$$0 < c_0 \leq b(D^k n_0) \leq \rho^k b(n_0) \quad (k \geq 0),$$

a contradiction. Thus every supporting face is a single point.

To express this as strict convexity, take distinct  $x, y \in \partial\mathcal{B}$  and  $0 < t < 1$ . Convexity gives  $\mu((1-t)x + ty) \leq 1$ . Equality would provide a supporting functional  $\ell \leq \mu$  at this point, and therefore

$$1 = (1-t)\ell(x) + t\ell(y), \quad \ell(x), \ell(y) \leq 1.$$

Both endpoint values would be one, so the entire segment  $[x, y]$  would lie in a nontrivial supporting face. This has been excluded. Therefore  $\mu((1-t)x + ty) < 1$  for the rate-one exponential law.

Theorem 1.2 with shape  $\kappa = 1$  gives Fréchet differentiability of this norm away from the origin and a  $C^1$  unit sphere, with a unique supporting line at every point. To pass to any rate  $\lambda > 0$ , couple the weights by  $\tau_e^{(\lambda)} = \lambda^{-1}\tau_e^{(1)}$ . Then every path cost, and hence every passage time, is multiplied by  $\lambda^{-1}$ . Thus

$$T_\lambda(x, y) = \lambda^{-1}T_1(x, y), \quad \mu_\lambda = \lambda^{-1}\mu_1, \quad \mathcal{B}_\lambda = \lambda\mathcal{B}_1.$$

Multiplication by a positive scalar preserves differentiability, strict convexity, and the  $C^1$  boundary property. This proves every assertion of the theorem.  $\square$

## 17 Geometric and geodesic consequences

The first two subsections concern the exponential law of arbitrary rate  $\lambda > 0$ . Theorem 1.1 supplies both regularity properties needed there. The final two subsections apply Theorem 1.2 to a quantitative midpoint estimate and to cone-tip competition for the whole Gamma family.

For a Euclidean unit vector  $u$ , put  $v_u = u/\mu(u)$  and  $\rho_u = \nabla\mu(u)$ . Homogeneity gives  $\rho_u \cdot v_u = 1$ , and the supporting inequality gives  $\rho_u \cdot x \leq \mu(x)$  for every  $x$ . Hence

$$P_u = \{x : \rho_u \cdot x = 1\}$$

is the supporting line at  $v_u$ . Strict convexity gives  $P_u \cap \mathcal{B} = \{v_u\}$ : two contact points would put their entire segment in the boundary. Thus the set of radial directions of contact with  $P_u$  consists of the single direction  $u$ .

### 17.1 A fixed deterministic direction

A finite geodesic is a path realizing the passage time between its endpoints. An infinite geodesic, or a bigeodesic, is respectively a one-sided or two-sided infinite path every finite segment of which is a finite geodesic. For the Gamma laws in this paper, finite geodesics exist and are almost surely unique. Indeed, choose  $\theta > 0$  so large that  $3\mathbb{E}e^{-\theta\tau_e} < 1$ . Counting simple paths shows that the probability of a simple path from a fixed vertex with at least  $m$  edges and cost at most  $t$  is bounded by  $Ce^{\theta t}(3\mathbb{E}e^{-\theta\tau_e})^m$ . Thus almost surely only finitely many simple paths from that vertex have cost below any fixed bound. Thus every passage ball  $\{z : T(x, z) \leq t\}$  is finite, and a minimum is attained between any two vertices. Two distinct simple paths with the same endpoints have different edge sets; conditioning on all but an edge in their symmetric difference shows that their costs are unequal almost surely. Countability gives uniqueness simultaneously for all lattice endpoints.

A sequence  $(z_n)$  is directed toward a unit vector  $u$  when  $\|z_n\|_2 \rightarrow \infty$  and  $z_n/\|z_n\|_2 \rightarrow u$ . An infinite geodesic has direction  $u$  when its vertex sequence has this property. Write  $\Gamma(x, y)$  for the unique finite geodesic between lattice vertices  $x$  and  $y$ . Local convergence of paths rooted at  $x$  means eventual agreement of each fixed finite initial segment.

**Corollary 17.1.** *Fix a deterministic unit vector  $u$ . There is an event  $\Omega_u$  of probability one on which all the following statements hold.*

1. *For every  $x \in \mathbb{Z}^2$  there is exactly one infinite geodesic  $\Gamma_x^u$  from  $x$  with direction  $u$ . For every sequence  $(z_n)$  directed toward  $u$ , including a sequence chosen from the edge configuration,  $\Gamma(x, z_n) \rightarrow \Gamma_x^u$  locally.*
2. *For every  $x, y \in \mathbb{Z}^2$ , the paths  $\Gamma_x^u$  and  $\Gamma_y^u$  have a common tail.*
3. *No bigeodesic has an end with direction  $u$ .*
4. *For every  $x, y \in \mathbb{Z}^2$ , the limit*

$$B_u(x, y) = \lim_{n \rightarrow \infty} (T(x, z_n) - T(y, z_n))$$

*exists and is independent of the choice of sequence directed toward  $u$ . Moreover,*

$$\mathbb{E}B_u(0, x) = \nabla\mu(u) \cdot x,$$

*and, for every  $\varepsilon > 0$ , only finitely many  $x \in \mathbb{Z}^2$  satisfy*

$$|B_u(0, x) - \nabla\mu(u) \cdot x| > \varepsilon\|x\|_1.$$

*Proof.* Damron–Hanson’s product-measure hypothesis A1’ holds: the common law is continuous, and the minimum  $Y$  of the four weights incident to the origin is exponential of rate  $4\lambda$ , whence  $\mathbb{E}Y^2 = 1/(8\lambda^2) < \infty$ . Their geometric assumptions (2.1)–(2.2) require differentiability at the chosen boundary point and at both endpoints of its supporting face. The preceding supporting-line calculation identifies that face with  $\{v_u\}$ . Thus their supporting sector is exactly the direction  $u$ . Theorems 1 and 2 of [10] give the convergence, coalescence, bigeodesic exclusion, and Busemann statements on one probability-one event for this fixed  $u$ . Their normalized supporting vector is  $\rho_u = \nabla\mu(u)$  by that calculation.

For completeness, uniqueness follows from the convergence assertion. If  $(x = x_0, x_1, \dots)$  is any infinite geodesic of direction  $u$ , then its segment ending at  $x_n$  is  $\Gamma(x, x_n)$  by uniqueness of finite geodesics. These segments converge both to that infinite geodesic and to  $\Gamma_x^u$ , so the paths agree.  $\square$

The Busemann limit also has an elementary pathwise description. If  $c$  is any common vertex after  $\Gamma_x^u$  and  $\Gamma_y^u$  have coalesced, local convergence implies that both finite geodesics to  $z_n$  pass through  $c$  for all sufficiently large  $n$ . Hence

$$B_u(x, y) = T(x, c) - T(y, c). \quad (17.1)$$

In particular the displayed difference defining  $B_u(x, y)$  is eventually constant for each fixed pair and each such target sequence. Taking limits in the triangle inequality and in differences of passage times gives

$$|B_u(x, y)| \leq T(x, y), \quad B_u(x, z) = B_u(x, y) + B_u(y, z).$$

The first inequality also gives integrability. If  $y$  lies on  $\Gamma_x^u$ , then  $B_u(x, y) = T(x, y)$ . Translation of the environment and of the targets shows that this additive random function is stationary.

The direction  $u$  is fixed before taking the probability-one event. One may intersect these events for a prescribed countable set of directions. The result does not assert simultaneous uniqueness in all directions, or simultaneous absence of all bigeodesics.

Independently of these shape arguments, the companion manuscript [30, Theorem 1.1] excludes all bigeodesics on one probability-one event for iid  $\text{Gamma}(\kappa, \lambda)$  edge weights, for each fixed  $\kappa, \lambda > 0$ . This includes the exponential laws here; these distributions satisfy its nonatomic minimum-second-moment hypotheses.

## 17.2 Passage through a prescribed vertex

**Corollary 17.2.** *For every nonzero  $v \in \mathbb{Z}^2$ ,*

$$\lim_{n \rightarrow \infty} \mathbb{P}(\lfloor n/2 \rfloor v \in \Gamma(0, nv)) = 0.$$

*More generally, let  $a_n, b_n, z_n$  be deterministic lattice vertices with*

$$\|a_n - z_n\|_2 \rightarrow \infty, \quad \|b_n - z_n\|_2 \rightarrow \infty, \quad \frac{b_n - z_n}{\|b_n - z_n\|_2} \rightarrow u$$

*for a deterministic unit vector  $u$ . Then  $\mathbb{P}(z_n \in \Gamma(a_n, b_n)) \rightarrow 0$ .*

*Proof.* Set  $p_n = a_n - z_n$  and  $q_n = b_n - z_n$ . Translation invariance gives  $\mathbb{P}(z_n \in \Gamma(a_n, b_n)) = \mathbb{P}(0 \in \Gamma(p_n, q_n))$ . If the asserted probabilities had positive limsup, the event that  $0 \in \Gamma(p_n, q_n)$  for infinitely many  $n$  would have positive probability. Intersect this event with  $\Omega_u$  from Corollary 17.1. Restrict to the infinite set of indices for which  $0 \in \Gamma(p_n, q_n)$ . On the intersection, the arm from 0 to  $q_n$  converges locally to  $\Gamma_0^u$ . Both arms have lengths tending to infinity, since their endpoints

tend to infinity. Local finiteness of the lattice permits a diagonal subsequence along which the other arm also converges to an infinite path. Every finite segment of the resulting two-sided path is a segment of one of the finite geodesics  $\Gamma(p_n, q_n)$ , and hence is geodesic. The two arms have no vertex in common except 0, because each finite geodesic is simple. This constructs a bigeodesic with an end of direction  $u$ , contradicting Corollary 17.1. The first assertion uses  $a_n = 0$ ,  $b_n = nv$ , and  $z_n = \lfloor n/2 \rfloor v$ .  $\square$

This is the compactness argument underlying [10, Theorem 2.1]. Ahlberg and Hoffman [2, Theorem 2.1] already prove the whole qualitative corollary for arbitrary deterministic endpoint sequences escaping from the tested site, assuming continuity and the minimum-of-four second moment. This includes all Gamma laws here and requires no shape regularity. The proof above records its connection with the fixed-direction consequences.

### 17.3 A quantitative midpoint bound for Gamma weights

Differentiability has a useful consequence even when strict convexity has not been established: a  $C^1$  planar convex body cannot have only finitely many extreme points. This verifies the geometric hypothesis of the quantitative midpoint theorem of Dembin, Elboim, and Peled.

**Corollary 17.3** (Quantitative midpoint bound for the Gamma family). *For every  $\kappa > 0$  there is  $C_\kappa < \infty$  such that, for every  $\lambda > 0$  and all  $a, b, z \in \mathbb{Z}^2$  with  $D = \min\{\|a - z\|_1, \|b - z\|_1\} \geq 1$ ,*

$$\mathbb{P}(z \in \Gamma_{\kappa, \lambda}(a, b)) \leq C_\kappa \frac{\log^2(D + 2)}{D^{1/16}},$$

where  $\Gamma_{\kappa, \lambda}$  denotes the unique geodesic for iid  $\text{Gamma}(\kappa, \lambda)$  edge weights.

*Proof.* Theorem 1.2 gives a  $C^1$  boundary. We verify that such a compact planar convex body has infinitely many extreme points. At a boundary point choose a supporting line. Its intersection with the body is a point or a compact segment, and its endpoints are extreme points of the body: any convex decomposition of an endpoint must stay in the supporting line, where extremality is immediate. Thus every boundary point is a convex combination of extreme points. Every interior point lies between two boundary points, so the whole body is the convex hull of its extreme points. If there were only finitely many, the body would be a polygon with nonempty interior. At a vertex of that polygon the two adjacent sides give distinct supporting lines, contradicting differentiability. Hence the number of extreme points exceeds forty.

The Gamma density is absolutely continuous, has no atom at zero, and has a finite exponential moment by Lemma 2.1. All the hypotheses of [13, Theorem 1.2] are therefore satisfied. Apply that theorem at rate one to obtain  $C_\kappa$ . The coupling  $t_e^{(\kappa, \lambda)} = t_e^{(\kappa, 1)}/\lambda$  leaves every geodesic unchanged and proves the assertion for all positive rates with the same constant.  $\square$

For Gamma laws, differentiability also verifies the tangent and supporting-face endpoint assumptions of [10]. Its conclusions then retain the supporting-sector formulation. The single-direction statements of Corollary 17.1 use the strict convexity established for exponential weights.

### 17.4 Cone-tip competition for Gamma weights

We apply the tangent theorem of Ahlberg, Deijfen, and Sfragara [1, Theorem 1] to their simultaneous cone-versus-tip initial configuration.

**Corollary 17.4** (Gamma and Richardson cone-tip survival). *Fix  $\kappa, \lambda > 0$ , a deterministic unit vector  $\theta \in S^1$ , and a geometric half-angle  $0 \leq \alpha \leq \pi$ . Define the closed cone*

$$C_\alpha^\theta = \{0\} \cup \{x \in \mathbb{R}^2 \setminus \{0\} : \angle(x, -\theta) \leq \alpha\},$$

$$I_1 = (C_\alpha^\theta \cap \mathbb{Z}^2) \setminus \{0\}, \quad I_2 = \{0\},$$

where  $\angle(x, -\theta)$  is the Euclidean angle between the two vectors. Use one iid  $\text{Gamma}(\kappa, \lambda)$  field on unoriented nearest-neighbor edges of  $\mathbb{Z}^2$ , with passage metric  $T$ . At time zero, start type 1 from every site of  $I_1$  and type 2 from  $I_2$ . Both types use this same realization of  $T$ , and each site is permanently claimed by the type that reaches it first. With  $T(I, z) = \inf_{x \in I} T(x, z)$ , interpreting the infimum of the empty set as  $\infty$ , the sites eventually claimed by type 2 are almost surely

$$\mathcal{E}_2 = \{z \in \mathbb{Z}^2 : T(0, z) < T(I_1, z)\}.$$

Then

$$\mathbb{P}(|\mathcal{E}_2| = \infty) > 0 \iff \alpha < \pi/2.$$

In particular, at  $\alpha = \pi/2$ , when the cone is the closed half-plane  $\{x : x \cdot \theta \leq 0\}$ , type 2 claims only finitely many sites almost surely. At  $\kappa = 1$ , this gives Benjamini's cone-tip prediction for equal-rate Richardson competition with any common rate  $\lambda > 0$  in the same initial configuration.

*Proof.* The Gamma law is atomless for every  $\kappa > 0$ , including  $\kappa < 1$ . For four independent samples,

$$\mathbb{E}[\min(\tau_1, \dots, \tau_4)^2] \leq \mathbb{E}\tau_1^2 = \frac{\kappa(\kappa + 1)}{\lambda^2} < \infty.$$

Thus the continuity and minimum-second-moment hypotheses of [1, Theorem 1] hold.

The infinite seed set causes no ambiguity in the arrival-time comparison. If  $I_1$  is nonempty, choose one seed as a finite upper bound for  $T(I_1, z)$ . Only finitely many seeds lie in the corresponding passage ball about  $z$ , so the infimum is attained. Simple paths from distinct seeds to  $z$  have different edge sets. Conditioning on all edge weights except one in their symmetric difference and using atomlessness rules out equal costs; countability gives this simultaneously for all such paths and sites. Thus the shared-metric competition is described by the displayed strict comparison defining  $\mathcal{E}_2$ .

Theorem 1.2 gives a  $C^1$  boundary. For the fixed  $\theta$ , put  $h(\theta) = \max_{x \in \mathcal{B}} x \cdot \theta > 0$ . The supporting line

$$L_\theta = \{x : x \cdot \theta = h(\theta)\}$$

is perpendicular to the positive  $\theta$ -ray and meets that ray at  $h(\theta)\theta$ . At any contact point with  $\mathcal{B}$ , the  $C^1$  boundary makes  $L_\theta$  the unique supporting line, hence a tangent in the sense of [1]; the contact point need not lie on the ray. For a seed set  $I \subset \mathbb{Z}^2 \setminus \{0\}$ , write  $\mathcal{E}_2(I) = \{z \in \mathbb{Z}^2 : T(0, z) < T(I, z)\}$ , with the same empty-set convention for  $T(I, z)$ . If  $I \subset J$ , then  $T(J, z) \leq T(I, z)$  and hence  $\mathcal{E}_2(J) \subset \mathcal{E}_2(I)$ . If  $\alpha < \pi/2$ , choose  $\alpha < \alpha' < \pi/2$ . The source's cone of half-angle  $\alpha'$  contains the closed  $\alpha$ -cone, regardless of its boundary convention, so its Theorem 1 gives positive survival for the closed  $\alpha$ -cone as well. If  $\alpha \geq \pi/2$ , the closed  $\alpha$ -cone contains the closed critical half-plane. Theorem 5 gives almost-sure finite type-2 growth for that half-plane, and monotonicity gives the same conclusion for the larger cone. This proves the stated closed-cone convention throughout  $[0, \pi]$ , without strict convexity, for every prescribed orientation.

When  $\kappa = 1$ , the shared exponential edge field gives the equal-rate two-type Richardson model. The coupling  $\tau_e^{(\lambda)} = \tau_e^{(1)}/\lambda$  multiplies both arrival times by the same positive constant and leaves every comparison defining  $\mathcal{E}_2$  unchanged. This proves the stated specialization.  $\square$

The probability assertion is for each fixed orientation, not a simultaneous pathwise statement over all directions. Below the threshold it asserts a positive survival probability without claiming that this probability is one. It does not address unequal rates, independent type-specific metrics, coexistence from two finite seed sets, or the rate or asymptotic shape of type-2 growth.

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